THE LEF MODEL · A SERIES OF TWELVE PAPERS
Lunar Elevation & Fission

7. The Poles

Why the Poles Are Crowded

Parts A–E

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Abstract

Crater density analysis: polar caps versus the equatorial belt

According to the Robbins catalogue data, a significant difference is recorded in the density of craters 20 km in diameter and larger between the polar regions and the equatorial belt of the Moon. In the raw count, the polar caps (within 30° of each pole) show a density of 219 craters per million km², while the equatorial belt shows a density of 166 craters per million km², giving a density ratio of 1.32. For craters 40 km in diameter and larger, this ratio rises to 1.51.

The raw count is affected by a structural bias, since most of the maria and basalt plains – characterised by young, crater-poor floors – are concentrated in the equatorial region. To make a neutral comparison between ancient surfaces only, a "clean count" procedure is applied, in which the areas of the maria and of the South Pole–Aitken (SPA) basin are removed from the calculation on both sides.

In the clean count, a density ratio of 1.20 ± 0.05 in favour of the poles is established for all craters larger than 20 km in diameter. This ratio shows a rising trend as a function of crater diameter, as detailed in the following table:

Crater diameter range (km)	Density ratio (poles vs. equator)
20–30	1.09
60–100	1.53

Dynamic mechanism: the model of flattening, corotation and rounding of the young Moon

The geophysical hypothesis presented here rejects the assumption that the craters were formed by a uniform fall of asteroids or of returning terrestrial material (as proposed in Paper 6), since a uniform fall cannot produce a bipolar density concentration along a single symmetric axis. The present pattern is explained through the morphological evolution of the young Moon during the stages of its separation from the Earth:

1. Stage of gathering and initial flattening: A low-density sialic mass of about 1,300 kg/m³ consolidated at the tip of a pear-like structure, began to move outward and flattened on an initial base 6,000 km wide.

2. Expansion toward the corotation line: As the mass moved outward, its base narrowed to a width of 3,000 km. The body remained rotationally locked with the Earth at a period of about 5.5 hours. Near the corotation line – the dynamic distance at which the rotational velocity equalled the free orbital velocity – the body was subject to two opposing stress vectors: the Earth's gravity pulled inward (on the near side), and the centrifugal force pulled outward.

3. Triaxial geometry and stretched volume: Under these stresses the body took on a flattened and stretched triaxial shape (axis ratio of roughly 1.8 : 1.4 : 1), defined as long toward the Earth, intermediate along the rotational path, and short along the polar axis. In this inflated stage the density of the body stood at about 1,700 kg/m³. Most of the large craters formed near the peak of this stage, as a result of material that was ejected and returned.

4. Dynamic release and rounding of a rigid crust: The moment the Moon's centre of mass crossed the corotation line, the compressive force on the base dropped to zero. The Moon was released from the Earth at a distance of about 210 km beyond the corotation line. The release of the stretching led to a self-driven decrease in volume, to the rounding of the body, and to a rise in density to 2,600 kg/m³, with the outer shell already rigid.

This geometric transition from a highly flattened body to a spherical body caused the surface to contract, which led to a mechanical crowding of the existing craters. The degree of crowding was maximal at the poles and minimal at the equator. Ancient craters that formed close to peak flattening underwent the maximum contraction and crowding (1.53), whereas later craters, formed when the body was already close to its spherical shape, were hardly crowded at all (1.09).

Empirical tests and statistical findings (Part D)

The following is a summary of the four main tests carried out in Part D to examine the validity of the contraction and rounding model:

Test number	Definition of the scientific test	Empirical findings and numerical values	Overall status
1	Pole/equator density ratio in the clean count	A ratio of 1.20 ± 0.05 for all craters ($\geq$ 20 km). Rising from 1.09 to 1.53 with diameter.	Passed (consistent with the formation-order prediction).
2	Change in crater diameter	The craters kept their original morphology and size on a rigid crust. The polar median is 1.06 times the equatorial median (confidence interval: 1.01–1.12).	Passed (confirms a richness of large craters at the poles).
3	Distribution of young (post-rounding) craters	Ratio values of 0.92 and 1.12 were measured in two independent samples.	Consistent (awaiting an enlarged dating sample; no significant evidence of an excess).
4	Symmetry between the two poles	After removing the SPA basin, 300 craters per million km² were counted at one pole versus 305 craters per million km² at the opposite pole.	Passed (difference within the counting error).

In addition, five complementary morphometric tests were carried out (examining the largest craters per equal area, the shape of the size distribution, a comparison of ancient versus fresh craters, the degree of ellipticity, and the structural transition from simple to complex craters), designed to check whether the craters themselves physically shrank together with the surface. Four tests established unequivocally that crater diameter was preserved, and not a single test showed signs of shrinkage of the craters themselves, a finding that confirms the craters were carried as rigid, intact elements on the contracting crust.

External rotational differentiation. We propose that the high crater density at the lunar poles is the record of a single process during the attachment to the Earth. As the pear grew, the body flattened, the outer shell at the poles was stretched and thinned, and material landed on the lunar surface in a uniform distribution. When the body rounded, the polar surface contracted toward the pole centres and the craters were crowded together. The same process left a thin, compacted crust beneath the poles, which is measured today (C.8, D.6).

What this paper proposes

Summary of the scientific proposal: the model of structural deformation and rounding of the young Moon

The paper proposes a geophysical and dynamic mechanism that explains the excess of craters at the Moon's poles compared with its equator. According to the model, the varying

crater density is not the result of non-uniform impact, but of a mechanical contraction of the lunar crust during the transition from a highly flattened triaxial body to a rigid spherical body.

1. The central claim: the morphological evolution of the Moon

- **Stage of maximum flattening:** The young Moon moved outward on the tip of a pear-like structure, while remaining rotationally locked with the Earth (a period of about 5.5 hours). Near the corotation line (a base width of about 3,000 km), a balance of opposing stresses acted on it: the Earth's gravity pulled the near side inward, and the centrifugal pull drew it outward. Under these stresses the body was stretched into a highly flattened triaxial shape (axis ratio of roughly $1.8 : 1.4 : 1$) and was in an inflated state with a low density of about 1,700 kg/m³. The craters formed uniformly on this flattened shape.
- **Stage of rounding and release:** When the centre of mass crossed the corotation line, the net compression on the base dropped to zero and the Moon began to round with a self-driven decrease in volume (release of the stretching). The body became smaller and denser. The outer shell froze and stiffened before the final release from the Earth, which took place in a round state (maximum axis ratio of 1.04).
- **The crater-crowding mechanism:** The geometric transition from a flattened body to a sphere caused the surface to contract in a highly non-uniform way – the mechanical crowding and contraction of the surface were maximal in the polar regions and minimal in the equatorial belt, which raised the density of the existing craters at the poles.
- **External rotational differentiation (the key statement):** We propose that the high crater density at the lunar poles is the outcome of a single process, which took place while the young Moon was attached to the Earth and rotating with it at the rate dictated by the Earth. As the pear grew and the material at its tip consolidated into a body, the body flattened, and the light material of the outer shell flowed gradually from the poles to the equator. As a result of the stretching, the shell at the poles remained thin throughout the time window between peak flattening and the onset of contraction, a window of tens of millions of years. During this time material landed on the lunar surface in a uniform distribution. In the last stage the body rounded up to its release on the corotation line; the polar surface contracted toward the pole centres, and the craters were compressed into a smaller area. The crater density at the poles, and the thin, compacted crust beneath them, are the record that the process left behind (C.8, D.6).

2. The numerical measures and the scientific predictions

- **Reading the shape from the crater record:** Counting the craters (in the diameter range ≥ 20 km) around the six ends of the body's three axes reconstructs a structural shape of $1.37 : 1.15 : 1$. A theoretical rounding of this shape predicts a clean density ratio of 1.21 between the poles and the equator, a value that matches closely the measured empirical ratio (1.19), without any parameter tuning.
- **Range of predictions by formation order:** The model does not predict a single fixed value, but a range of values that depends on the timing of crater formation relative to peak flattening:

- **Craters formed at peak flattening (large craters):** underwent the maximum contraction (rounding from a shape of $1.8 : 1.4 : 1$), and therefore a density ratio of about 1.5 in favour of the poles is predicted for them.
- **Craters formed late in the rounding stage:** formed on a body that had already begun to round (a shape of about 1.20, near a density of 2,600 kg/m³ and a rotation period of 5.77 hours), and therefore a density ratio of about 1.17–1.2 is predicted for them.
- **Craters formed after release:** formed on a fully round body, and therefore their density ratio is 1.0 (no polar excess).

- **Agreement with the empirical data (the clean count):** Removing the maria and the SPA basin (30° caps versus the equatorial belt $|\phi| < 15^\circ$) gives an overall ratio of 1.20 ± 0.05 for craters of diameter ≥ 20 km, showing a consistent and clear rise as crater diameter increases (a finding confirming that the large craters are the most ancient, formed close to the peak):

Crater diameter (km)	Empirical density ratio (poles vs. equator)
20–30	1.09
30–40	1.15
40–60	1.26
60–100	1.53
100–300	1.84 ± 0.38

- **Independent density checks:** The shape of the Moon at peak flattening ($b/c \approx 1.4$) matches precisely the Maclaurin spheroid, which represents the equilibrium of a uniform fluid rotating with a period of 5.5 hours for a body with a density of 1,600–1,750 kg/m³. To allow the ends of the Earth-facing axis to contract during rounding, the physics dictates that the peak density must be lower than 1,800 kg/m³.

3. The structural and mechanical requirements of the model

For the proposed mechanism to hold, the geophysical system is required to meet the following mechanical conditions:

Requirement area	Mechanical description of the requirement	Reference in the paper
Radial and structural	An interior soft enough to allow flow and rounding in the attached stage, beneath a viscous/rigid outer crust.	C.7
Morphological preservation	A crust (shell) rigid enough to preserve the integrity of the craters and their original diameter while the surface beneath them shortened at the poles.	C.7

Requirement area	Mechanical description of the requirement	Reference in the paper
Mechanical support at release	A crust with the mechanical strength to carry and stabilise the body during the geometric transition from a shape of 1.20 to a round body at release.	C.1
Resistance to tensile stresses	The crust is required to withstand the residual tensile stresses that remained beyond the corotation line, up to a dynamic distance of about 20,100 km.	C.6
Absorption of area (closure)	The closure of the surface during rounding – estimated at about a quarter of the average area and about half of the polar area – was taken up and absorbed within the crustal structure.	C.7, E.3
Non-contraction of free ends	The end facing away from the Earth (the summit of the dome) acted mechanically as a free end and did not contract, and therefore it stands consistently with the far-side measures.	A.9
The polar shell	During flattening the outer shell at the poles was thinned by the outward flow of light material, and the contraction of the rounding stage was concentrated in it; the polar crust therefore remains thinner than the equatorial crust today.	C.8, D.6

- **Dynamic note on the crater excess in the trailing hemisphere:** The excess in this hemisphere is explained by two combined mechanisms: structural vents on that side itself, which ejected returned material that created craters around them, and the landing of material that lagged behind the Moon's rotational velocity and came down behind (on the trailing side).
- **Limitation of the numerical values:** The author notes that the values of the peak shape and of the density path were derived from a preliminary and limited calculation, and that focused research may lead to an estimated update of the exact numerical values.

The computational environment (a): body data and orbital dynamics

The following defines the physical parameters, distances and dynamic states that serve as the basis for computing the model of the forming Moon (the LEF model) in the attached stage:

Scientific quantity or condition	Numerical value	Physical meaning and dynamic rationale	Source / reference
Rotation period in the attached stage (maximum flattening)	About 5.5 hours (5.77 at release)	The Earth and the attached body rotate together owing to a mechanical constraint. The elongated pear exerts a torque that slows the Earth, while the centralising core speeds it up. This high rotation rate is the physical mechanism that flattened the young, soft body.	Paper 2
Rotation period after orbital locking	≈ 6.2 hours at release; 14 hours at 30,000 km; 41 hours at 60,000 km	Post-release: the spin follows the orbit and slows as the body recedes. At this stage the body is already round and its shell rigid; the tidal and rotational forces raise only a small bulge (axis ratio 1.03 at 30,000 km, and 1.00 at 60,000 km). The early flattening disappeared entirely before release.	Paper 4
Corotation line radius	16,331 km (period of 5.5 hours); 16,399 km ($15,818$ km with a two-body correction at a period of 5.77 hours)	An equilibrium point at which the Earth's pull equals the centrifugal contribution in the rotating frame. Beyond this distance (≈ 2.6 Earth radii), the net stress vector points outward and the body moves at free-orbit velocity.	Paper 2
Density of the body through the stages	$1,300 \rightarrow \approx 1,700 \rightarrow 2,600 \rightarrow 3,344$ kg/m³; equivalent-sphere radii: $2,380 \rightarrow 2,180 \rightarrow 1,890 \rightarrow 1,737$ km	A continuous transition from a light sialic mass on the tip of the pear $(1,300)$, to a stretched, inflated, voluminous body at peak flattening $(1,700)$, which shrinks and stiffens at release $(2,600)$ down to the present density. A light body is more susceptible to flattening at the same rotation rate.	Paper 2; Paper 3 §B.2; Appendices C.1, C.3
Rigid Roche limit	$\approx 13,000$ km at a density of 1,300 kg/m³ (10,314 km at a density of 2,600; varies as a function of $\rho^{-1/3}$)	The distance beyond which the tidal forces are weaker than the body's self-gravity, so that the body can hold its own shape by gravity and mechanical strength. The body flattens on a base that narrows from 6,000 to 3,000 km. A free fluid body with a density of 2,600 kg/m³ would require a distance of 19,971 km ($k < 0.09$ of about 20,100 km) to hold itself together – a distance not reached before release, when the body was held by its rigid shell.	Paper 2; Appendices C.2, C.5
Mechanical release radius	The corotation line at $\approx 16,400$ km at the equator (17,287 km at 22.5°); release about 210 km beyond it	The onset of separation occurs when the centre of mass reaches the corotation line. The final release takes place about 210 km beyond it, when about 58% of its mass has crossed the line. The mechanical contact offers slight resistance and the body is drawn outward. At this stage the body is rigid and round (axis ratio ≤ 1.04).	Papers 2 and 4

Scientific quantity or condition	Numerical value	Physical meaning and dynamic rationale	Source / reference
Range of flattening and rounding	Flattening: from the tip until the base narrows to 3,000 km near corotation. Rounding: from the centre of mass crossing the line until release.	The body flattens during its outward motion and reaches peak flattening near the corotation line. On crossing the line, the compressive force on the base drops to zero, the stretching is released, and the body rounds, shrinks in volume and detaches. Late in the rounding about 75% of the flattening had disappeared, and at release the body is round.	Paper 3; the present discussion
Source pressure and lifting of material	Source pressure: 150 GPa; pressure attributed to lifting: 140 GPa	Internal pressure in the Earth generated by rotation and density sorting, which feeds and drives material through the pear structure to the Moon. The mechanical detachment cuts off the pressure flow, exposes the vents of the pear, releases the energy and prevents its further elongation.	Paper 3 §A; Appendix C.6

The computational environment (b): geometric record data and the empirical count

The following defines the measures, databases and basic geometric assumptions used to compute surface areas and to convert crater counts into normalised densities:

Scientific quantity or condition	Numerical / geometric value	Physical meaning and dynamic rationale	Source / reference
Static mass of the Earth	5.972×10^{24} kg	The target mass of the Earth used in the gravity, orbit and tide equations, after the completion of the mass transfer to the lunar body.	Paper 2
Fixing of the Moon's shape (fossil flattening)	Distance range: before 16 Earth radii; fossil flattening: $f \approx 0.001$	Defines the orbital distance at which the Moon's global shape froze. This row does not dictate a simultaneous solidification of the entire lunar interior. The fossil flattening represents the Moon's present deviation from a perfect sphere, preserved as a result of the early rapid rotation.	Appendix 4 §C.3
Crater record database	Robbins catalogue (2019): 6,973 craters ($D \geq 20$ km); 2,329 craters ($D \geq 40$ km)	The two morphometric control groups used to compare densities per unit area. The base count is made without geological masks; the "clean count" procedure (removal of maria and basins) is defined separately in section A.11.	Paper 11 §B.8

Scientific quantity or condition	Numerical / geometric value	Physical meaning and dynamic rationale	Source / reference
Axis system and definition of the poles	Sub-Earth point: (0° N, 0° E); geographic poles: $\pm 90^\circ$	The angular-distance belts and the spatial geometry are computed directly from the point permanently facing the Earth. The polar caps are defined mathematically around the north and south poles to give an absolute separation between the counting regions.	Paper 11 §B.8; map of Paper 3
Lunar radius for area calculations	1,737.4 km	The official mean radius (IAU), used as a fixed geometric constant to convert raw crater numbers into standard density units (craters per million km²).	The paper's area calculations

Dynamics of the detachment stages: morphological evolution and stress balance

The following is a chronological and mechanical layout of the stages in the development of the young Moon – from the consolidation of the initial mass on the tip of the pear structure to its modern state – defining the balance of forces, the density changes and the geometric transformation:

Dynamic stage	Approximate orbital distance (km)	Density ($\mathrm{kg/m^3}$)	Spin period (hours)	Geometry and axis ratio ($a:b:c$ or a/c)	Mechanical state and stress balance in the system
Mass gathered at the tip of the pear	$\approx 8,991$	1,300	5.50	—	Sialic material sits on the tip of a long extension of the Earth (a base 6,000 km wide); the start of outward motion and the development of initial flattening.
Independent attached body (the rigid Roche limit)	$\approx 12,994$	$\rightarrow 1,300$	5.50	Rising trend in flattening	The body reaches the distance at which its self-gravity and strength overcome the tidal forces ($k > 0.09$). The body holds its own shape but remains attached; it flattens and is stretched outward while the base of the pear narrows from 6,000 to 3,000 km.

Dynamic stage	Approximate orbital distance (km)	Density ($\mathrm{kg/m^3}$)	Spin period (hours)	Geometry and axis ratio ($a:b:c$ or a/c)	Mechanical state and stress balance in the system
Peak flattening	Near the corotation line	$\approx 1,700$	5.50	$1.8:1.4:1$	The body is stretched, inflated and reaches maximum geometric deformation. It is subject to an extreme balance of stresses: the Earth's gravity pulling inward (on the near side) against the centrifugal pull outward. Mechanically compressed on a base of 3,000 km.
Onset of rounding (crossing the line)	Beyond the corotation line	1,700	5.55	1.40	The centre of mass crosses the corotation line. The net compressive force on the base drops to zero; the egg part (axis a) is released, and the global volume begins to shrink.
Intermediate rounding stage (a)	Intermediate stage	2,000	5.62	1.30	Continuous release of the tensile stresses; the body thickens and becomes gradually more compact.
Intermediate rounding stage (b)	Intermediate stage	2,300	5.70	1.24	Continued inward mechanical collapse of the stretched axes and a rise in the density of the core material.
Late in rounding	Near the release line	2,600	5.77	1.20	About 75% of the maximum excess flattening of the long axis over the polar axis has disappeared (a contraction of about 0.6 of the radius). The outer shell freezes, stiffens and fixes the crater record.
Mechanical release (detachment)	$\approx 16,610$ (about 210 beyond the line)	2,600	5.77	≤ 1.04	Opening of the contact layer and final separation from the Earth. The body is released in an almost fully round state, protected by a rigid crust that carries the craters intact.
Receding stage (a)	30,000	2,600	14.00	1.03	The spin locks to the orbit and slows gradually. The 1.03 is not fossil flattening but a small elastic bulge raised on the rigid body by tidal and rotational forces that are still fast.

Dynamic stage	Approximate orbital distance (km)	Density ($\mathrm{kg/m^3}$)	Spin period (hours)	Geometry and axis ratio ($a:b:c$ or a/c)	Mechanical state and stress balance in the system
Receding stage (b)	60,000	2,600	41.00	1.00	Decay of the dynamic bulge toward a perfect sphere as the distance increases and the spin slows.
The modern state (today)	$\approx 384,400$	3,344	27.3 days	1.00	The body is round (carrying a minor fossil flattening of $f \approx 0.001$ that froze before a distance of 16 Earth radii).

Chronology of events and the crater-formation window

The geodynamic model defines an overall timeline of about 600 million years from the start of the formation of the original bulge to the final release stage (dated to 3.873 billion years before present), divided into three sub-segments that form the exclusive morphogenetic window of the craters in this paper:

- **Initial pear stage (about 400 million years):** the uplift of the sialic mass on the extension of the Earth's material and the growth of the bulge.
- **Dynamic flattening stage (about 100 million years):** the stage of extreme triaxial deformation and elongation of the axes near the corotation line.
- **Attached rounding stage (about 100 million years):** the stage of mechanical release of the stretching, thickening of the body, stiffening of the crust and final detachment.

Basic definitions and mechanical measures for reading the table

- Tidal axis a: the spatial axis running along the symmetric line that directly connects the forming Moon with the centre of the Earth.
- Motion axis b: the axis perpendicular to the tidal axis, directed along the body's orbital velocity vector.
- Polar axis c: the geographic axis perpendicular to the plane of the system, connecting the poles (the shortest axis during flattening).
- **"Holds itself together":** the mechanical ability of the body to maintain integrity and a closed geometric shape within the equations of a fluid body, under the influence of self-gravity and structural strength (it does not mean neutralising the external tidal forces).
- **"Detachment":** physical separation and full mechanical opening of the contact layer and interface between the Moon and the pear material of the Earth.
- Physical confirmation of the peak density ($\approx 1,700\ \mathrm{kg/m^3}$): this value is supported by two independent dynamic checks:
  1. The pita part of the body ($b/c \approx 1.4$) matches mathematically the exact equilibrium shape of a Maclaurin spheroid for a homogeneous fluid rotating with a period of 5.5 hours at a density of 1,600–1,750 $\mathrm{kg/m^3}$.

2. The upper mechanical limit for the contraction of the tidal-axis ends during rounding physically requires a peak density lower than 1,800 $\mathrm{kg/m^3}$.

- The rigid Roche limit ($10,314$ and $12,994$ km): computed from the structural equation $1.26R_{\oplus}(\rho_{\oplus}/\rho)^{1/3}$. A lighter body ($1,300\,\mathrm{kg/m^3}$) has a weaker self-gravitational hold, and therefore its Roche limit lies farther from the source ($12,994$ km) – where it first succeeds in stabilising an independent shape while still attached.

The free-fluid stability criterion (the k measure)

The mechanical stretching is assessed using the system measure:

$$\mathbf{k} = \frac{\mathbf{M_E}}{\rho \mathbf{r^3}}$$

- **Fluid stability threshold:** set at $k < 0.09$. When the value of the measure is below this threshold, a free fluid body can maintain a closed, stable shape without breaking apart under the Earth's tidal forces (M_E).
- **Parametric sensitivity:** Since the orbital distance (r) appears to the third power, the value of the measure k falls rapidly as the body recedes. Conversely, since the density (ρ) is in the denominator, a body of higher density will bring the value of k below the critical threshold earlier and closer to the source. A free fluid body at the release density would require a distance of about 20,000 km to reach stability – a distance the Moon did not reach in the attached stage, and hence the structure held together thanks to its rigid shell alone.

Introduction — the question and how it is tested

1. Research methodology: definition of the counting regions and the empirical measures

To carry out the morphometric comparison and to prevent biases arising from incomplete identification of small craters, a firm minimum threshold is set for the diameter of the counted craters. The data are based on the Robbins catalogue (2019) and are analysed by the following geometric units:

- **Normalised density measure:** defined as the total number of craters in a given region divided by its static area. The measure is expressed in craters per million km², and thus allows a neutral comparison between area cells of different geometric dimensions.
- **The polar caps:** two geometric caps of equal area, defined by an angular radius of 30° from the Moon's centre around the rotation poles (perpendicular to the equatorial plane).
  - **The northern cap:** extends from latitude $60^\circ\ \mathrm{N}$ to $90^\circ\ \mathrm{N}$.
  - **The southern cap:** extends from latitude $60^\circ\ \mathrm{S}$ to $90^\circ\ \mathrm{S}$.
- **The equatorial belt:** a global band defined by the latitude range between $15^\circ\ \mathrm{S}$ and $15^\circ\ \mathrm{N}$.

- **Spatial assignment:** each crater is catalogued and assigned to the relevant counting region according to the coordinates of its geometric centre point.

2. Findings of the raw count versus the clean-count procedure

The initial empirical count shows a significant density excess in the polar regions, which becomes more pronounced as the diameter of the examined craters increases. However, geological analysis shows that the raw count suffers from severe structural biases that require mathematical correction:

- **The equatorial-belt bias (maria and basalt plains):** The basalt and maria areas (the LROC feeding system) are young floors on which the ancient craters were melted or covered by thermal activity and by crust melted by the Earth's heat. These areas make up 22% of the equatorial belt but only 1% of the polar caps. Including them in the raw count artificially lowers the equatorial density and inflates the pole/equator ratio.
- **The polar-cap bias (the South Pole–Aitken basin):** The SPA basin physically erased the craters at its centre. Since 24% of the area of the polar caps (mainly in the south) lies inside the basin, including it makes the poles look emptier than they really are and lowers the ratio.
- **The clean-count procedure:** removing the areas of the maria and the SPA basin from both sides of the equation uniformly. This is the only procedure that allows ancient surfaces (Highlands) to be compared with ancient surfaces, and from it is derived the firm system ratio on which the study rests: 1.20 ± 0.05.

Crater measure examined	Density in the northern cap	Density in the southern cap	Density in the equatorial belt	Density ratio (poles vs. equator)	Methodological status
Diameter ≥ 20 km (raw)	227 per million km² (577 craters on ~2.54M km²)	211 per million km²	166 per million km²	1.32 (combined density: 219)	Biased (includes maria and SPA)
Diameter ≥ 40 km (raw)	—	—	—	1.51	Biased (trend sharpens for large craters)
All craters (clean count)	—	—	—	1.20 ± 0.05	The official normalised count

3. The dynamic starting assumption and the mechanical contraction mechanism

- **The assumption of uniform material distribution:** Within the LEF model (Paper 6), the ancient crater population was created by the landing of returning terrestrial material on the crust while it was still spread as a bulge above the parent body (the Earth). For the purpose of the analysis, a working assumption is set of a completely uniform distribution of material per unit area. This assumption neutralises the question of source asymmetry: any non-uniformity in density seen on the surface today must stem from later mechanical processes that acted on the crust that carried the craters.

- **Rejection of alternative hypotheses:** In the course of the study, six different physical mechanisms were examined as possible explanations for the density difference; all six explanations were rejected and set aside.
- **The seventh explanation (rounding and volume decrease):** The young Moon behaved physically as a flattened, squashed body (like a balloon pressed between hands that gradually loses volume). At peak flattening, owing to the symmetric rotational stresses around the rotation axis, the body was wide and flat at its ends and narrow at its equator. The craters were imprinted uniformly on this morphology.
- **Crowding of the poles:** The moment the centre of mass crossed the corotation line and the stretching was released while the body was still attached, the mechanical hands "moved apart" and the body tends toward a spherical shape with a decrease in volume. In this geometric process, the flat, wide parts converged inward toward two small polar caps. Since the rotation axis passes through the poles and the flattening was symmetric around it, every meridian retreated toward the two end points, which the rotation never moved dynamically (C.3).
- **Morphometric preservation of the craters:** The Moon's outer shell was rigid enough (C.7) to carry the craters as intact, isolated elements. The craters themselves kept their original diameter and size; it was the elastic surface between them that shrank and contracted (the surface closed by a quarter of the average area and by about half of the polar area), which brought a direct rise in the density measure in the polar regions compared with the equatorial region (where the surface converged the least).

1. Predictions of the geometric model and the differential crowding mechanism

The degree of mechanical crowding of the craters is not a direct function of chronological age alone, but a geometric record of the body's shape and dynamic flattening at the moment of crater formation. Since material was ejected and returned throughout all the stages of the transition, the level of structural deformation was preserved within the crustal morphology according to the following principles:

- **Formation at peak flattening:** Craters that formed when the body was at the peak of triaxial deformation ($1.8 : 1.4 : 1$) were carried through the entire rounding sequence. As a result, they underwent the mechanical maximum of surface contraction, and today show a predicted density ratio of 1.49 in favour of the polar caps.
- **Formation in late rounding stages:** Craters that formed when the Moon had already begun to collapse inward and had rounded to a shape of 1.20 (near the shell-stiffening stage) experienced less residual surface contraction, and therefore their predicted density ratio drops to 1.17.
- **Post-release formation:** Craters that formed on a body that had already completed the transition to a spherical shape experience no surface contraction at all, and therefore their predicted ratio is 1.00 (a completely uniform distribution).
- **Correlation between crater diameter and timing of formation:** Most of the largest craters formed near the peak of the stage of ejection and return of the terrestrial material. Therefore, the model consistently predicts that the density ratio will rise as crater diameter increases.

2. Empirical correlation: the model prediction versus the clean-count measures

The following is a firm, normalised comparison between the predicted theoretical density ratios (derived from the Moon's shape at the time of formation) and the empirical measures actually obtained in the clean count (removal of the maria and the SPA basin) by diameter class:

Assigned dynamic stage	Theoretical body shape ($a:b:c$)	Predicted density ratio (caps/belt)	Crater diameter class (km)	Measured empirical density ratio	Directly reconstructed shape (inverse reading)
Structural peak flattening	$1.8:1.4:1$	1.49	100–300	1.84 ± 0.38	$2.24:1.74:1$
Crossing the corotation line (release of the egg)	1.40	1.34	60–100	1.53 ± 0.19	$1.70:1.42:1$
Middle rounding stage	1.30	1.26	40–60	1.26 ± 0.12	$1.34:1.21:1$
Late rounding stage	1.20	1.17	30–40	1.15 ± 0.11	$1.11:1.07:1$
Post-release (rigid round body)	1.0	1.0	20–30	1.09 ± 0.07	$1.05:1.03:1$
Whole population (≥ 20 km)	$1.37:1.15:1$	1.21	All craters ≥ 20	1.20 ± 0.05	—

- **Analysis of trends and uncertainty:** The measured empirical values span the theoretical range precisely and preserve its ordering (a decrease in the excess as the craters become smaller). The whole group (≥ 20 km) shows an almost complete match between the mechanical prediction of the six axes (1.21) and the finding on the surface (1.20 ± 0.05). The largest crater class (100–300 km) carries a wide statistical uncertainty (± 0.38) owing to a small sample.
- **Productivity of the method:** The row-by-row pairing serves as a physical guide to reading the morphological trend, and is not a claim that each crater was assigned chronologically and exclusively to the specific stage.

3. Preservation of crater diameter on a contracting crust

One of the central questions in developing the geophysical model is: if the polar surface contracted mechanically during rounding, why did the craters themselves not contract with it (which would have taken them below the 20 km counting threshold and made the effect disappear)?

- **The isolated rigid-crust mechanism (C.7):** Within the physical model, the craters kept their morphological integrity and their original diameter. They were carried as rigid elements on a solid upper crust, while the contraction of the surface was accomplished by the closing and

narrowing of the elastic spaces between the craters. Therefore, the crater diameter measured today reflects precisely the original diameter at the moment of impact.

- **Mathematical evidence from the size distribution (D.2):** Had the craters contracted physically together with the surface, the crowding effect would have become statistically invisible at any fixed counting threshold. The rise in the number of craters per unit area would have been entirely neutralised by the drop in diameter of the borderline craters below the 20 km threshold (the two opposite effects in the size distribution of lunar craters cancel each other under homogeneous contraction). The very existence of the significant measured excess (1.20) is direct empirical proof that the craters kept their original size.

4. Structure and order of the parts of the paper

The remainder of the scientific analysis in the paper is organised in the following firm structure:

- **Part A (the empirical observation):** presentation of the full count data by latitude, by longitude, within the polar caps and around the six geometric axes of the body (in both the raw and the clean versions).
- **Part B (rejection of alternative hypotheses):** a detailed analysis of the six alternative scientific explanations that were examined and the mechanical reasons for their rejection.
- **Part C (the physical mechanism):** the mathematical calculation of the differential area changes that accompany the transition from a flattened triaxial body to a perfect sphere.
- **Part D (the four structural tests):** presentation of the results of the four statistical tests of the model's validity (density ratios, median diameters, young populations and polar symmetry).
- **Part E (summary and remainders):** definition of the topics and open questions left for future research.

PART A

Part A — The empirical observation

Part A presents the raw empirical findings as they are counted on the surface, before applying any physical interpretation or formation models. The observation is based on a normalised count: the number of craters in each area cell is divided by the geometric area of that cell.

In this part, four independent counts are made on the same database and with the same methodology:

- **A.1 (latitudinal section):** division of the Moon into latitude bands parallel to the equator, to examine the density difference between the poles and the equator.
- **A.2 (longitudinal section):** division of the Moon into longitudinal segments from pole to pole, to examine the density difference between the different sides of the Moon (a separate and distinct pattern).
- **A.3 and A.4 (clean procedure):** an exact repetition of sections A.1 and A.2 after removing young floors (maria, basalt plains) and large basins (SPA) and recalculating the remaining

areas.

- **A.5 (summary of observations):** a compilation of the four spatial tests and their conclusions.

A.1 Crater density by latitude (raw data)

The spatial analysis in this section is designed to determine whether there is a non-uniformity in crater density along the geographic latitude axes.

Methodological criteria and calibration of the count:

1. Completeness of the database (Robbins catalogue): The count is limited to craters with a fitted diameter of 20 km and larger. This limit is firm and guarantees full counting completeness at all latitudes: it neutralises optical biases (low illumination and long shadows at the poles that hide small craters) and filters out secondary craters (clusters and chains of debris smaller than 20 km in diameter, which would produce misleading local densities).

2. Control group and confirmation of the trend: A second series is made for craters 40 km in diameter and larger, to verify that the spatial pattern does not rest on the population of the smallest borderline craters.

3. No initial masks: The first count presents the surface as it is, without artificially omitting the maria, on the view that they form part of the structural prediction (the effect of removing them is analysed separately in Part A.3).

4. Calculation of differential geometric areas: The Moon is divided into 12 latitude bands of fixed width 15°. The area of each band is computed by the spatial equation:

$$\text{Area} = 2\pi R^2 (\sin \phi_2 - \sin \phi_1)$$

where the adopted mean radius is $R = 1,737.4\ \text{km}$ (IAU standard), and ϕ_1, ϕ_2 are the boundaries of the band.

According to this equation, the area of a band on a sphere is determined not by its angular width but by how far it extends along the polar axis (like the height of a label on a cylinder). Near the equator the band is tall along the axis, while near the poles the surface curves sharply and the circumference of the ring is small. As a result, the band areas vary considerably: the equatorial band ($0-15^\circ$) extends over 4.909 million km², while the polar band ($75-90^\circ$) extends over only 0.646 million km² (a ratio of about 7:1). This geometric difference requires the use of the density measure (craters per 10^6 km²) rather than absolute numbers.

Summary of crater-count data by latitude band (A.1)

The following are the full numerical and relative data measured in the 12 global latitude bands, without applying geological masks:

Sector	Geographic description of the region	Latitude band (°)	Computed area (10^6 km^2)	Craters ($\geq$ 20 km)	Density per 10^6 km^2	Ratio to the equator ($\geq$ 20 km)	Craters ($\geq$ 40 km)	Density per 10^6 km^2	Ratio to the equator ($\geq$ 40 km)
•	South — adjacent to the pole	75–90° S	0.646	139	215	1.29	53	82	1.54
•	South — outer part of the polar cap	60–75° S	1.895	397	210	1.26	132	70	1.31
•	South — next to the polar cap	45–60° S	3.014	693	230	1.38	230	76	1.44
•	South — mid-latitudes	30–45° S	3.928	806	205	1.23	272	69	1.30
•	South — low latitudes	15–30° S	4.574	817	179	1.07	287	63	1.18
•	Equator — southern half-band	0–15° S	4.909	873	178	1.07	283	58	1.08
•	Equator — northern half-band	0–15° N	4.909	761	155	0.93	239	49	0.92
•	North — low latitudes	15–30° N	4.574	693	151	0.91	233	51	0.96
•	North — mid-latitudes	30–45° N	3.928	692	176	1.06	207	53	0.99
•	North — next to the polar cap	45–60° N	3.014	525	174	1.05	171	57	1.07
•	North — outer part of the polar cap	60–75° N	1.895	383	202	1.21	157	83	1.56
•	North — adjacent to the pole	75–90° N	0.646	194	300	1.80	65	101	1.89

- **Setting the reference point (normalisation):** A base value of 1.00 is defined as the combined density of the full equatorial band (15° S $-$ 15° N). In the ≥ 20 km group the reference density

stands firmly at 166.436 craters per million km² (a total of 1,634 craters over a total area of 9.818 million km²). In the $\geq 40\ \mathrm{km}$ group the reference density is 53.170. The halves of the equatorial band show an internal asymmetry (0.93 in the north versus 1.07 in the south), and only their combination yields the normalised reference value.

- **Ranges of the density scales:** For the $\geq 20\ \mathrm{km}$ group the values range between 151 and 300; for the $\geq 40\ \mathrm{km}$ group the range is between 49 and 101 craters per million km². The ratios are computed from raw, unrounded count and area values.

Analysis and conclusions from the spatial profile

Reading the density columns and the graphic profile yields the following main findings:

- **A firm bipolar upturn:** Both ends of the table show the maximum density values in the system. The band adjacent to the north pole ($75{-}90^\circ\ \mathrm{N}$) reaches a peak density of about 300 craters per million km², a value 1.80 times that of the equator. The band adjacent to the south pole ($75{-}90^\circ\ \mathrm{S}$) stands at a density of 215, 1.29 times that of the equator.
- **The equatorial minimum zones:** The most crater-poor rows lie north of the geographic equatorial belt, in the ranges $0{-}15^\circ\ \mathrm{N}$ and $15{-}30^\circ\ \mathrm{N}$, with minimum density values of 155 and 151 respectively.
- **Sharpening of the trend at larger sizes:** In the control group of craters of diameter $\geq 40\ \mathrm{km}$, the bipolar density difference sharpens and shows extreme ratio values of 1.89 in the north and 1.54 in the south relative to the equatorial reference.
- **Mid-hemisphere asymmetry:** The profile does not rise in a perfectly linear and smooth way from the equator to the poles. The southern hemisphere is richer in crater density than the northern hemisphere at mid-latitudes (a value of 230 in the $45{-}60^\circ\ \mathrm{S}$ sector versus 174 in the $45{-}60^\circ\ \mathrm{N}$ sector). The gradient rises and falls in different places, but the structural conclusion is firm: the end bands adjacent to both poles carry an exceptional density excess that no intermediate band reaches. Part of the low value at the centre stems from the concentration of maria at the equator, a factor treated and neutralised in section A.3.



Colour scale: left, “low density” (soft yellow); right, “high density” (soft red).

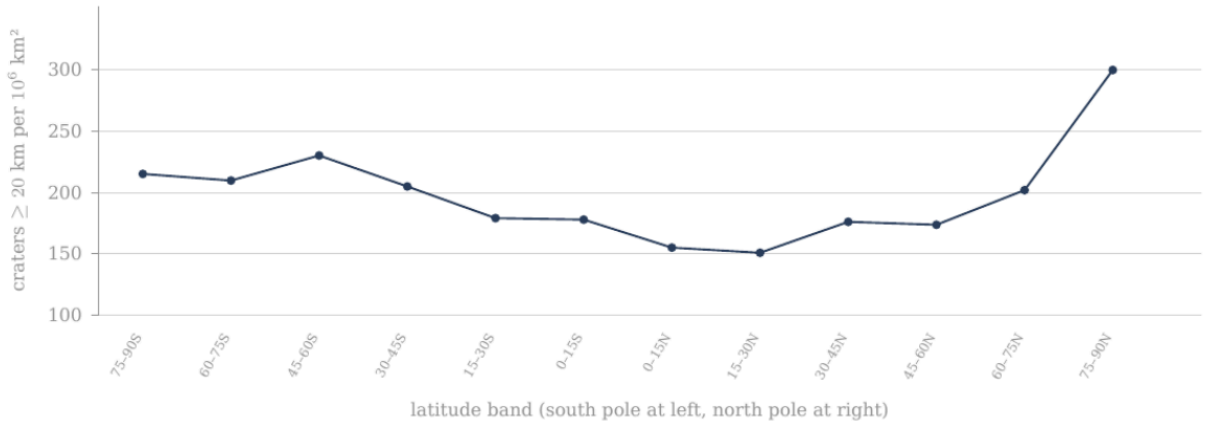


Figure 1. Crater density by latitude, craters of 20 km and larger, in twelve bands of 15° (A.1); the whole catalogue, without a mask. Both poles stand highest. (In the plot: vertical axis — craters ≥ 20 km per 10^6 km², 100–300; horizontal axis — latitude band, the south pole on the left and the north pole on the right, from 75–90S to 75–90N; the dark-blue line — the density in each band.)

A.2 Crater density by longitude (raw data)

Whereas section A.1 found a significant density excess in the polar regions, this section examines the uniformity of the spatial distribution in the perpendicular direction – along the Moon's global circumference. The aim of the analysis is to isolate and prevent a statistical mixing between the latitude effect (the central topic of the study) and potential asymmetries between different hemispheres along the longitude axes.

Methodological criteria and segment geometry:

1. Structure of the counting cells: The lunar surface was divided into 12 longitude bands (segments) of fixed width 30°. Each band extends continuously from pole to pole and includes all latitudes (like a segment of an orange).

2. Built-in equality of areas: Unlike the latitude bands, all longitude segments on a sphere have exactly equal areas. This geometric form removes the need for correction factors or normalisation by cell size, and allows a direct comparison between the absolute crater numbers (the raw counts) and the density ratios.

3. Surface dimensions: The area of each band is computed exactly by:

$$A = \frac{4\pi R^2}{12}$$

For a radius of $R = 1,737.4\ \mathrm{km}$, the area of each longitude segment is fixed at 3.161 million km² (1/12 of the total lunar surface).

4. The dynamic coordinate system: The band covering the sector $345^\circ – 15^\circ$ east contains the sub-Earth point ($0^\circ\ \mathrm{N}, 0^\circ\ \mathrm{E}$) – the centre of the face permanently turned toward the Earth owing to tidal locking.

5. Definition of the orbital elements ("leading" vs. "trailing"): The terms are applied according to the accepted mechanical definitions:

- **The leading point:** the apex of the Moon's orbital motion vector, located at $(0°, 90° \text{ W})$. It faces forward in the direction of the orbit (the front of the body).
- **The trailing point:** the point facing backward along the direction of orbital flight, located at $(0°, 90° \text{ E})$.

6. Fixed normalisation base: The reference point (1.00) remains exactly the same as that set in section A.1 – the density of the full raw equatorial band $(15° \text{ S} - 15° \text{ N})$. Its value is 166.436 craters per million km² for the $\geq 20 \text{ km}$ group, and 53.170 for the $\geq 40 \text{ km}$ group. Keeping this fixed base allows a direct and fair comparison between the four analysis tables of Part A (from A.1 to A.4).

Summary of crater-count data by longitude band (A.2)

The following are the full numerical data measured in the 12 global longitude segments, without applying geological masks:

Sector	Geographic and morphological definition of the sector	Longitude band (° east)	Computed area (10^6 km^2)	Craters ($\geq$ 20 km)	Density per 10^6 km^2	Ratio to the equator ($\geq$ 20 km)	Craters ($\geq$ 40 km)	Density per 10^6 km^2	Rat to t equ ($\geq$ 40
•	Around the meridian of the sub-Earth point	15–345	3.161	479	152	0.91	168	53	1.0
•	Near side — east of the sub-Earth point	45–15	3.161	505	160	0.96	135	43	0.8
•	Near side — toward the trailing meridian	75–45	3.161	611	193	1.16	170	54	1.0
•	Around the trailing meridian (90° east)	105–75	3.161	750	237	1.43	269	85	1.6
•	Far side — beyond the trailing meridian	135–105	3.161	793	251	1.51	285	90	1.7
•	Far side — toward the anti-Earth point	165–135	3.161	667	211	1.27	245	78	1.4

Sector	Geographic and morphological definition of the sector	Longitude band (° east)	Computed area (10^6 km^2)	Craters ($\geq$ 20 km)	Density per 10^6 km^2	Ratio to the equator ($\geq$ 20 km)	Craters ($\geq$ 40 km)	Density per 10^6 km^2	Rat to t equ ($\geq$ 40
◖	Around the meridian of the anti-Earth point	195–165	3.161	711	225	1.35	233	74	1.3
◖	Far side — beyond the anti-Earth point	225–195	3.161	736	233	1.40	239	76	1.4
◖	Far side — toward the leading meridian	255–225	3.161	643	203	1.22	205	65	1.2
◖	Around the leading meridian (90° west)	285–255	3.161	487	154	0.93	170	54	1.0
◖	Near side — beyond the leading meridian	315–285	3.161	324	102	0.62	111	35	0.6
◖	Near side — west of the sub-Earth point	345–315	3.161	267	84	0.51	99	31	0.5

Analysis and conclusions from the global longitudinal profile

Examining the measures and the graphic profile yields the following main findings:

- **Severe non-uniformity around the body's circumference:** The lunar surface shows an extreme asymmetry along the meridians. The far side and the sectors surrounding the trailing meridian (90° E) show consistently high densities, ranging from 1.22 to 1.51 times the equatorial reference value in the ≥ 20 km group.
- The global peak and minimum zones:
  - **Maximum density:** recorded in the eastern sector of the far side, in the range $105^\circ - 135^\circ$ east, with 793 craters and a density of 251 per million km² (a ratio of 1.51).
  - **Minimum density:** recorded on the western near side, in the range $315^\circ - 345^\circ$ east, with only 267 craters and a density of 84 per million km² (a ratio of 0.51).
- **Magnitude of the longitudinal pattern:** The extreme variance ratio between the peak segment and the minimum segment stands at almost a factor of three ($251/84 = 2.97$ for the ≥ 20 km group, and $90/31 = 2.88$ for the ≥ 40 km group). This difference is considerably higher than the raw density differences measured between the poles and the equator in section A.1.

This finding confirms that this is an independent and entirely separate geographic pattern, which requires methodological isolation.

- **Geophysical and morphological context (the LEF model):** The extreme minimum sector ($315^\circ - 345^\circ$) hosts the Mare Imbrium basin. Within the LEF model, this structure acted as the main feeding and heat-conducting vent from the Earth into the lunar interior, and not as a site of ejection and throwing out of material. This sector, together with the extensive western maria concentrated in it, is characterised by crust that was melted and thermally reset, and therefore carries few craters today. Conversely, the prominent crater excess on the trailing side is read in the model through the distribution of returning ejecta that lagged behind the rotational velocity and landed behind (Paper 6).



Colour scale: left, “low density” (soft yellow); right, “high density” (soft red).

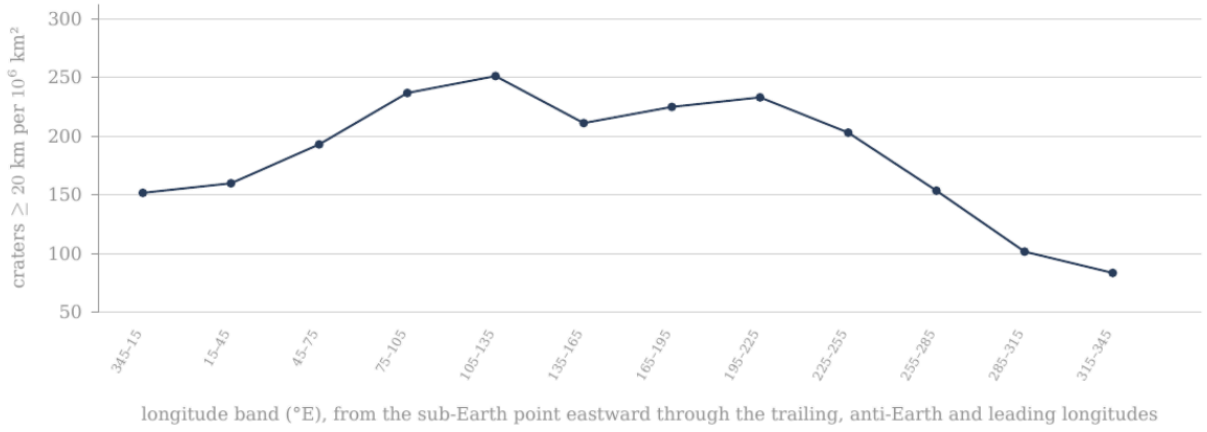


Figure 2. Crater density by longitude, craters of 20 km and larger, in twelve bands of 30° (A.2); the whole catalogue, without a mask. The peak lies slightly east of the trailing meridian, and the minimum slightly west of the sub-Earth point. (In the plot: vertical axis — craters ≥ 20 km per 10^6 km², 50–300; horizontal axis — longitude band (° east), from the sub-Earth point eastward through the trailing meridian, the anti-Earth point and the leading meridian, from 345–15 to 315–345; the dark-blue line — the density in each band.)

A.3 Crater density by latitude after removing the basins and maria

Whereas sections A.1 and A.2 counted the global surface in full, this section applies a firm geological mask to remove area units whose crater record was affected by endogenic factors (thermal and volcanic activity and mechanical erasure) unrelated to geographic latitude.

Rationale for removing the geological units:

1. The system of maria and basalt plains (LROC mask): The mare floors represent young feeding vents above magmatic domes (Paper 5), in which the ancient craters were melted. The basalt plains (such as Procellarum, Tranquillitatis, Fecunditatis, Insularum, Frigoris) represent crust

that was melted by the global heat radiated by the Earth, without a local feeding dome. In both cases, the lack of craters stems from a thermal reset of the record. Since these units are concentrated on the near side and at low latitudes, they artificially lower the equatorial density.

2. The large basins ($\geq 500\ \mathrm{km}$): these impact structures extend over large portions of the latitude bands (mainly in the polar regions). Their floors and rims have a distinct morphological history that independently affects the statistics of the band.

3. Aim of the procedure: to isolate the historically most stable crustal parts (Highlands), in which the record was preserved uniformly, in order to obtain a reliable measure of the original crater distribution.

Methodology for applying the geological mask

- **Definition of the mask:** a combination of the LROC global mare polygons (LROC_GLOBAL_MARE_180) and the 20 largest basins with outer rim diameter $\geq 500\ \mathrm{km}$ from the Neumann et al. (2015) catalogue (Table 1).
- **Basin geometry:** each basin is represented as a spherical cap centred on the source coordinates, with an angular radius of $D/(2R_{\mathrm{Moon}})$.
- **Filtering rule and residual area:** a crater is removed from the count if its geometric centre point falls within a basin or a mare polygon. Spatial overlaps are removed only once. The residual area is computed on a high-resolution global grid of $0.05^\circ \times 0.05^\circ$ using exact spherical cell areas based on great-circle distance.
- **Keeping the reference point for comparison:** to allow the eye to follow the movement of the bands, the static reference base of section A.1 is kept (value 1.00 = density of the full raw equatorial belt without a mask). Its value is 166.436 (for $\geq 20\ \mathrm{km}$) and 53.170 (for $\geq 40\ \mathrm{km}$).
- Definition of the density-change measure ($\Delta\rho$): a measure that compares each band with itself before and after the mask:

$$\Delta\rho = \left(\frac{\rho_{\mathrm{after}}}{\rho_{\mathrm{before}}} - 1\right) \times 100\%$$

A positive value indicates that the removed area was crater-poor relative to the rest of the band (and therefore its removal raised the residual density).

Summary of masked crater-density data by latitude (A.3)

The following are the exact numerical values obtained in the 12 latitude bands after applying the filtering mask:

Sector	Geographic description of the region	Latitude band (°)	Residual area (10^6 km^2)	Craters ($\geq$ 20 km)	Density per 10^6 km^2	Ratio to reference ($\geq$ 20 km)	Density change ($\Delta\rho_{20}$)	Craters ($\geq$ 40 km)	Density per 10^6 km^2
•	South — adjacent to the pole	75–90° S	0.275	84	305	1.83	+42.0%	34	124
•	South — outer part of the polar cap	60–75° S	1.033	238	230	1.38	+10.0%	71	69
•	South — next to the polar cap	45–60° S	1.544	380	246	1.48	+7.1%	109	71
•	South — mid-latitudes	30–45° S	2.067	476	230	1.38	+12.2%	150	73
•	South — low latitudes	15–30° S	2.194	455	207	1.25	+16.1%	156	71
•	Equator — southern half-band	0–15° S	2.976	634	213	1.28	+19.8%	215	72
•	Equator — northern half-band	0–15° N	2.880	582	202	1.21	+30.3%	186	65
•	North — low latitudes	15–30° N	2.153	484	225	1.35	+48.4%	171	79
•	North — mid-latitudes	30–45° N	2.411	580	241	1.45	+36.6%	171	71
•	North — next to the polar cap	45–60° N	1.639	375	229	1.37	+31.4%	123	75
•	North — outer part of the polar cap	60–75° N	1.752	366	209	1.25	+3.3%	148	84
•	North — adjacent to the pole	75–90° N	0.646	194	300	1.80	0.0%	65	101

- **Numerical colour scales in the table:** in the ≥ 20 km group the range is between 202 and 305; in the ≥ 40 km group the range is between 65 and 124 craters per million km².

Analysis of the spatial effects and the built-in bias of the static base

Comparing the masked profile with the raw profile yields the following scientific insights:

1. Behaviour of the edge cases in the change column:

- Zero polar change in the north ($75{-}90^\circ\ \mathrm{N}$): this band does not physically touch the mask (it contains no maria or large basins). Its area ($0.646 \times 10^6\ \mathrm{km}^2$) and its crater count (194) remained constant, and therefore $\Delta\rho = 0.0\%$.
- Maximum equatorial change in the north ($15{-}30^\circ\ \mathrm{N}$): this band loses more than half of its geometric area (a drop from 4.574 to 2.153 million km²) owing to the concentration of near-side maria. The residual density on what remains jumps by +48.4% for the 20 km group, and by +55.9% for the 40 km group (the sharpest rise in the system).

2. Robustness of the endogenous polar excess: Applying the mask significantly raises the low and middle latitudes (especially in the north), and thus flattens part of the spatial difference. However, the two bands adjacent to the poles remain the densest regions on the lunar body. In the $\geq 20\ \mathrm{km}$ group, the south pole reaches 305 and the north 300 per million km², while the 10 intermediate bands are lower (values of 202 to 246). In the $\geq 40\ \mathrm{km}$ group, the south pole leads the entire system with a peak value of 124 (a ratio of 2.32). This finding proves that the polar excess is not an artefact of the presence of maria at the equator; it is embedded in the solid ground itself.

3. Methodological clarification on a like-for-like comparison: The ratio column in this table divides a masked band (a numerator that has lost its maria) by a raw equator (a denominator that has kept them), and thus produces a mathematical exaggeration of the predicted excess (ratio values of 1.40–1.48). If the two masked polar caps (882 craters on 3.706 million km², giving a density of 238.0) are compared with the equatorial belt masked with exactly the same mask (1,216 craters on 5.856 million km², giving a density of 207.7), the true ratio value is 1.15. Therefore, for the final scientific test of the paper, the clean count of sections A.11 and D.1 is applied (simultaneous removal of the maria and the SPA basin from both sides), yielding the firm normalised ratio value: 1.20 ± 0.05.



Colour scale: left, “low density” (soft yellow); right, “high density” (soft red).

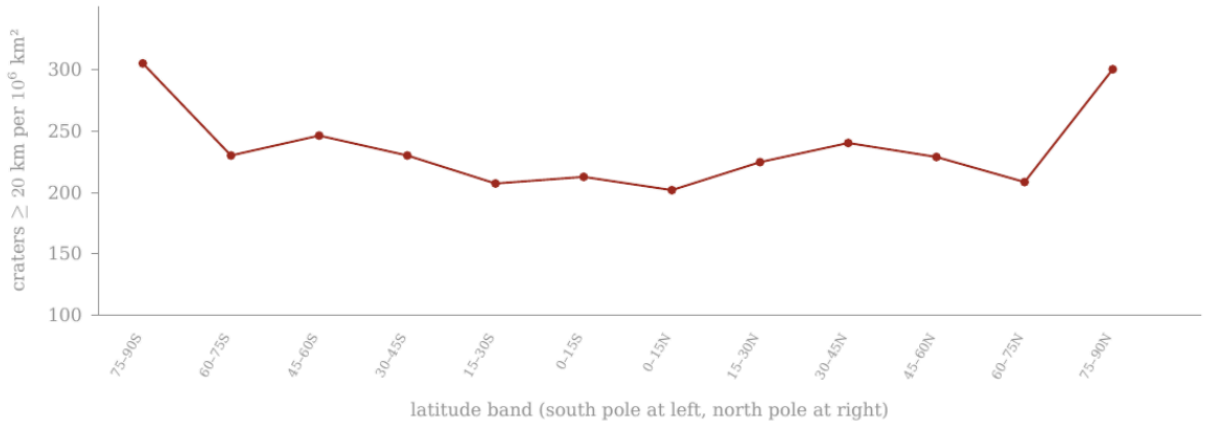


Figure 3. The same twelve bands, after removing the twenty basins of 500 km and larger and the maria and subtracting their area (A.3). The mask raises the equator and leaves the poles. (In the plot: vertical axis — craters ≥ 20 km per 10^6 km², 100–300; horizontal axis — latitude band, the south pole on the left and the north pole on the right; the red line — the masked profile.)

A.4 Crater density by longitude after removing the basins and maria

This section applies the firm geological mask (the 20 large Neumann basins and the LROC mare polygons) to the 12 longitude bands of section A.2. Its main aim is to determine which part of the extreme east–west asymmetry stems directly from the presence of the young maria and the basins, and which part is embedded in the ancient highland crust itself.

Area-calculation methodology and calibration of the system:

1. Loss of equal areas in the longitude bands: Unlike the raw count, in which all longitude segments are equal in size, applying the mask changes the band areas differentially. A band that held a large portion of maria or basins loses more area. The actual residual areas now range widely, between 1.073 and 2.729 million km², which requires relying exclusively on the density measure and not on absolute counts.

2. Consistency and mathematical closure of the mask: Summing all 12 bands gives full agreement with the data of section A.3: both geometric sectioning methods (latitudinal and longitudinal) divide exactly the same residual crustal area, which makes up slightly more than half of the Moon (21.570 million km²) and contains exactly the same number of craters (4,848 in the ≥ 20 km group, and 1,599 in the ≥ 40 km group).

3. Keeping the fixed reference base: The reference point (value 1.00) remains completely fixed – the density of the full raw equatorial belt without a mask (166.436 for ≥ 20 km, and 53.170 for ≥ 40 km).

Summary of masked crater-density data by longitude (A.4)

The following are the exact numerical values obtained in the 12 longitude segments after applying the filtering mask:

Sector	Geographic and morphological definition of the sector	Longitude band (° east)	Remaining area (10^6 km^2)	$N \geq 20$	Density ≥ 20	Ratio ≥ 20	Change ≥ 20 ($\Delta\rho_{20}$)	$N \geq 40$	Density ≥ 40
◖	Around the meridian of the sub-Earth point	15–345	1.879	420	224	1.34	+47.5%	149	79
◖	Near side — east of the sub-Earth point	45–15	1.510	351	232	1.40	+45.5%	92	61
◖	Near side — toward the trailing meridian	75–45	1.837	438	238	1.43	+23.3%	121	66
◖	Around the trailing meridian (90° east)	105–75	1.692	370	219	1.31	-7.8%	131	77
◖	Far side — beyond the trailing meridian	135–105	2.729	725	266	1.60	+5.9%	251	92
◖	Far side — toward the anti-Earth point	165–135	1.995	457	229	1.38	+8.6%	166	83
◖	Around the meridian of the anti-Earth point	195–165	1.345	356	265	1.59	+17.7%	96	71
◖	Far side — beyond the anti-Earth point	225–195	1.904	480	252	1.51	+8.3%	165	87
◖	Far side — toward the leading meridian	255–225	1.813	369	204	1.22	+0.1%	121	67
◖	Around the leading meridian (90° west)	285–255	2.044	417	204	1.23	+32.4%	152	74
◖	Near side — beyond the leading meridian	315–285	1.748	276	158	0.95	+54.0%	90	51

Sector	Geographic and morphological definition of the sector	Longitude band (° east)	Remaining area (10^6 km^2)	$N \geq 20$	Density ≥ 20	Ratio ≥ 20	Change ≥ 20 ($\Delta\rho_{20}$)	$N \geq 40$	Density ≥ 40
◖	Near side — west of the sub-Earth point	345–315	1.073	189	176	1.06	+108.5%	65	61

Dynamic analysis of the effect of the mask on the longitudinal asymmetry

Examining the residual measures and the geological change column yields the following insights:

1. Filling of the extreme western trough: The longitude band $345°-315°$ east – which hosts the Mare Imbrium basin and most of the western near-side maria – shows the sharpest rise in the system: a density jump of +108.5% (the density more than doubles, from 84 to 176 per million km²). This finding proves that the extreme minimum measured in section A.2 stemmed directly from the resetting of the record by the young basalt floors.

2. Anomalous behaviour in the trailing region: The sector around the trailing meridian ($105°-75°$ east) is the only band in the system in which the residual density decreases after applying the mask (a drop of -7.8% in the 20 km group, and -9.0% in the 40 km group). This means that the parts removed in this sector (local basins) were richer and denser in craters than the crust that remained around them.

3. Reduction of the magnitude of the longitudinal pattern: The filtering mask dramatically reduces the east–west asymmetry and draws the profile into a narrower range. The extreme ratio between the peak segment and the minimum segment drops from 2.97 to 1.68 for craters ≥ 20 km (a peak of 266 in the $135°-105°$ sector versus a minimum of 158 in the $315°-285°$ sector), and from 2.88 to 1.79 for craters ≥ 40 km (92 versus 51).

4. Robustness of the residual circumferential pattern: Despite the mathematical reduction, the basic spatial pattern does not disappear. The far side beyond the trailing meridian firmly keeps its status as the densest region in both control groups, while the near side beyond the leading meridian remains the poorest. This residual pattern is analysed later in the paper using the six-axis model (A.9). For solving the polar puzzle, the isolated measurement itself and the separation of the longitudinal effects provide the required methodological foundation.



Colour scale: left, “low density” (soft yellow); right, “high density” (soft red).

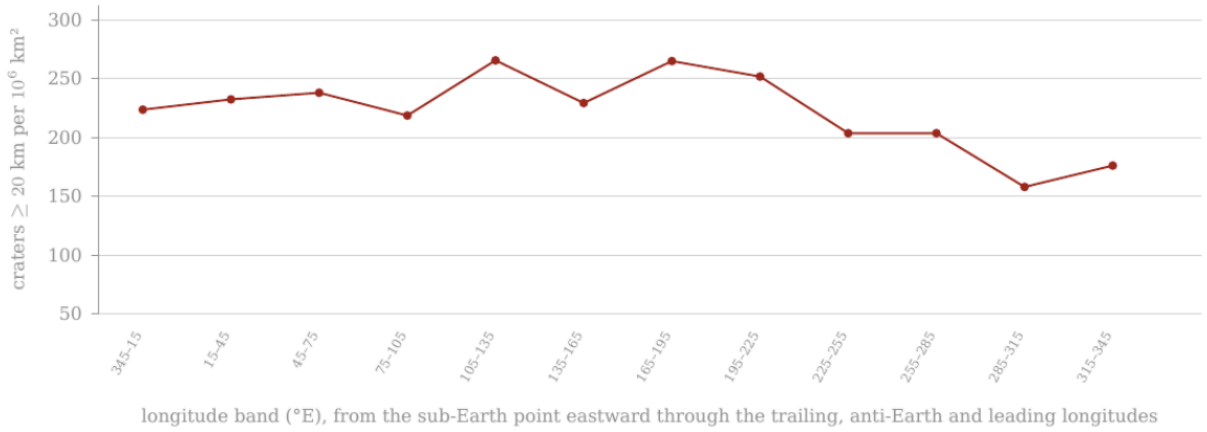


Figure 4. The same twelve longitude bands, after removing the twenty basins of 500 km and larger and the maria and subtracting their area (A.4). The western minimum rises; the far-side peak remains. (In the plot: vertical axis — craters ≥ 20 km per 10^6 km², 50–300; horizontal axis — longitude band (° east), from the sub-Earth point eastward through the trailing meridian, the anti-Earth point and the leading meridian; the red line — the masked profile.)

A.5 Summary and conclusions of the four band tests

This section compiles and analyses the combined findings of the four spatial band tests carried out in sections A.1–A.4. The analysis is designed to answer a firm, structural question: is the excess crater density in the polar regions a real geophysical property of the lunar crust, or is it an artificial statistical bias arising from the way the area is divided or from the presence of large local geological formations?

The methodology is based on cross-checking four complementary tests:

1. Distribution by latitude without a mask (A.1): examining the equator-versus-poles trend on the raw surface.

2. Distribution by longitude without a mask (A.2): examining the global circumferential trend (east versus west) on the raw surface.

3. Masked distribution by latitude (A.3): cleaning out endogenic effects (maria, basins ≥ 500 km) along the latitude axes.

4. Masked distribution by longitude (A.4): cleaning out endogenic effects along the longitude axes.

The combination of the four tests makes it possible to separate two perpendicular sectioning directions (latitude versus longitude) and at the same time to control how far the record depends on the young basalt surfaces and the giant basins.

A.5.1 Separating the patterns: two independent spatial arrays

The first scientific insight arising from the summary is that there is no single spatial profile able to explain the global crater distribution. The latitude axes and the longitude axes reveal two entirely

separate and distinct geographic patterns:

1. Characteristics of the latitude profile (bipolar trend):

- **The extreme density difference:** In the raw count (without a mask), in the diameter group ≥ 20 km, the band adjacent to the north pole ($75-90^\circ$ N ◖) shows a peak density of about 300 craters per million km², while the minimum band ($15-30^\circ$ N ◖) stands at about 151 per million km² – a difference of almost a factor of two (ratio 1.80) [A.1]. In the control group ≥ 40 km a similar ratio is recorded: about 101 per million km² at the north pole versus about 51 per million km² in the low-latitude zones ($15-30^\circ$ N) [A.1].
- **Non-monotonic behaviour:** The spatial profile does not rise in a smooth, continuous and consistent gradient from band to band along the way from the equator to the poles; it includes local fluctuations, rises and falls [A.1].
- **The restricted structural conclusion:** The scientific argument is not about a smooth linear gradient, but about a firm edge property: the end bands adjacent to both poles reach exceptional density values that no intermediate band along the way can attain [A.1]. An observer moving from the equator toward the pole will see a fluctuating count, but the final band, where the polar section takes place, will show a density roughly twice that of the poorest band measured along the way [A.1].

2. Characteristics of the longitude profile (circumferential asymmetry):

- **The extreme density difference:** In the raw count, in the ≥ 20 km group, the density ranges from an absolute minimum of about 84 per million km² in the western near-side sector ($315-345^\circ$ E ◖) to a global peak of about 251 per million km² in the eastern far-side sector ($105-135^\circ$ E ◖) [A.2]. In the ≥ 40 km group the range is between about 31 and about 90 craters per million km² [A.2].
- **The non-uniformity measure (peak/minimum ratio):** The variance factor (the density of the richest segment divided by the density of the poorest segment) stands at 2.97 for the 20 km group, and at 2.88 for the 40 km group [A.2].
- **Comparing the magnitudes of the effects:** The degree of non-uniformity along the circumference (a difference of almost a factor of three between east and west) is sharper and wider than that measured along the latitude axes (a factor of two between the poles and the equator) [A.1, A.2]. This means that circumferential movement around the equator produces more extreme density changes than vertical movement from the equator to the poles [A.2].

Methodological analysis: ruling out a mixing of the patterns

The fact that the asymmetry along the meridians reaches a range of a factor of three could lead to the claim that the polar excess (which is smaller, a factor of two) is merely a by-product or a statistical distortion caused by the larger longitudinal pattern. The study rules out this claim through the following geometric explanation:

- **Complete structural perpendicularity:** The two profiles are computed and measured in directions fully perpendicular to each other.

- **The geometric offsetting mechanism:** Each latitude band is a complete ring encircling the Moon over 360°. This means that each latitude band takes in exactly all the global meridians – the richest meridians (the far and trailing sides) and the poorest meridians (the western near side) enter the statistical denominator of every latitude band in the same fixed geometric proportion.
- **Isolation of the effect:** The simultaneous existence of two different profiles proves that the presence of a crater-rich or crater-poor region along the circumferential axis is not, in itself, an explanation or a source of the density excess embedded in the polar latitudes. The longitudinal asymmetry is an independent and separate pattern treated within the six-axis model (A.9) [A.2]. The present analysis successfully isolates the polar pattern in order to examine precisely what remains of it under uniform control conditions.

A.5.2 The mask as a control test

Sections A.3 and A.4 are not a recount of a new crater population, but an exact methodological repetition of the two band tests (latitudinal and longitudinal) under the filtering conditions of a uniform geological mask. This mask removes from the calculation the 20 basins of diameter ≥ 500 km and the LROC mare polygons [A.3, A.4]. A crater whose geometric centre falls within the mask boundaries is omitted from the count, and the area of the sector is reduced accordingly in the density denominator [A.3, A.4].

The scientific rationale for the choice of the filtered surfaces:

- **Maria and basalt plains (thermal reset):** These units are geologically young floors. A mare acts as a vent above a magmatic feeding dome that melted the early craters, while a basalt plain represents crust melted under the general heat load radiated by the Earth. Their paucity of craters stems from resurfacing and not from a low impact rate. Since they are statistically concentrated in the equatorial region, including them distorts the latitude measure [A.3].
- **The giant basins (mechanical erasure):** The formation of these impact structures physically erased the crater population that existed within them before them [A.5.2].
- **The like-for-like comparison rule:** To determine whether the density difference stems from a latitude-dependent geodynamic effect, physics requires comparing crustal units with a uniform history (ancient ground versus ancient ground). Including partially resurfaced areas would measure the rate and distribution of the thermal resurfacing activity and not the effect of the body's geometry [A.5.2].

The mathematical dynamics of the density measure:

A decrease in the absolute number of craters (N) following the application of the mask does not necessarily imply a decrease in the residual density [A.3]. The density equation is governed by the changing ratio between numerator and denominator on the same ground [A.3]:

$$\rho_{\text{after}} = \frac{N_{\text{after}}}{\text{Area}_{\text{remaining}}}$$

If the mask removes a relative share of area larger than the share of craters it omits (a situation typical of crater-empty mare areas), the overall density of the residual surface will show a rising trend [A.3]. The mechanism works like computing a trees-per-dunam measure in a forest containing a lake: removing the lake area (empty of trees) from the denominator of the total area will mathematically raise the density value of the remaining land, even though not a single physical tree has been added to the system [A.5.2].

Applying the same analytical operation to both perpendicular sectioning axes produces a powerful control tool: had the two spatial effects (latitudinal and longitudinal) been the exclusive product of the presence of the maria and basins, applying the mask should have changed both profiles in a coherent and similar way. The empirical findings rule this out and show a clearly differential behaviour [A.5.2].

A.5.3 Response of the longitude profile to the filtering mask

Unlike the latitude profile, the effect of the mask on the longitude profile is dramatic and fundamentally changes the raw amplitude (Mesa) of the system [A.5.3]:

1. Convergence of the extreme values and reduction of the amplitude along the circumference:

- The western minimum segment ($315^\circ - 345^\circ$ east): in the raw count (without a mask), this segment showed a minimal density of about 84 craters per million km² [A.2]. With the mask applied and the areas of Mare Imbrium and the western maria cleaned out, the density computed on the remaining highland surface jumped to about 176 per million km² (a rise of +108.5% relative to its raw state) [A.4].
- **The global non-uniformity measure (peak/minimum ratio):** Following the rise in the minimum values, the overall variance factor along the longitude axes was considerably reduced: for the ≥ 20 km group, the ratio dropped from 2.97 in the raw count to 1.68 in the masked count [A.2, A.4]. In the control group ≥ 40 km a parallel behaviour was recorded, with the ratio falling from 2.88 to 1.79 [A.2, A.4]. This consistency in both size groups confirms that the original wide amplitude in longitude stemmed mainly from a geological bias in the distribution of the maria [A.5.3].

2. Collapse of the asymmetry between the trailing and leading meridians:

The statistical analysis of the full longitude bands (stretching from pole to pole) reveals the collapse of the significance of the orbital asymmetry at band level [A.5.3]:

- **In the raw count:** the trailing band ($75^\circ - 105^\circ$ east) numbered 750 craters ($\rho = 237$) versus 487 craters ($\rho = 154$) in the leading band ($255^\circ - 285^\circ$ east) [A.2]. This difference gave a significant ratio of 1.54, a value representing 7 counting standard deviations (7σ), which entirely rules out the probability of a random fluctuation (the natural variance of a crater count scales as the square root of the count $\sqrt{N}$, and in a sample of hundreds of craters the uncertainty is a few per cent) [A.5.3].
- **After applying the mask:** the leading band kept a larger residual area ($2.044 \times 10^6\ \mathrm{km}^2$, 417 craters) than the trailing band ($1.692 \times 10^6\ \mathrm{km}^2$, 370 craters) owing to a more extensive

concentration of maria on the trailing side [A.4]. As a result, the masked density values converged to 219 versus 204 craters per million km², giving a residual ratio of only 1.07 [A.4]. Since the inherent counting uncertainty of this ratio is about 7%, the remaining difference equals only one standard deviation (1σ) and does not hold statistical significance at band level (for the ≥ 40 km group a residual value of 77 versus 74 was recorded) [A.4, A.5.3].

3. The geometric effect: averaging over bands versus concentration in caps:

Despite the collapse of significance at the level of the long band, the longitudinal pattern does not vanish entirely from the solid crust [A.4]. The reason lies in the geometric limitation of full longitude bands: a band stretching from pole to pole tends to dilute and obscure local excesses by averaging them with ordinary ground along the whole meridian [A.5.3].

When the geometric method is changed and focused caps with a radius of 30° are examined around the two orbital end points (section A.9), the masked population keeps density values of 252 per million km² on the trailing side versus 174 on the leading side [A.5.3]. This difference yields a ratio of 1.45, representing 5 standard deviations (5σ), and holds strong significance. This finding proves that the crater excess that survives the mask is concentrated locally around the core coordinates of meridian 90° east, and is not spread homogeneously along the whole band [A.5.3].

Methodological summary:

Applying the control mask has a twofold result: it changes and reduces a considerable part of the raw longitudinal amplitude (confirming the role of the young maria), but it does not entirely cancel the residual non-uniformity along the circumference [A.5.3]. However, for solving the polar puzzle and confirming the paper's model of rounding and mechanical contraction, it is enough that the longitudinal pattern has been measured, defined and statistically separated, and therefore the physical explanation of the east–west asymmetry remains outside the central argument of the present study [A.5.3].

A.5.4 Response of the latitude profile to the filtering mask: stability of the polar excess

Unlike the longitude profile, in which the mask erased a considerable part of the asymmetric amplitude [A.5.3], the latitude profile responds to the filtering procedure in a way that confirms the robustness of the polar effect in the ancient rocky crust [A.5.4]. The behaviour of the end bands reflects the mechanics of the residual density equation [A.3, A.5.4]:

- **The band adjacent to the south pole (75°-90° S ◖):** in the raw count (without a mask), 139 craters were counted in this band (ρ ≈ 215 per million km²) [A.1]. Applying the mask removed a huge portion of area from the sector, leaving only 84 craters [A.3]. However, since the South Pole–Aitken (SPA) basin – the giant structure covering a considerable part of this band – is characterised by a crater-poor floor owing to mechanical erasure (the "dry pool" effect), the mask removed a relative share of area much larger than the number of craters omitted [A.5.4]. As a result, the available area shrank to about 0.275 × 10⁶ km², and the density computed on it jumped to about 305 craters per million km² [A.3].

- **The band adjacent to the north pole (75°-90° N ◖):** in the raw count, 194 craters were counted in this sector, giving a density of about 300 per million km² [A.1]. Since this northern band is almost unaffected geographically by the mask polygons (no large basins or extensive maria near the north pole), the area and numerator data remained unchanged, and the residual density settles at about 300 craters per million km² [A.3].
- **Convergence of the bipolar end values:** Cleaning the geological records leads to a fundamental mathematical correction. In the raw count, the southern hemisphere gave a misleading appearance of paucity relative to the northern one (215 versus 300) [A.1]. After applying the mask, this artificial asymmetry collapses, and the two bands adjacent to the poles converge almost to the same firm, high value: 305 in the south and 300 in the north [A.3].

A.5.5 Scientific insights from the comparison of the poles and the equal-surface test

The convergence of the values between north and south is one of the central methodological findings of the control test, and provides a firm lesson in processing planetary data [A.5.5]:

- **Collapse of the raw north–south asymmetry:** The original raw comparison was flawed in that it set a northern polar band made of pristine, ancient, almost undamaged crust against a southern polar band most of which is made of the floor of a giant impact basin that erased the early record [A.5.5]. This comparison represented not a real property of the latitudes but a property of the type of ground [A.5.5].
- **Establishing a like-for-like comparison:** Once a uniform control mask is applied that omits the same geological units from both sides, the two bands are measured on the same type of surface (pristine Highlands), and the internal asymmetry between them drops to zero in the ≥ 20 km group [A.5.5].
- **The morphological conclusion:** Neutralising the raw difference does not produce one dense pole and one sparse opposite pole, but reveals two polar caps of similar, stable density, considerably higher than the rest of the surface [A.5.5].

A.5.6 Summary of stage one of the spatial observations and the geometric limitations of the bands

The four band tests (A.1–A.4) fully settle the question of the validity of the morphological finding and allow the scientific problem to be narrowed down to four firm conclusions and one structural limitation [A.5.6]:

Established conclusions:

1. The effect is global and bipolar: The high polar density is not a local feature of the northern hemisphere alone; it is shared by both polar end bands after the biases are cleaned out [A.5.6].

2. The excess does not depend on the mask: The effect does not disappear or fade after the removal of the large basins and the maria. On the contrary, at the south pole cleaning the area brought out and raised the true density of the remaining crust [A.5.6].

3. Complete isolation from the longitudinal asymmetry: The polar density is not an indirect expression of the east–west asymmetry. The longitude profile shows a different amplitude, a separate geometric behaviour and an entirely opposite response to the mask (collapse of most of the longitudinal amplitude at band level versus preservation and enhancement of the density at the polar ends) [A.5.3, A.5.6].

4. No requirement for a smooth monotonic profile: The empirical data on the surface do not support the claim of a smooth gradient climbing continuously from the equator to the poles. The precise scientific finding – and the physically stronger one – establishes that the bands adjacent to the poles settle significantly at the absolute upper end of the global latitude distribution [A.5.6].

The built-in geometric limitation of bands (The Ring Averaging Limitation):

Despite settling the initial question and confirming the robustness of the finding, the use of full latitude bands (75°-90°) runs into a clear analytical barrier:

- A latitude band encircles the Moon over 360° around the rotation axis, and thus it compresses and averages into a single statistical number all the global meridians contained within it [A.5.6].
- The geometry of the band cannot indicate how the density is organised and arranged within the polar area cell itself [A.5.6]. An average value of 300 per million km² can arise physically from several different morphological scenarios: a very dense point accumulation right next to the polar apex, a circumferential ring structure at a certain angular distance, an asymmetric concentration on one flank of the cap, or a uniform, homogeneous distribution [A.5.6].
- **The analytical method is like determining the average number of people per m² in a round hall:** the global figure gives the overall degree of crowding, but it cannot tell whether the crowd stands concentrated at the centre of the room, spread along the walls or distributed evenly [A.5.6]. Resolving this limitation and decoding the internal structure of the caps requires moving to the next methodology of the study: the analysis of focused caps and the count around the six axes of the body (A.6–A.9) [A.5.3].

A.5.7 — End of the band stage and the move to caps

At this point, the analytical work that latitude and longitude bands can perform in the system has come to its full conclusion [A.5.7].

Summary of the findings in one line:

The existence has been established of an exceptionally high crater density at both polar ends of the latitude profile, fully isolated from the east–west spatial pattern, and standing stable and firm even after the effects of the basins and maria have been cleaned out [A.5.7].

In broader methodological detail, the band analysis fulfilled four main control aims [A.5.7]:

1. Spatial location: defining the exact location of the density anomaly relative to the latitude axes.

2. Separation of patterns: complete isolation of the polar effect from the orbital asymmetry along the meridians.

3. Geological response: quantitative examination of how the two patterns change under the constraint of a fixed, uniform geological mask.

4. Bipolar balance: revealing the fact that after cleaning out the geological biases, two polar bands (northern and southern) are obtained with similar, high and balanced density values in the system.

Change of analytical method: the move to the geometric question

Despite the robustness of the findings, the mechanics of bands runs into a structural barrier and cannot answer the next scientific question [A.5.7]. Up to this stage, the questions examined were: are the polar latitudes characterised by a higher crater density, and does this finding survive spatial control procedures [A.5.7]?

From here on, the scientific argument changes its character and becomes a firm geometric and morphological problem [A.5.7]:

How is the crater density arranged and organised within the polar space itself?

To decode the internal structure of the anomaly, the study fundamentally changes the form of measurement and the geometric sectioning of the crust [A.5.7]:

- **Eliminating circumferential averaging:** abandoning the division into closed global bands that encircle the body over 360° and average the meridians [A.5.6, A.5.7].
- **The focused-cap method:** defining each geographic pole as the exclusive structural centre of a spherical area cell (a polar cap) [A.5.7].
- **Distance-vector analysis:** examining and mapping the distribution of crater density as a direct function of the angular and radial distance from the polar apex [A.5.7].

Therefore, the next section – section A.6 (the polar caps) – is not a statistical repetition or a further count of the same population [A.5.7]. It represents the next stage in the development of the argument, following directly from the mechanical limit of the band test: a sharp move from proving the very existence of a high polar density to decoding the spatial geometry and internal arrangement of the craters within the polar cap [A.5.7].

A.6 The polar caps, without a mask

In this section, the study formally changes the geometry of the measurement and moves from circumferential bands to the analysis of six concentric (nested) spherical caps around each pole, with angular radii of: $10^\circ, 15^\circ, 20^\circ, 30^\circ, 45^\circ$, and 60° [A.6]. The aim of the test is to decode the internal structure of the anomaly and to examine how the cumulative density changes as the analytical space converges inward, toward the end point of the rotation pole [A.6].

Principles of the geometric structure of the caps:

1. Definition of angular radius (θ): a cap of radius θ takes in all craters whose arc distance from the polar apex satisfies $\leq \theta$ (for example: a cap of 30° extends from latitude 60° to 90° at the pole) [A.6].

2. The differential area equation: the area of each spherical area cell is computed exactly by:

$$A(\theta) = 2\pi R^2(1 - \cos\theta)$$

For a radius of $R = 1,737.4\ \mathrm{km}$, the area factor grows non-linearly: the narrow core cap (10°) extends over 0.288 million km², while the wide cap (60°) extends over 9.483 million km² (an area more than 30 times larger) [A.6].

3. Cumulative (nested) structure: The rows in this table do not represent isolated, independent annular regions, but a cumulative summation profile [A.6]. The mechanism is like the rings on a dartboard: the cap of 30° is not only the outer ring, but the whole board from the bullseye (0°) out to 30° [A.6].

- **If the craters are physically concentrated near the pole:** the small caps will show high density values, and the density will fall consistently as the cap widens and takes in ordinary ground [A.6].
- **If the distribution is homogeneous:** all the nested caps will show similar density values, independent of radius [A.6].

4. Keeping the static reference base: The reference point (value 1.00) remains completely fixed – the density of the full raw equatorial belt without a mask ($15^\circ\ \mathrm{S} - 15^\circ\ \mathrm{N}$). Its value is 166.436 (for $\geq 20\ \mathrm{km}$) and 53.170 (for $\geq 40\ \mathrm{km}$), which allows a direct comparison with the data of sections A.1–A.4 [A.6].

5. Sample sensitivity and the size factor: The inner caps (10°) cover small area cells and contain a limited number of craters, and therefore their confidence intervals are wide and carry the highest statistical uncertainty (the full analysis is given in D.4) [A.6]. The control group ($\geq 40\ \mathrm{km}$) represents an older crater population; a higher ratio compared with the 20 km group indicates a sharpening of the effect in the ancient craters (discussed in A.8.5) [A.6].

Summary of crater-density data in six nested caps (A.6)

The following are the numerical and relative values measured in the concentric caps around both poles, without applying geological masks:

Geographic pole	Angular cap radius (°)	Cap area ($10^6\ \mathrm{km}^2$)	Craters ($\geq 20\ \mathrm{km}$)	Density per $10^6\ \mathrm{km}^2$	Ratio to reference ($\geq 20\ \mathrm{km}$)	Craters ($\geq 40\ \mathrm{km}$)	Density per $10^6\ \mathrm{km}^2$	Ratio to reference ($\geq 40\ \mathrm{km}$)
North pole	60°	9.483	1,794	189	1.14	600	63	1.19
North pole	45°	5.555	1,102	198	1.19	393	71	1.33
North pole	30°	2.541	577	227	1.36	222	87	1.64
North pole	20°	1.144	309	270	1.62	119	104	1.96
North pole	15°	0.646	194	300	1.80	65	101	1.89

Geographic pole	Angular cap radius (°)	Cap area (10^6 km^2)	Craters ($\geq$ 20 km)	Density per 10^6 km^2	Ratio to reference ($\geq$ 20 km)	Craters ($\geq$ 40 km)	Density per 10^6 km^2	Ratio to reference ($\geq$ 40 km)
North pole	10°	0.288	86	298	1.79	31	108	2.02
South pole	10°	0.288	72	250	1.50	26	90	1.70
South pole	15°	0.646	139	215	1.29	53	82	1.54
South pole	20°	1.144	237	207	1.24	83	73	1.36
South pole	30°	2.541	536	211	1.27	185	73	1.37
South pole	45°	5.555	1,229	221	1.33	415	75	1.41
South pole	60°	9.483	2,035	215	1.29	687	72	1.36

- **Colour scales:** in the ≥ 20 km group the range is between 189 and 300; in the ≥ 40 km group the range is between 63 and 108 craters per million km² [A.6].

Morphological analysis of the raw radial profile

Reading the measures and the graphic behaviour of the two hemispheres yields the following findings:

1. Dominance of the overall polar excess: No polar cap, at any diameter and in either hemisphere, shows paucity relative to the equatorial belt [A.6]. The minimum ratio value in the system is 1.14 (the widest northern cap of $60°$), which proves that the interiors of all the caps are in a built-in density excess relative to the raw equator [A.6].

2. The differential behaviour of the north pole: The northern profile (the dark-blue line) shows a classic inward mechanical convergence and concentration [A.6]. As the measurement radius narrows, the density climbs continuously: from 189 in the wide cap ($60°$), through 227 in the average test cap ($30°$), up to a peak of 300 craters per million km² in the $15°$ cap [A.6]. In the extreme core cap ($10°$) the profile settles at an almost identical value (298), giving a ratio of 1.79 [A.6]. In the control group of large craters (≥ 40 km), the northern concentration trend sharpens and reaches a peak ratio of 2.02 in the $10°$ cap [A.6].

3. The flat behaviour of the raw south pole: The southern profile (the red line) shows an entirely different trend at first sight [A.6]. Over most of the measurement range ($15°$ to $60°$), the density curve remains flat, horizontal and almost independent of distance from the pole, with constant values in the narrow range of 207–221 per million km² [A.6]. Only in the innermost core cap ($10°$) is a sharp rise recorded, toward a value of 250 (a ratio of 1.50) [A.6].

The limit of geophysical interpretation at this stage:

The raw empirical findings seem to show a deep structural difference between the two poles: the north is characterised by a sharp inward concentration, while the south shows a flat, homogeneous distribution [A.6]. However, the physics of the model recognises that the southern profile is cut and averaged over the floor of the giant South Pole–Aitken basin, which erased a large part of the

record [A.6, A.11]. To determine which part of the variance stems from the deformation of the body's shape and which part is the distortion of a single impact basin, the geological filtering mask must be applied [A.6].



Colour key: a continuous transition from soft yellow (“low density”) to soft red (“high density”).

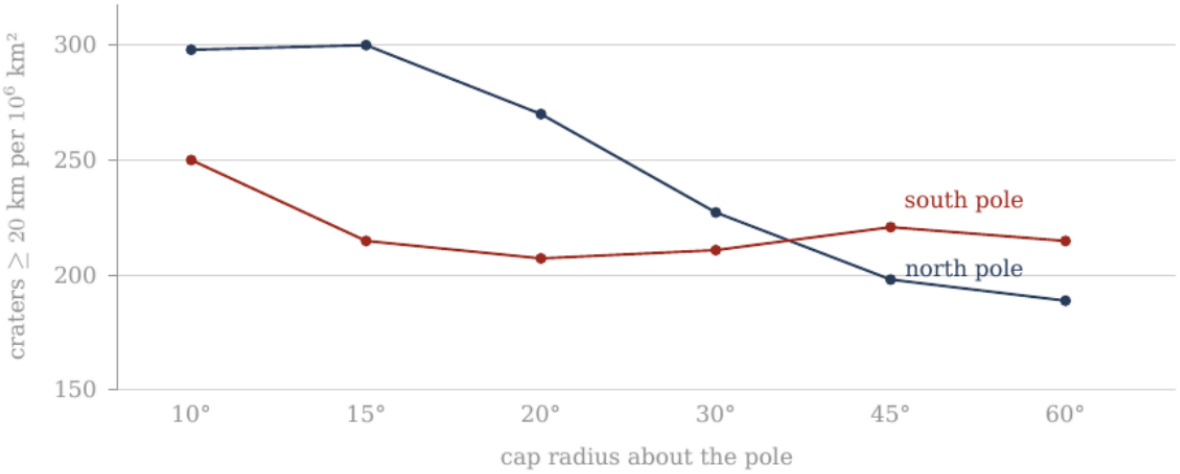


Figure 5. Density of craters 20 km in diameter and larger in caps of increasing radius around the two poles (A.6), the whole catalogue. The north rises toward its pole; the south is flat — the floor of South Pole–Aitken, which makes up most of the 30° southern cap (A.3, D.4). (In the plot: horizontal axis — "cap radius around the pole", 10°, 15°, 20°, 30°, 45°, 60°; vertical axis — "craters ≥ 20 km per 10^6 km²", 150–300; the dark-blue line — "north pole"; the red line — "south pole".)

A.7 The polar caps after removing the basins and maria

This section examines whether the differential behaviour between the two hemispheres observed in section A.6 (a concentrated northern profile versus a flat southern profile) is a property of the original crater population, or a distortion caused by the type of ground and by the resurfaced geological floors [A.7]. To this end, the firm filtering mask (the 20 Neumann basins of diameter ≥ 500 km and the LROC mare polygons) is applied to the six concentric caps [A.7]. This procedure ensures a precise, like-for-like comparison on an ancient, uniform surface outside the sites of thermal and mechanical erasure [A.7].

Measures and methodological calibration:

1. Computing the residual area and numerator: Craters whose centre point falls within the mask are omitted, and the area of the mask is subtracted exactly from the denominator [A.7]. The measurement grid is based on exact spherical cells at a resolution of $0.05^\circ \times 0.05^\circ$, with overlaps cleaned out only once [A.7]. Where the sectioning geometries overlap (the caps of 30° versus the

latitude bands $60^\circ - 90^\circ$ in A.3), full consistency and mathematical closure of the area data and counts are obtained [A.7].

2. Setting the fixed reference scale: The reference point (value 1.00) remains unchanged – the density of the full raw equatorial belt without a mask (166.436 for 20 km, and 53.170 for 40 km) [A.7]. This choice is deliberate and serves as a fixed static scale for quantitatively tracking the degree of geological change of each cap, and not as a measure of the final polar excess [A.7].

3. The density-change measure ($\Delta\rho$): compares each concentric cap with itself in its raw state in A.6:

$$\Delta\rho = \left(\frac{\rho_{\text{after}}}{\rho_{\text{before}}} - 1 \right) \times 100\%$$

A positive value indicates a rise in the density of the remaining area following the removal of crater-poor units [A.7].

Summary of masked crater-density data in six nested caps (A.7)

The following are the numerical measures and relative values obtained in the masked concentric caps:

Geographic pole	Angular cap radius (°)	Remaining area (10^6 km^2)	$N \geq 20$	Density ≥ 20	Ratio ≥ 20	Change ≥ 20 ($\Delta\rho_{20}$)	$N \geq 40$	Density ≥ 40	Ratio ≥ 40	Chang ≥ 40 ($\Delta\rho_{40}$)
North pole	60°	6.448	1,515	235	1.41	+24.2%	507	79	1.48	+24.3%
North pole	45°	4.037	935	232	1.39	+16.7%	336	83	1.57	+17.6%
North pole	30°	2.399	560	233	1.40	+2.8%	213	89	1.67	+1.6%
North pole	20°	1.144	309	270	1.62	0.0%	119	104	1.96	0.0%
North pole	15°	0.646	194	300	1.80	0.0%	65	101	1.89	0.0%
North pole	10°	0.288	86	298	1.79	0.0%	31	108	2.02	0.0%
South pole	10°	0.104	36	348	2.09	+39.1%	14	135	2.54	+49.8%
South pole	15°	0.275	84	305	1.83	+42.0%	34	124	2.32	+50.7%
South pole	20°	0.532	149	280	1.68	+35.2%	52	98	1.84	+34.7%
South pole	30°	1.308	322	246	1.48	+16.7%	105	80	1.51	+10.3%
South pole	45°	2.852	702	246	1.48	+11.3%	214	75	1.41	+0.5%
South pole	60°	4.918	1,178	240	1.44	+11.6%	364	74	1.39	+2.2%

- **Colour scales:** in the ≥ 20 km group the range is between 232 and 348; in the ≥ 40 km group the range is between 74 and 135 craters per million km² [A.7].

Geodynamic analysis of the masked radial profile

Examining the residual measures and the shared graphic behaviour (Figure 6) yields the following insights:

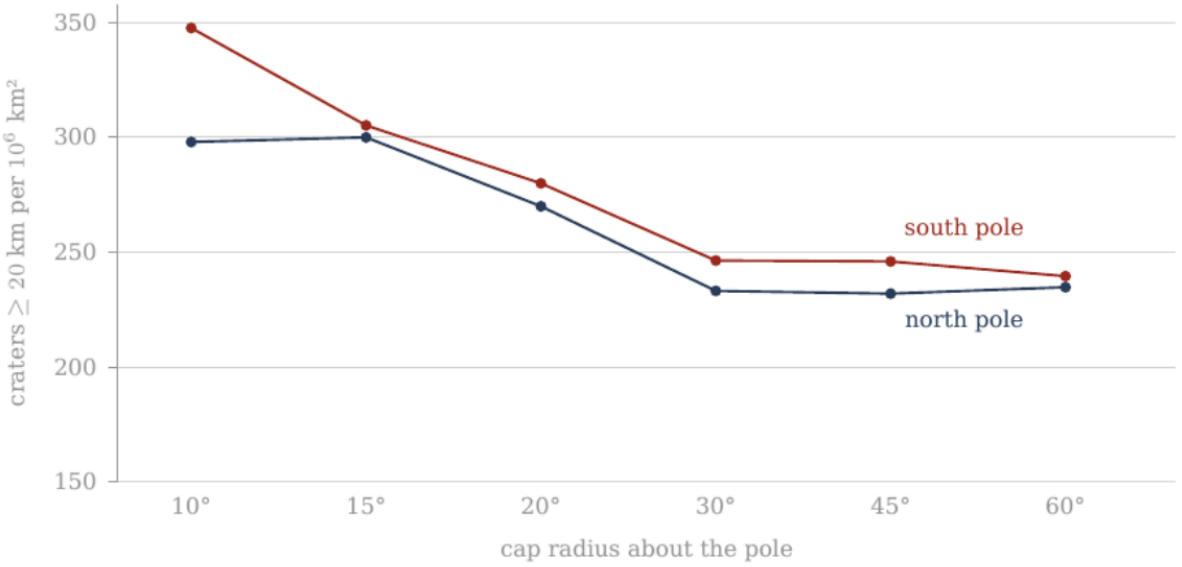


Figure 6. The same six caps around the two poles after removing the basins and maria (A.7), craters of 20 km and larger. The northern line does not change within 20°; the southern line rises in its inner caps, and the two draw closer. (In the plot: horizontal axis — "cap radius around the pole", 10°, 15°, 20°, 30°, 45°, 60°; vertical axis — "craters ≥ 20 km per 10^6 km²", 150–350; the dark-blue line — "north pole"; the red line — "south pole".)

1. Zero geological effect in the northern core: The three inner caps in the north ($10^\circ, 15^\circ, 20^\circ$) show constant change values of 0.0% [A.7]. This proves mathematically that there are no Neumann basins or basalt maria within 20° of the north polar apex; therefore, the high core profile measured there (300 per million km²) reflects precisely the pristine state of the crust [A.7].

2. The extreme effect in the southern core and the erasure by the SPA basin: The smallest caps in the south show the effect of the extreme mechanical erasure by the SPA basin [A.7]. In the southern 10° cap, the measurement area shrinks by more than half (a drop from 0.288 to 0.104 million km²), indicating that most of the inner southern core was covered by the basin floor [A.7]. Cleaning out this floor raises the true density of the residual highland crust by $+39.1\%$ in the 20 km group (a value of 348, a ratio of 2.09) and by $+49.8\%$ in the 40 km group (a value of 135, a ratio of 2.54) [A.7].

3. Morphological convergence to a uniform concentration curve: Comparing Figure 6 with Figure 5 reveals a morphological reversal [A.7]. Whereas in the raw state the southern line was completely flat, applying the control mask equalised the type of crust and raised the inner caps of the south [A.7]. Now the southern (red) curve changes its shape entirely and climbs steeply inward toward the pole, like the northern (dark-blue) curve [A.7]. Once both poles are counted on the same type of surface, both show a single firm correlation: craters becoming denser as the polar apex is approached, a finding that confirms the geodynamic surface-contraction model [A.7].

Methodological clarification on the clean excess values:

The ratio column in this table compares a masked numerator with a raw denominator (which keeps the empty equatorial maria), and therefore serves only as a tracking scale [A.7]. To derive the true clean polar excess, the paper applies a like-for-like comparison (simultaneous removal of the maria and the SPA basin from both sides) on the standard caps of 30° versus the equatorial belt ($|\phi| < 15^\circ$) [A.7]:

- **Full raw catalogue (including maria):** a ratio of 1.32 [A.7].
- **Removal of maria only, from both sides:** a ratio of 1.11 ± 0.04 [A.7].
- **Removal of maria and the SPA basin from both sides (the clean count):** a ratio of 1.20 ± 0.05 [A.7].

The standard deviation (± 0.05) reflects the random fluctuation of the count; the confidence interval of 95% is twice as wide, and establishes with full statistical confidence that the true ratio lies in the range between 1.10 and 1.30 – a value clearly above the uniform-distribution baseline (1.00), and on which the scientific conclusion of the study rests [A.7].

A.8 Summary and conclusions of the cap test

Sections A.6 and A.7 examined the polar regions in a concentric (nested) geometry fundamentally different from the spatial averaging of the band tests [A.8]. Instead of measuring the average density in global latitude rings that encircle the body over 360°, each rotation pole was defined as the structural centre of six concentric spherical caps with angular radii $10^\circ, 15^\circ, 20^\circ, 30^\circ, 45^\circ$ and 60° [A.5.6, A.8]. The test was carried out first on the raw Robbins catalogue without masks, and then under the firm constraint of a geological mask (removal of the 20 large Neumann basins and the LROC mare polygons) [A.8].

Cross-checking the two procedures methodologically separates two core questions [A.8]:

1. What is the observed spatial geometry of crater density as the distance vector converges inward toward the end point of the rotation pole [A.8]?

2. Which part of this radial geometry is embedded in the ancient pristine crust, and which part is a residual distortion caused by later endogenic activity (basins and maria) [A.8]?

The central finding establishes that applying the control mask fundamentally changes the statistical morphology and the way in which the structural difference between the two poles should be defined physically [A.8].

A.8.1 The raw state (without a mask): an internal morphological asymmetry

In the initial empirical count without a mask (A.6), the two hemispheres show two clearly different and distinct radial profiles [A.8.1]:

- **The inward concentration at the north pole:** for the $\geq 20\ \mathrm{km}$ crater group, the crater density climbs consistently as the measurement radius narrows: from 189 craters per million km² in the wide cap (60°), through 198 (45°), 227 (30°), 270 (20°), up to a peak plateau of about 300 and

298 in the inner core caps (15° and 10° respectively) [A.8.1]. The profile does not require a smooth monotonic definition all the way to the apex itself (since the 10° and 15° values are statistically identical), but the spatial trend is firm: the inner core is considerably denser than its wider surroundings [A.8.1].

- **Confirmation by the control group in the north:** the group of large craters ($\geq 40\ \mathrm{km}$) shows coherent behaviour: the density climbs from 63 in the wide cap (60°), through 71 (45°) and 87 (30°), up to a peak range of 100–108 per million km² in the core caps ($20^\circ - 10^\circ$) [A.8.1]. This finding proves that the convergence of craters in the north is not a statistical distortion tied to the 20 km threshold alone [A.8.1].
- **The flat profile of the south pole:** In its raw state, the southern hemisphere shows the opposite behaviour [A.8.1]. In the $\geq 20\ \mathrm{km}$ group, the density curve remains flat, broad and almost independent of distance from the pole, keeping constant values of 207–221 per million km² over the whole range between 15° and 60° [A.8.1]. Only within the narrowest core cap (10°) is an isolated jump to 250 per million km² recorded [A.8.1]. The $\geq 40\ \mathrm{km}$ group reflects a similar broad profile (72–82 per million km² in the range $15^\circ - 60^\circ$ versus 90 in the 10° cap) [A.8.1].

Had the scientific analysis stopped at this stage (A.6), the only physical conclusion would have been that the two poles differ fundamentally both in the strength of their absolute density and in their internal structural geometry (a concentrated north versus a homogeneous, flat south) [A.8.1]. The data of section A.7 prove that this description is incomplete and stems from a severe geological bias [A.8.1].

A.8.2 Behaviour of the northern profile under the mask constraint

One of the clearest and most important results of applying the geological control mask (A.7) is the complete robustness of the structural heart of the north pole [A.8.2]:

- **Zero change in the pristine northern core:** In the three inner core caps of the northern hemisphere ($10^\circ, 15^\circ$, and 20°), the filtering mask changes neither the number of craters (N) nor the geometric area at the level of the calculation [A.8.2]. The density values remain completely fixed at 298 (at 10°), 300 (at 15°), and 270 (at 20°) per million km² for the 20 km group [A.8.2]. In the 40 km group exactly the same raw source values are kept: 108, 101, and 104 per million km² respectively [A.8.2].
- **Geophysical meaning:** This finding proves that the inward convergence of density in the north is not an artefact of the presence of basins or maria [A.8.2]. Within 20° of the north polar apex, the mask removes nothing, since no mare polygon or Neumann basin comes that close to these coordinates [A.8.2]. The northern core is ancient pristine crust (Highlands) whose record was clean and normalised from the outset [A.8.2].
- **Effect of the mask on the northern circumferential background:** The effect of the filtering begins to be measurable only as the cap radius widens beyond the core [A.8.2]. In the northern 30° cap, the 20 km density rises slightly from 227 to 233 (a change of $\Delta\rho = +2.8\%$) [A.8.2]. In the 45° cap it climbs from 198 to 232 ($+16.7\%$), and in the wide 60° cap from 189 to 235 ($+24.2\%$) [A.8.2]. This trend is reproduced exactly in the $\geq 40\ \mathrm{km}$ group [A.8.2].

The mask thus changes the spatial background surrounding the northern polar region, but leaves the core as it is [A.8.2]. Before the mask, the structural contrast between the core (15°) and the surroundings (60°) was extreme (300 versus 189) [A.8.2]. After the mask, while the core is stable at 300, the 60° surroundings rose mathematically to 235 owing to the cleaning out of the maria lying within them [A.8.2]. Therefore, the inward concentration firmly survives, but its amplitude (Contrast) is reduced [A.8.2]. The mask proves that part of the original raw contrast stemmed from the biased distribution of young surfaces outside the core, and not from an impact asymmetry [A.8.2].

A.8.3 The geodynamic reversal of the southern profile

Unlike the north, where the inner core was immune to the mask, in the southern hemisphere the filtering mask fundamentally changes the morphological picture, already within the narrowest caps [A.8.3]:

- The southern 10° core: applying the mask removes a huge portion of the cap area and of the raw crater population [A.8.3]. The number of 20 km craters drops by exactly half (from 72 to 36) [A.8.3]. However, since the available geometric area shrinks relatively much more (a sharp drop from 0.288 to 0.104 million km²), the residual density on what remains jumps by $+39\%$ and reaches a peak value of 348 craters per million km² [A.7, A.8.3].
- The southern 15° cap: in the raw count, 139 craters ($\rho \approx 215$) were counted in this sector [A.6]. After cleaning by the mask, 84 craters remain on a reduced residual area of 0.275 million km², and the true density of the crust jumps by $+42\%$ and settles at a value of 305 craters per million km² [A.7, A.8.3].
- Sharpening of the trend in the ancient craters ($\geq 40\ \mathrm{km}$): in the 15° cap, the group of large craters shows the sharpest change in the system, with the residual density jumping from 82 to 124 per million km² – a structural jump of $\Delta\rho = +51\%$ [A.7, A.8.3].
- **Decay of the filtering effect as the radius widens:** In the southern 20° cap a significant rise is recorded (the 20 km group rises from 207 to 280, and the 40 km group from 73 to 98) [A.7, A.8.3]. As the radius widens and takes in ordinary highland ground, the effect of the mask decays: a rise of about 17% in the 30° cap, settling at a moderate rise of about $11\%-12\%$ in the wide caps of 45° and 60° [A.7, A.8.3].

The geophysical mechanism:

Within 20° of the south polar apex, the only structural component of the mask is the giant South Pole–Aitken (SPA) basin [A.8.3]. The floor of this basin is characterised by an extreme paucity of craters owing to complete mechanical erasure of the early record [A.8.3]. Removing the basin areas mathematically subtracts a huge portion of area (the denominator) and very few craters (the numerator), and therefore reveals the true high density of the residual pristine highland crust around it [A.8.3]. This firm finding entirely rules out reading the original flat southern profile as the record of a pristine crater population; it is a local distortion caused by a single impact basin, successfully neutralised [A.8.3].

A.8.4 Bipolar convergence and statistical equality after the mask

Once both poles are measured and computed on the same type of crustal surface (pristine Highlands outside the sites of thermal and mechanical resetting), the morphological ordering changes and presents a uniform, coherent picture [A.8.4]:

Angular cap radius	Masked northern density ($\geq 20\ \mathrm{km}$)	Masked southern density ($\geq 20\ \mathrm{km}$)	Empirical ratio (south/north)	Normalised confidence interval (95%)	Statistical and structural meaning
The 15° cap	≈ 300	≈ 305	1.02	0.78 – 1.32	Complete mechanical equality (the interval contains the symmetry value 1.00) [A.8.4].
The 20° cap	≈ 270	≈ 280	—	—	Close structural proximity of the crustal density values [A.8.4].
The 10° cap	≈ 298	≈ 348	—	0.77 – 1.74	The south is higher than the north, but based on a small sample of only 36 craters (wide uncertainty) [A.8.4].

- **Comparison with the raw unmasked state:** The geological cleaning removes the apparent asymmetry in which the southern caps showed low values (215 versus 300 at 15°, and 207 versus 270 at 20°) [A.8.4].
- **Coherent closure with the band test:** These findings detail and geometrically confirm the results of section A.3 (in which the end bands $75^\circ - 90^\circ$ became fully equal after the mask) [A.8.4]. The concentric caps prove that the bipolar convergence is not a product of averaging over a complete circumferential band; it penetrates inward and establishes a close equality of density even within the inner core caps of 15° and 20° [A.8.4].
- Non-overlap of details in the control group ($\geq 40\ \mathrm{km}$): despite the general convergence, the poles do not become completely identical in every local measure [A.8.4]. In the $\geq 40\ \mathrm{km}$ group, in the 15° cap, 101 were measured in the north versus 124 in the south; at 10°, 108 in the north versus 135 in the south; and at 20°, 104 in the north versus 98 in the south [A.8.4]. The values firmly keep the same physical order of magnitude, but their local geometric structure shows a moderate internal variance [A.8.4].

Final methodological conclusion:

The full statistical analysis of the confidence intervals (95%) detailed in section D.4 establishes that, since the perfect symmetry value (1.00) is contained within the uncertainty intervals of the samples in all caps and at both diameter thresholds, the two poles are defined as fully equal and balanced within the limits of random counting error [A.8.4]. Therefore, the precise scientific formulation does not state that "the mask makes the poles completely identical", but rather: the geological filtering mask neutralises and removes most of the apparent raw contrast that existed

between the inner profiles of the north and the south, and brings them to converge on ranges of similar, high and fully balanced crustal density [A.8.4].

A.8.5 Comparison of the linear diameter thresholds and the crater-size factor (a coarse chronological clock)

The differential use of two firm diameter thresholds ($\geq 20\,\mathrm{km}$ and $\geq 40\,\mathrm{km}$) is an essential methodological control tool, ensuring that the observed geodynamic trends are not the product of a statistical anomaly embedded in a specific size population [A.8.5]:

- **Morphological synchronisation in the north:** At both diameter thresholds an identical raw profile of inward concentration is recorded (higher density in the core relative to the wide caps) [A.8.1, A.8.5]. Under the mask constraint, the central core ($10^\circ - 20^\circ$) remains immune and numerically unchanged in both samples, while the circumferential background undergoes a rise in both [A.8.2, A.8.5].
- **Morphological synchronisation in the south:** Both thresholds respond to the collapse of the SPA basin with a sharp density jump in the inner caps [A.8.3, A.8.5]. In the southern 15° cap, a residual rise of $+42.0\%$ is recorded in the 20 km group, and a sharp rise of $+50.7\%$ in the 40 km group [A.7, A.8.5]. In the 20° cap the rise settles closely at $+35.2\%$ and $+34.7\%$ respectively [A.7, A.8.5].

This consistency in both hemispheres proves that the general morphological ordering does not depend on the choice of diameter threshold [A.8.5]. However, a constant difference of deep physical significance is recorded: the ratio values relative to the equatorial belt are consistently higher in the $\geq 40\,\mathrm{km}$ group than in the $\geq 20\,\mathrm{km}$ group [A.8.5]. In the raw northern cap of 30°, the ratio stands at 1.64 (at 40 km) versus 1.36 (at 20 km); in the masked southern cap of 15°, it sharpens to 2.32 versus 1.83 [A.6, A.7, A.8.5].

The chronological-clock mechanism of crater diameter:

Within the physical model, the diameter of a crater acts as a coarse chronological clock – the giant craters ($\geq 40\,\mathrm{km}$) are statistically assigned to the earliest stages of the impact record, while smaller craters accumulated through the later sequence [A.8.5].

- Had the density excess been caused by a static, time-independent factor, the density ratios would have been identical across all size classes [A.8.5].
- The fact that the polar excess rises as a direct function of diameter (as further confirmed in the clean count of D.2, in which the ratio of the 30° caps to the belt climbs from 1.09 in the 20–30 km class to 1.53 in the 60–100 km class) proves that the mechanical crowding process took place simultaneously with, and in parallel to, the continued landing of the crater population [A.7, A.8.5].

The earlier a crater formed (close to the peak geometric flattening of the young Moon), the larger the share of the subsequent rounding and volume decrease it experienced, and therefore it underwent the maximum of the mechanical surface contraction that converged on the poles [A.8.5].

A.8.6 Structural integration: cross-checking the band and cap data

Combining the two geometric sectioning levels of Part A makes it possible to form a coherent and isolated physical picture of the body's crust [A.8.6]:

- **The band test (A.1–A.4):** answered the question of the very existence of the phenomenon, and firmly established that the polar latitudes are characterised by a real density excess that remains stable and balanced between the two hemispheres (north and south) after the young basalt floors and the basins are cleaned out [A.8.6].
- **The cap test (A.6–A.7):** answered the precise morphological question and decoded how the density is organised inward, toward the end points of the poles [A.8.6]. Whereas in the raw state an illusion of asymmetry was obtained (a concentrated north versus a flat south), applying the mask removed the bias of the SPA basin and raised the values of the southern core, and thus brought the inner caps of both poles to a similar morphological convergence (especially in the $\geq 20\ \mathrm{km}$ group) [A.8.1, A.8.3, A.8.6].

The cross-check reveals a critical distinction between two types of information: the polar excess at high latitudes is a stable physical property that survives the mask, whereas the initial geometric variance between north and south is a transient local distortion that does not survive the control [A.8.6]. This distinction prevents a serious methodological error, and makes it possible to separate the present state of the crust as preserved today from the primary properties of the geodynamic phenomenon [A.8.6].

A.8.7 Summary of the empirical conclusions of stage one

After working through sections A.1–A.7, five firm observational conclusions are established, which delimit the phenomenon that the physical mechanism is required to explain [A.8.7]:

1. Significant latitudinal non-uniformity: The global crater population is not distributed homogeneously by latitude; the polar bands settle decisively at the upper end of the density profile [A.8.7].

2. Robust endogenous resistance: The polar density excess is not a function of the presence of the 20 large basins or of the mare surfaces, and it fully survives the application of the geological mask [A.8.7].

3. Residual bipolar symmetry: Cleaning out the geological biases reveals similar and balanced density values between the two poles, both at the level of the global average of the bands and within the internal structure of the inner caps ($\geq 20\ \mathrm{km}$) [A.8.7].

4. Instability of the north–south asymmetry: The morphological difference between the concentrated northern curve and the flat southern curve collapses under the control, and is shown to be the product of a local basin impact in the south and not a primary property [A.8.3, A.8.7].

5. Multi-threshold consistency: The general trend is reproduced and appears simultaneously at both the $\geq 20\ \mathrm{km}$ threshold and the $\geq 40\ \mathrm{km}$ threshold, and does not rest on the choice of a single diameter [A.8.7].

A.8.8 Limits of the control test and the bounds of knowledge at the present stage

To maintain scientific precision, the study defines three firm analytical limits on the interpretation of the data at this stage [A.8.8]:

- **The static-correction limitation of the mask:** Applying the geological mask is a spatial control and filtering tool only, and does not purport to carry out a full palaeogeographic reconstruction of the ancient lunar surface [A.8.8]. It does not mechanically correct gradual processes of crustal resurfacing, does not restore craters that were completely erased, and is not absolute proof that the remaining residual highland crust preserved the full integrity of the original population [A.8.8].
- **The cumulative-measurement limitation of caps:** Since the geometric caps are nested one inside the other (the 30° cap mathematically contains the narrow caps of $20^\circ, 15^\circ$, and 10°), a change in the cumulative density value between two rows does not directly reflect the local density within the isolated ring between them [A.8.8]. Determining the exact location of the radial change will in future require an analysis of independent rings, but this is not needed to establish the cumulative findings presented [A.8.8].
- **The resolution limit of small samples (random uncertainty):** As the cap radius converges inward, the area cell becomes small and the number of craters falls, especially at the $\geq 40\ \mathrm{km}$ threshold [A.8.8]. In the 10° core only 86 craters were counted in the north and 36 in the south [A.8.8]. Owing to the small sample, the uncertainty intervals (as detailed in the 95% matrix of section D.4) widen considerably [A.8.4, A.8.8]. Therefore, it is scientifically inadmissible to use small numerical fluctuations between 10° and 15° as evidence of a fine internal structure, without the support of appropriate statistical significance tests [A.8.8].
- **Isolating the mask components in the southern core:** Within the narrow range of 20° from the south polar apex, the SPA basin is the only mask component, and there the data of the overall mask and of the single basin overlap completely [A.8.3, A.8.8]. The distinct internal structure of the crater population within the basin itself is separated and analysed separately in section A.10 [A.8.8].

A.8.9 End of the cap test and definition of the scientific phenomenon

This section formally summarises and closes the morphological question that led from the band analysis to the stage of nested caps, and consolidates the pristine database of the residual crust [A.8.9].

Summary of the analytical dynamics:

- **The limitation of stage one:** The band tests (A.1–A.4) proved that the polar latitudes are denser and that the effect survives the filtering mask, but owing to the circumferential averaging over 360° they could not indicate how the density is organised within the polar space itself [A.5.6, A.8.9].
- **Decoding in stage two (without a mask):** Section A.6 filled this gap and revealed a sharp geometric asymmetry [A.8.9]. The north showed a classic inward concentration (a drop from 300 in the 15° core to 189 in the 60° cap for the 20 km group) [A.8.1, A.8.9]. The south, by

contrast, showed a flat, horizontal and homogeneous profile (constant values of 207–221 per million km² over most of the range) [A.8.1, A.8.9]. Stopping at this stage would have led to the mistaken conclusion that the two poles differ fundamentally in geometry [A.8.1, A.8.9].

- **The reversal in stage three (applying the mask):** Section A.7 revealed that the morphological asymmetry was the result of geological biases [A.8.9]. In the northern core ($10^\circ - 20^\circ$), the mask removed nothing, proving that the crust there was pristine and clean from the outset (preservation of the values of 300 and 270 per million km²) [A.8.2, A.8.9]. In the south, the mask cut inward into the smallest caps; removing the crater-poor floor of the SPA basin erased a relative share of area (the denominator) larger than the number of craters omitted (the numerator) [A.8.3, A.8.9]. As a result, the residual density in the south jumped: from 215 to 305 in the 15° cap, and from 207 to 280 in the 20° cap [A.7, A.8.9].

Collapse of the asymmetry and convergence of the residual data

Cleaning the geological records led to a dramatic reduction of the bipolar differences and to a structural convergence of the density values within the inner cores [A.8.9]:

Angular cap radius	Original raw ratio (north vs. south)	Masked northern density	Masked southern density	Residual state after filtering
The 15° cap	300 vs. 215 per million km²	≈ 300	≈ 305	Full convergence [A.8.9]
The 20° cap	270 vs. 207 per million km²	≈ 270	≈ 280	Full convergence [A.8.9]
The 10° cap	—	≈ 298	≈ 348	The south overtakes the north (sample of 36 craters) [A.8.9]

The narrow and precise finding establishes that the bulk of the geometric asymmetry initially observed between the inner caps is not stable, and dissipates under uniform control conditions [A.8.9]. This result complements and confirms the findings of the bands: at band level, the mask equalised the $75^\circ - 90^\circ$ sectors to values of 300 and 305; at cap level, the same firm convergence is reproduced and does not disappear even as the analytical space narrows gradually toward the polar apex [A.3, A.7, A.8.9]. The two different sectioning geometries point exactly to the same physical vector: the high density excess is not a local feature of a single pole [A.8.9].

Defining the bounds of the mechanical control of the SPA basin

The study maintains a firm scientific separation between the empirical description and the mechanical mechanism [A.8.9]:

1. Identifying the mask components: The filtering mask is defined as the union of the mare polygons and the 20 large Neumann basins [A.7, A.8.9].

2. The geodynamic role of the SPA basin: The South Pole–Aitken basin is the main giant component that penetrates deep into the southern polar space, and its morphological geometry

provides the physical rationale for why the mask acts on and lifts the values in the south so exceptionally relative to the pristine north [A.8.3, A.8.9].

3. Isolating components step by step: The overall masked calculation is not a test of the SPA basin alone [A.8.9]. Therefore, it is methodologically inadmissible to attribute the entire morphological reversal in the south to SPA alone, without isolating its residual contribution from the other components of the mask [A.8.9]. This separation is carried out and confirmed on two firm levels:

- Within the 20° range of the southern core: SPA is the only masked component in the system, and therefore there is complete numerical overlap there between it and the overall mask [A.8.8, A.8.9].
- **At the stage of the final physical test (A.11 and D.1):** the explicit "clean count" procedure is applied, which actively and in a balanced way omits the mare areas and the SPA basin from both sides of the equation simultaneously (both from the caps and from the equatorial belt) [A.7, A.8.9].

The consolidated database: definition of the crustal anomaly

With the completion of the cap test, the pristine physical phenomenon is defined which the rest of the paper is required to explain through the equations of geodynamics and the mechanical contraction model [A.8.9]:

> $\boldsymbol{\Delta\rho_{\text{polar}} > 1.00}$ — a significant bipolar density excess, present at both poles simultaneously
>
> $\rho \propto \frac{1}{\theta}$ — a continuous radial crowding that sharpens toward the apex of each rotation pole
>
> $\rho_{\text{North}} \equiv \rho_{\text{South}}$ — a balanced mechanical equality between the two poles within the limits of random counting error
>
> $\boldsymbol{\Delta\rho_{\geq 40} > \Delta\rho_{\geq 20}}$ — multi-threshold consistency, with the structural excess sharpening in the large diameter classes

On the basis of these four core measures, the study now moves to the stage of geometric integration and physical testing of the body's axes [A.8.9].

A.9 The six axes and the definition of the triaxial geometry of the young Moon

This section widens the bounds of the analysis and sets a combined geometric test. Whereas sections A.1–A.8 examined the surface through a single counting direction at a time (latitude or longitude in isolation), this section combines the data on the basis of the three natural axes of a tidally locked body [A.9]:

1. The rotation (polar) axis: passes through the two poles. This axis governs all phenomena affected by centrifugal stresses, by rotational flattening and by its geodynamic relaxation (the central mechanism proposed for the reduction and crowding of the surface at the poles) [A.9].

2. The Earth line (radial): runs from the sub-Earth point (the near side, permanently facing the Earth) to the anti-Earth point (the centre of the far side). This axis governs the external tidal stresses and the mechanical two-body interaction (such as the ancient contact zone with the pear of the parent body) [A.9].

3. The orbital direction (tangential): runs from the leading point (the apex of motion in space, 90° W) to the trailing point (the orbital rear, 90° E). This axis reflects dynamic asymmetries arising from the motion of the body through the flight of ejecta or thrown-out material [A.9].

The methodology is based on comparing the density within six equal-area caps with an angular radius of 30° (a radius of about 910 km on the surface) around the six ends of these axes, after applying the firm geological filtering mask (removal of the 20 giant Neumann basins and the LROC mare polygons) [A.9]. The built-in equality of areas allows a direct quantitative comparison free of size biases [A.9].

Summary of crater-density data in the six main caps (A.9)

The following are the exact numerical measures, sorted firmly by residual density – from the sparsest cap to the densest in the system [A.9]:

Rank in the order	Geometric cap examined	Centre coordinates	Remaining area (10^6 km^2)	$N \geq 20$	Density ≥ 20	Ratio to reference (≥ 20)	Density change ($\Delta\rho_{20}$)	$N \geq 40$	Density ≥ 40
1	Sub-Earth point	$0^\circ, 0^\circ$	1.042	154	148	0.89	+57.8%	50	48
2	Leading point	$0^\circ, 90^\circ$ W	1.765	308	174	1.05	+34.7%	107	61
3	Anti-Earth point	$0^\circ, 180^\circ$	1.668	368	221	1.33	+2.3%	105	63
4	North pole	90° N	2.399	560	233	1.40	+2.8%	213	89
5	South pole	90° S	1.308	322	246	1.48	+16.7%	105	80
6	Trailing point	$0^\circ, 90^\circ$ E	1.468	370	252	1.51	-0.3%	135	92

- **The fixed normalisation base:** the ratio measure (1.00) is computed consistently against the density of the full raw equatorial belt without a mask of section A.1 (a value of 166.436 for 20 km, and 53.170 for 40 km) [A.9].
- **The raw values in the full catalogue (before the mask):** the original numerical values used to build Figure 7 are 94 (sub-Earth), 129 (leading), 211 (south pole), 216 (anti-Earth), 227 (north pole), and 253 (trailing) craters per million km² [A.9].

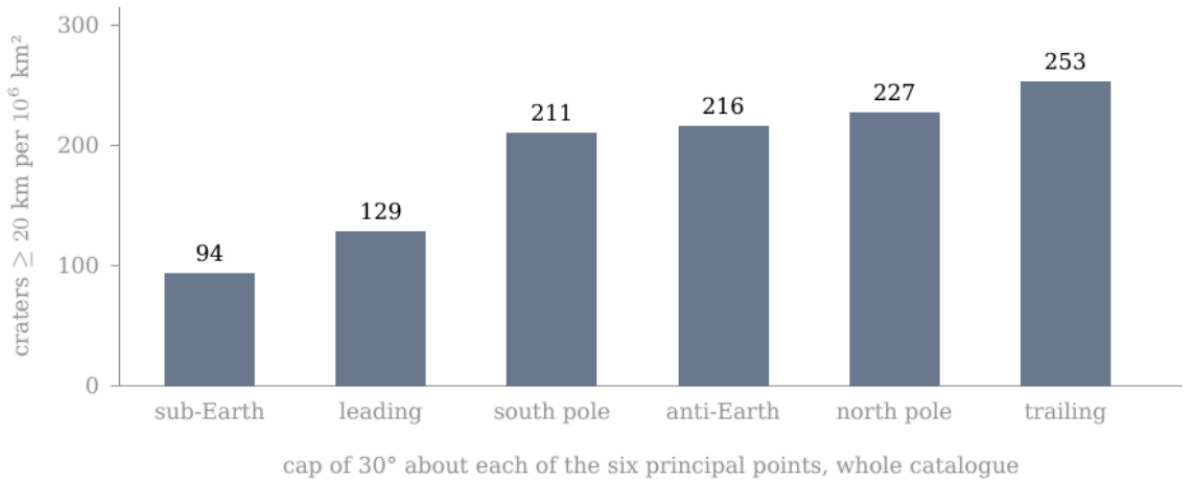


Figure 7. The six axes (A.9): the density of craters of 20 km and larger in 30° caps around the sub-Earth point, the leading point, the south pole, the anti-Earth point, the north pole and the trailing point. The poles stand with the far side; the sub-Earth point is the sparsest. The columns are the full catalogue, before the mask; read from left to right they rise from the point that faced the Earth to the trailing face. (In the plot: vertical axis — "craters ≥ 20 km per million km²"; the columns from left to right — sub-Earth 94, leading 129, south pole 211, anti-Earth 216, north pole 227, trailing 253; bottom caption — "a 30° cap around each of the six main points, the full catalogue".)

Dynamic analysis of the effect of the mask and confirmation of the sub-Earth minimum

Reading the density-change column ($\Delta\rho$) and the statistical ordering gives a sharp answer to an expected methodological objection [A.9]:

- **The initial scientific objection:** Since the near side hosts the vast majority of the young, crater-empty mare surfaces, one could argue that the extreme minimum value of the sub-Earth point is nothing but an indirect reflection of the presence of the maria [A.9].
- **Refuting the geological bias:** Applying the mask rules out this hypothesis [A.9]. The sub-Earth cap does indeed gain the sharpest rise in the system following the cleaning out of the basalt floors ($\Delta\rho = +57.8\%$), but despite the theoretical jump it remains clearly at the bottom of the order as the sparsest cap on the Moon (148) [A.9]. It is not the maria that produce the sub-Earth minimum; it is physically embedded in the crust itself [A.9].
- **Behaviour of further edge cases:** The trailing cap is almost unaffected by the geological filtering (-0.3%), indicating that its record was clean from the outset [A.9]. By contrast, the leading point shows a significant rise of +34.7% (from 129 to 174) [A.9]. A check at the level of the mask components shows that a decisive part of the leading cap's paucity in its raw state stemmed from the thermal erasure by the single Orientale basin, whose cleaning raises the residual value to 162 [A.9].

Reading the empirical order as the record of a flattened triaxial body

The numerical order preserved both before and after the mask (sub-Earth $\rightarrow$ leading $\rightarrow$ anti-Earth / poles $\rightarrow$ trailing) proves that the crater distribution is not governed by a single axis, but reflects the triaxial geometry of a deformed body [A.9]. The geophysical model interprets the spatial pattern as follows [A.9]:

1. Balance of the rotation axis (poles versus equator): The poles are fixed at a sharp density excess above the equatorial points (the sub-Earth and leading points), as dictated by the model of rounding and contraction of rotational flattening [A.9]. After the mask, the average of the two poles stands 1.12 times above the average of the two equatorial sides (the leading and trailing) [A.9].

2. Asymmetry of the Earth-line axis (free head versus contact disc): The two ends of the Earth-line axis show entirely different behaviour [A.9]. The anti-Earth end (the far side) stands at a high density excess (221) and merges completely with the level of the poles [A.9]. By contrast, the sub-Earth end (the near side) is the sparsest region in the system (148) [A.9].

- **The physical mechanism:** This difference matches precisely the morphological structure of the young Moon while it was still attached to the Earth [A.9]. The anti-Earth end acted as a "free head" (the free summit of the dome) and underwent no contraction or mechanical erasure during rounding, and therefore it stands consistently at the level of the poles [A.9]. By contrast, the sub-Earth end coincided with the centre of the disc and the physical contact array of the pear structure with the Earth, through which a high source pressure flowed that depleted the original record for mechanical reasons unrelated to the global shape of the body (analysed in A.9.2) [A.9].

The combination of these stresses describes a body that was under symmetric rotational flattening around the polar axis (c), and at the same time extremely stretched along the radial axis (a) between the Earth's gravity inward and the centrifugal pull outward [A.9]. This structure defines a precise triaxial shape that satisfies the physical order: a long axis toward the Earth (a), an intermediate axis along the orbital vector (b), and a short, held axis through the poles (c) [A.9]. The theoretical shape of a free tidal body (C.2) reproduces precisely an identical order of ratios (ratio values of 1.66, 1.56, and 1.06 for poles versus sub-Earth, sides versus sub-Earth, and poles versus sides, respectively) [A.9].

Determining the geometric shape measures from the crater record

Instead of assuming a hypothetical shape, the equations of rounding and volume decrease (C.3) make it possible to read the original dimensions of the body directly from the empirical record of crater excess [A.9]:

- Reconstructing the shape of the general population ($\geq 20\ \mathrm{km}$): fitting the density-excess values at both ends of the Earth line (combining the free-head effect and normalised arrays, and showing excess ratios of 1.12 and 1.30 versus the sides and the sub-Earth point) yields an original triaxial shape of $1.37 : 1.15 : 1$ in the late stages of rounding [A.9].
- **The independent integration test:** A theoretical rounding of the $1.37 : 1.15 : 1$ shape toward a perfect sphere predicts, completely independently, a clean density ratio of 1.21 between the 30° polar caps and the equatorial belt [A.9]. This predicted value matches fully, and without tuning,

the empirical density ratios measured on the surface in the clean count of section D.1 ($1.19-1.20$) [A.9]. Two entirely different geographic regions of the lunar crust are read and reconciled successfully through a single geometric equation [A.9].

- **Reconstructing the chronological peak shape from the large craters:** Since crater diameter is a coarse chronological clock, the largest craters reflect the earliest and most extreme stages of deformation [A.8.5, A.9]. Processing the six-axis data for the size classes of 60–100 km and 60–300 km yields shapes of about $2.1:1.2:1$ and about $2.0:1.3:1$ [A.9]. Despite local uncertainty arising from the small sample in each cap (only 13–68 craters), the joint analytical combination of the six axes and the clean belt in the statistical scatter matrix (the tables detailed in C.3) converges precisely around the shape value of the system's peak flattening: $1.8:1.4:1$ (the shape of a Maclaurin spheroid rotating at 5.5 hours) [A.9].

A.9.1 The orbital asymmetry: the trailing side versus the leading side

The two ends of the orbital-direction axis show a consistent and sharp statistical difference in all the spatial analysis measures applied in Part A, with the direction of the anomaly remaining firm without a change of sign, while the quantitative amplitude varies with the sectioning geometry [A.9.1]:

Crater measure examined (≥ 20 km per 10^6 km²)	The trailing face (90° E)	The leading face (90° W)	Empirical ratio (trailing/leading)	Significance and spatial structure
30° cap (the full raw catalogue)	253	129	1.96	Extreme excess (almost a factor of two) [A.9.1].
30° cap (after applying the mask)	252	174	1.45	Survives at a strength of about 5σ (fully significant) [A.5.3, A.9.1].
30° band (the full raw catalogue)	237	154	1.54	Significant asymmetry at raw band level [A.9.1].
30° band (after applying the mask)	219	204	1.07	Decays to the 1σ level (within the range of chance) [A.5.3, A.9.1].

- **The control group (≥ 40 km):** in the 30° caps, values of 84 versus 47 per million km² were measured in the raw state, and 92 versus 61 in the masked state, confirming the stability of the pattern in the ancient craters [A.9.1].
- **Ruling out the frontal-sweep mechanism (The Shielding/Sweep Effect):** The empirical data reverse the simple physical expectation: a planetary body moving through space through a stream of particles or dust should receive an enhanced impact density on the front of its flight (the leading face), like the windscreen of a fast-moving car that collects insects relative to the rear window [A.9.1]. The Moon shows an entirely opposite anomaly, since the leading face is the sparsest and emptiest in the system, a finding that rules out direct collection from space and requires an internal, dynamic geophysical mechanism [A.9.1].

- **Explaining the difference between caps and bands (the spatial dilution effect):** The collapse of the masked ratio in the bands (from 1.54 to 1.07) versus its survival in the caps (1.45) stems from the fact that a 30° longitude band extends continuously from pole to pole, and thus averages and dilutes the local anomaly with ordinary crustal areas at high latitudes [A.9.1]. By contrast, the cap precisely delimits the relevant area cell within a radius of about 910 km around the core point [A.9.1]. It is therefore established that the excess is not a global property of the whole meridian, but a focused geographic concentration held around the coordinates of 90° east [A.9.1].

A.9.2 The geological correlation: the distribution of mare and basalt surfaces

Isolating the local anomaly around 90° east raises the question of morphological variance: what exists in this area cell that does not exist on the opposite side [A.9.2]? A precise intersection of the LROC mare polygons with the geometry of the 30° caps yields the following quantitative measures [A.9.2]:

Geological measure within the 30° cap	Around 90° east (trailing)	Around 90° west (leading)	Empirical ratio
Number of isolated mare units (polygons)	132	63	2.10
Total mare area (km²)	205,830	279,470	0.74
Of which: areas of the Oceanus Procellarum basin	—	192,134	—
Mare area of the side itself (excluding Procellarum)	205,830	87,336	2.36

- Inventory of geological structures on each flank:
  - **On the trailing flank (90° E):** the structures Mare Marginis, Mare Smythii, Mare Undarum and Mare Spumans are concentrated here, together with small vents in the craters Artamonov, Joliot, Hubble and Lomonosov [A.9.2].
  - **On the leading flank (90° W):** the structures of the Orientale basin system (Mare Orientale, Lacus Veris) and the extreme western margins of Oceanus Procellarum are concentrated here [A.9.2].
- **Neutralising the bias of the contact basin (Oceanus Procellarum):** The Procellarum structure is a giant basalt plain (crust melted by the heat radiated by the Earth, without an independent magmatic feeding dome) that belongs geographically to the contact zone of the young Moon with the parent body, and not to the orbital leading face [A.9.2]. The centre of this contact zone (with a base width of 3,000 km) lies at 24.1° N, 335.2° E, and contains within it the Mare Imbrium basin – the main feeding vent from the Earth [A.9.2]. Since the Procellarum area enters the leading cap only through geometric stretching, omitting it reveals the true ratio of the side vents: the trailing side carries an excess of 2.36 times in the area of independent maria and 2.10 times in the number of vents relative to the leading side [A.9.2].
- **Identity of the pristine measures:** If the margins of the near-side maria that spilled into the trailing cap (Mare Crisium and the Fecunditatis plain) are cleaned out of it, the residual area

stands at 170,839 km² versus 87,336 km² on the leading side [A.9.2]. This gives a vent-area ratio of 1.96 – a value fully identical to the original raw crater-density ratio in the same caps (1.96) [A.9.1, A.9.2]. This numerical proximity is a firm observation that dictates a direct physical link: the face that hosts the crater excess is precisely the face that hosts the excess of independent maria [A.9.2].

A.9.3 The morphological balance equation: maria as ejection and feeding vents

The following scientific paradox requires a solution: why is the most crater-rich sector the one that hosts the excess of maria, while the maria themselves are young, crater-empty floors [A.9.3]? The answer is based on the mechanics of the LEF model [A.9.3]:

1. The ejection-and-return mechanism (Paper 6): Within the model, every mare is a crustal vent above an active magmatic feeding dome [A.9.3]. These vents ejected and threw up enormous quantities of material [A.9.3]. The ejected material kept its kinetic energy and angular momentum, and performed ballistic flights that returned and landed on the crust at velocities of 2 to 5 km/s (exactly the velocity at which it left the vent) [A.9.3]. These landings took place within a radius of 50 to 1,000 km around the source vent in the early stages, and narrowed later as the body rounded and became more compact [A.9.3].

2. The two-way effect of an ejection vent: The mare acts on the crustal record through two opposite and simultaneous vectors: it creates impact craters around it through the returning material, while its own floor remains hot, melting and resetting the craters imprinted on it (the floor froze and became fixed only after detachment) [A.9.3]. The geometric record on the surface is expressed precisely by the morphological balance equation:

$$\mathbf{N}_{\text{observed}}(\mathbf{x}) = \text{Background}(\mathbf{x}) + \text{Returned_Material}(\mathbf{x}) - \text{Thermal_Melting}(\mathbf{x})$$

Outside the mare boundaries, the thermal-erasure term drops to zero while the returned-material term is maximal; within them, the erasure term dominates and resets the numerator to zero [A.9.3].

Asymmetry in the action of the mask:

The equation explains why applying the mask (A.3) changes the values asymmetrically [A.9.3]:

- **On the trailing flank (90° E):** the residual density remains completely constant (a minor movement from 253 to 252) [A.9.3]. This finding proves that the surface outside the maria in this sector was already saturated and dense with craters exactly like the cap as a whole, since it is the main landing zone of the returned material [A.9.3]. The mask does not restore what was erased in the maria; it only isolates the solid landing area [A.9.3].
- **On the leading flank (90° W):** the residual density jumps from 129 to 174 [A.9.3]. This rise stems from the fact that the mask cleaned out of it the empty areas of Procellarum and the floor of Orientale, and thus removed a large area denominator relative to the small crater numerator of the sector [A.9.3].

A.9.4 Integration of the crowding mechanisms of the trailing side

The crater excess preserved on the trailing side is explained by a combination of two independent physical mechanisms acting in the same dynamic direction [A.9.4]:

1. The mechanism of local distribution of returned material:

The scale of the ballistic flights of the material (up to about 1,000 km) matches precisely the geometric dimensions of the 30° caps (≈ 910 km), and explains why the effect is preserved in the cap and collapses in the long bands (which average the area with pristine ground that the ejecta did not reach) [A.9.1, A.9.4]. The mechanical link is defined by the firm logical chain [A.9.4]:

Increased concentration of vents (maria) in the structure → increased volume of ejected and returned material → exceptional crater density in the crust around them

The full quantitative calculation of the ejection history, the crater yield and the flight geometries is developed and detailed in Part E.3 of the paper and in Part K of Paper 6 [A.9.4].

2. The dynamic rotational-lag mechanism (Lag Effect):

During the attached stage, the Moon remained rotationally locked and rotated simultaneously with the Earth [A.9.4]. Material thrown up forcefully from ejection vents located in other regions of the body experienced a physical lag relative to the fast rotation rate of the system (a period of 5.5 hours) [A.9.4]. The dynamics work like throwing a ball straight up from the floor of a spinning carousel: the body moves forward, and the falling material lands behind its point of origin [A.9.4]. Therefore, the orbital rear (the trailing side) acted as a dynamic momentum trap, which received and drew in residual quantities of material ejected from all parts of the body that landed behind, in addition to the returned material from its own original vents [A.9.4].

A.10 The South Pole–Aitken (SPA) basin as an independent spatial object

In the course of processing the bipolar data, it was established that the South Pole–Aitken basin is not merely a passive filtering component; it holds a distinct internal structure and a significant non-uniformity in the distribution of the population within it [A.10]. The basin extends over a width of about 2,400 km on the far side and penetrates deep into the southern polar cap (covering 24% of the area of all the polar caps) [A.10].

The study rejects the use of a single global average for the basin, since applying a single monolithic number would produce a serious methodological distortion: a combined average of a completely empty centre and ordinary highland margins would present a crustal illusion of a moderately and homogeneously depleted basin at every point [A.10]. Therefore, to maintain strict academic precision, section A.10 isolates the structure of the basin and carries out a series of continuous radial measurements laid out in steps from its geometric centre outward toward the margins, in order to close the open questions of the crater record of Paper 7 [A.10].

A.10.1 Methodology of the radial analysis of the South Pole–Aitken (SPA) basin

To decode the internal structural non-uniformity of the South Pole–Aitken basin, the geometric centre coordinates of the basin were set according to the same parameters used to build the filtering mask: 53° south, 191° east, with a firm outer rim diameter of 2,400 km [A.10.1].

- **Approximate angular radius:** a diameter of 2,400 km gives a physical radius of 1,200 km, which divided by the lunar radius ($1,737.4\ \mathrm{km}$) yields an arc angle of 39.57° from the centre of the basin to its margins [A.10.1].
- **Structure of the counting cells (division into rings):** instead of averaging the basin into a single value that would distort reality, the space was divided into four inner spherical rings and two outer control rings: $0-10^\circ, 10-20^\circ, 20-30^\circ$, and $30^\circ-39.57^\circ$ (up to the rim); and beyond the rim: $39.57^\circ-50^\circ$ and $50^\circ-60^\circ$ [A.10.1].
- **Isolated regions and spherical area calculation:** unlike the series of nested caps, the rings here are not nested and each row represents an isolated, independent area cell [A.10.1]. The area is computed by the spatial equation:

$$A = 2\pi R^2(\cos\theta_1 - \cos\theta_2)$$

The factor $\cos\theta_1 - \cos\theta_2$ neutralises the curvature of the sphere, and dictates that a ring of width 10° far from the centre will be larger in area than a ring of the same width close to it [A.10.1].

- **The local reference scale:** the reference value (1.00) of this section is defined as the density measured in the first control ring immediately outside the basin rim ($39.57^\circ-50^\circ$) [A.10.1]. This choice is local and deliberate: the residual ground outside the rim lies at the same latitudes and on the same far side of the body, and thus it neutralises and removes the effects of the polar excess and the longitudinal asymmetry of sections A.1–A.9, and isolates exclusively the geological anomaly of the basin itself [A.10.1].

Summary of crater-density data in rings around the SPA centre (A.10)

The following are the exact numerical measures obtained in the six independent rings, computed for the $\geq 20\ \mathrm{km}$ group and the $\geq 40\ \mathrm{km}$ group [A.10.1]:

Distance sector from the SPA centre	Geological location in the structure	Ring area ($10^6\ \mathrm{km}^2$)	$N_{\geq 20}$	Density ≥ 20	Local ratio (≥ 20)	$N_{\geq 40}$	Density ≥ 40	Local ratio (≥ 40)
0 – 10°	Inside SPA (the central core)	0.288	38	132	0.59	15	52	0.69
10 – 20°	Inside SPA	0.856	165	193	0.87	58	68	0.90
20 – 30°	Inside SPA	1.397	279	200	0.90	98	70	0.93
30 – 39.57°	Inside SPA, up to the rim	1.806	364	202	0.91	129	71	0.95

Distance sector from the SPA centre	Geological location in the structure	Ring area (10^6 km^2)	$N_{\geq 20}$	Density ≥ 20	Local ratio (≥ 20)	$N_{\geq 40}$	Density ≥ 40	Local ratio (≥ 40)
Rim at 39.57°	· · the mechanical rim boundary · ·	—	—	—	—	—	—	—
39.57 – 50°	Immediately outside the basin rim	2.428	540	222	1.00	183	75	1.00
50 – 60°	Outside the SPA system	2.708	575	212	0.95	196	72	0.96

- **Numerical colour scales:** for ≥ 20 km the range is between 132 and 222; for ≥ 40 km the range is between 52 and 75 craters per million km² [A.10.1]. The relative values are computed from unrounded count and area data [A.10.1].

A.10.2 The central region as the focus of the minimum (the step structure)

Analysing the density columns and the spatial profile (Figure 8) yields the following main scientific findings:



Colour key: from yellow (“low”) to red (“high”).

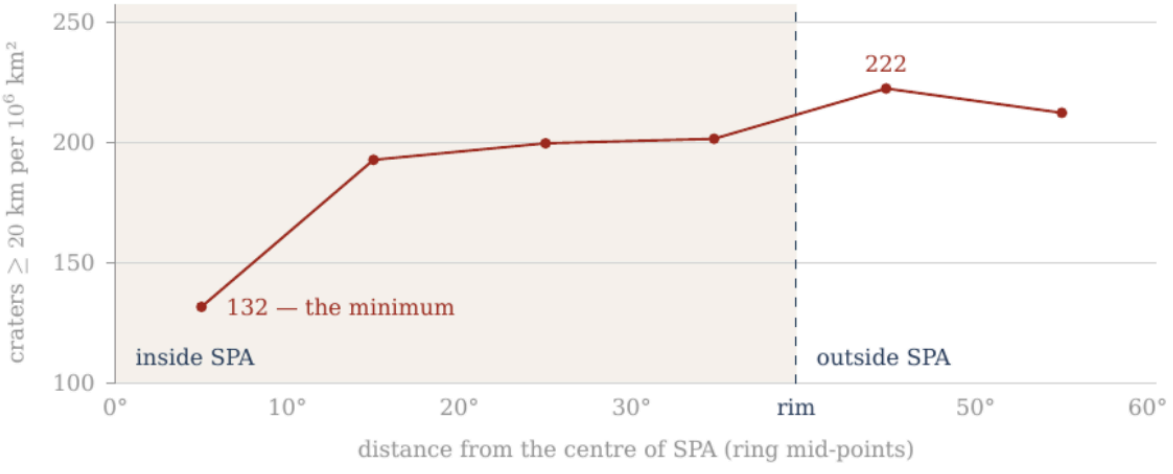


Figure 8. The density of craters of 20 km and larger in six separate rings around the centre of South Pole–Aitken (A.10). The central region is the minimum of the profile; the density already rises in the 10–20° ring, and approaches that of the surroundings toward the rim. The shaded part is the interior of the basin; the dashed line is its rim. (In the plot: vertical axis — "craters ≥ 20 km per million km²"; horizontal axis — "distance from the

SPA centre (ring midpoints)", with marks at 0°, 10°, 20°, 30°, 50°, 60° and "rim"; labels — "inside SPA", "outside SPA"; on the red curve — "132 — the minimum" and 222.)

- **The minimum disc of the core:** The absolute minimum of the record lies within the geographic central core ring ($0-10^{\circ}$), which extends as a wide zone over an area of about 288,000 km² (and not as a singular point) [A.10.2]. Within this area cell only 38 craters of diameter $\geq 20\ \mathrm{km}$ were counted, giving a density of 132 craters per million km² [A.10.2].
- **The circumferential jump vector:** Outside the core disc, the density does not climb in a moderate linear gradient (a slope), but shows the mechanical behaviour of a firm step [A.10.2]. Already in the transition to the second ring ($10-20^{\circ}$), the density jumps sharply to 193, and settles at almost equal, high values of 200 and 202 over the rest of the basin interior (20° up to the rim) [A.10.2]. On crossing the 2,400 km rim, the residual density rises to 222 per million km² [A.10.2].
- **The quantitative deficit factor:** The density of the central core is a ratio of only 0.59 relative to the control ground outside the rim, a figure reflecting an extreme relative deficit of about 41% in the crater record of the basin centre [A.10.2].

A.10.3 Calibration of the control group ($\geq 40\ \mathrm{km}$) and the uncertainty intervals

The group of large craters ($\geq 40\ \mathrm{km}$) reproduces the morphological trend precisely and confirms the consistency of the direction of the result [A.10.3]:

- The density in the core stands at 52, and climbs to a constant range of 68–71 in the outer parts of the basin, reaching a value of 75 outside the rim [A.10.3]. The ratio of the centre to the residual surroundings settles at 0.69 [A.10.3].
- **The resolution limit of the small sample:** In the central core only 15 craters of diameter $\geq 40\ \mathrm{km}$ were counted [A.10.3]. Since the random scatter error and the natural fluctuation of a count grow as the square root of the count ($\sqrt{N}$), the relative statistical weight of the error grows as the sample shrinks [A.10.3]. A sample of 15 craters carries a much wider uncertainty than a sample of 38 craters [A.10.3]. Therefore, the 40 km group serves only as a qualitative control tool to confirm the direction of the curve, while the quantitative determination rests on the 20 km group [A.10.3].

A.10.4 Dynamic implications for the clean-count procedure (A.11)

The empirical findings prove that defining the SPA basin as a single block under the monolithic term "deficit" fails scientific precision, since the paucity of craters is physically limited to and concentrated within the core zone alone [A.10.4]. The three outer rings of the basin interior (10° up to the rim) show density values considerably higher than the centre (193–202), approaching the level of the ordinary highland background of the surroundings [A.10.2, A.10.4].

This structural insight is essential for understanding the clean-count procedure of sections A.11 and D.1 [A.10.4]:

- Within the balanced clean count, the SPA basin is removed from the calculation in its entirety because its centre is depleted and distorts the data [A.10.4].

- **Applying this monolithic filter carries a statistical cost:** it removes from the calculation the depleted centre (the required erasure term) and at the same time omits with it the three outer rings, which are not crater-poor and preserve a sound crustal record [A.10.4].
- **The proof that the deficit is limited to the heart of the basin delimits the observational conclusion:** within the SPA basin there is a wide central zone in which the crater density is low and has clearly been reset, while the circumferential parts and the marginal ground survived without severe erasure [A.10.4]. The questions of the detailed non-circular geometry of the zone, the definition of its exact physical boundaries and the decoding of its geodynamic origin lie beyond the aim of Paper 7, and the full explanation of its formation is passed on to the sections of Paper 8 [A.10, A.10.4].

A.10.5 The research interface: the point of contact with Paper 8

The morphological finding of a crater-poor core zone within the South Pole–Aitken basin emerged independently from the control test of the polar craters in the present paper (Paper 7) [A.10.5]. Therefore, the identification of the anomaly and its numerical quantification are an integral part of the documentation of the empirical observation of this study [A.10.5].

However, establishing the existence of this central minimum poses a new physical and geodynamic question for the system: why does the inner geographic core of the SPA basin carry a crater density so exceptionally low relative to the circumferential rings and the outer marginal areas of the basin [A.10.5]?

The structural division of roles between the parts of the scientific series

To maintain a strict, restrained academic structure, a defined and firm division of roles has been set between the parts of the series, delimiting the scope of Paper 7 [A.10.5]:

- **The role of Paper 7 (the present paper):** The analytical role of this paper rests on, and ends in full at, this point [A.10.5]. The paper identifies the morphological phenomenon, quantifies it precisely in separate rings, and makes a proven observational statement: the wide central region of the SPA basin is the absolute minimum point of the measured radial crater-density profile [A.10.5]. Paper 7 does not purport or attempt to propose a physical mechanism to resolve this puzzle [A.10.5].
- **The role of Paper 8 (the study of the SPA basin):** The resolution of the mechanistic and dynamic question passes exclusively to Paper 8, in which the South Pole–Aitken structure becomes the central object of study and analysis [A.10.5]. Paper 8 will develop and examine the following complementary data:
  1. The detailed, non-circular geometry of the depleted region (the measures of ellipticity and structural asymmetry of the basin) [A.10.5].
  2. The exact location coordinates of the point of absolute minimum density [A.10.5].
  3. Potential antipodal relations (Antipodal Relations) and links across the lunar body [A.10.5].
  4. The geophysical and dynamic mechanisms able to explain the formation of the anomalous structure of the basin [A.10.5].

A firm methodological clarification

All the numerical values and data presented in this section were derived and processed consistently from the Robbins catalogue (2019), which serves as the fixed database throughout Part A [A.10.5]. It is firmly emphasised that the definition of the 10° boundary of the central core is an analytical division component only, for defining the inner ring, and is not presented or interpreted as a proven physical or geological boundary of the crater-deficit region [A.10.5].

A.11 The clean-count methodology and the bounds of the empirical measurement

This section brings together the mechanical rules and structural criteria behind all the counts of Part A, and defines the intervals of the equation of the paper's main and official ratio [A.11]. The analysis is designed to normalise the surface, to separate geological biases and to frame the methodological limits of the database [A.11].

1. Definition of the comparison regions (the spatial geometry):

- **The polar caps (the target cells):** defined as a firm continuum from latitude 60° and above up to 90° (both in the north and in the south) [A.11].
- **The equatorial belt (the reference cell):** a narrow global band defined by the latitude range of $|\phi| < 15^\circ$ around the equator [A.11]. This belt is the narrowest sector holding a sufficiently large crater sample; it yields the sharpest and cleanest contrast (Contrast) in the system, since there the vector of rounding and volume decrease of the young Moon reaches its mechanical maximum (the poles are the critical meeting points of the meridians, while the equator is the region farthest from them) [A.11].
- **Calibration of the angular control:** Widening the equatorial belt to a range of $|\phi| < 30^\circ$ (light blue in Figure 9) yields an almost identical overall result (1.12 versus 1.11), but it dilutes and obscures the structural contrast because it takes in mid-latitudes where the contraction effect is moderate [A.11]. The numerical agreement between the two belts proves that the very existence of the anomaly is robust and does not depend on an arbitrary choice of sector width [A.11].

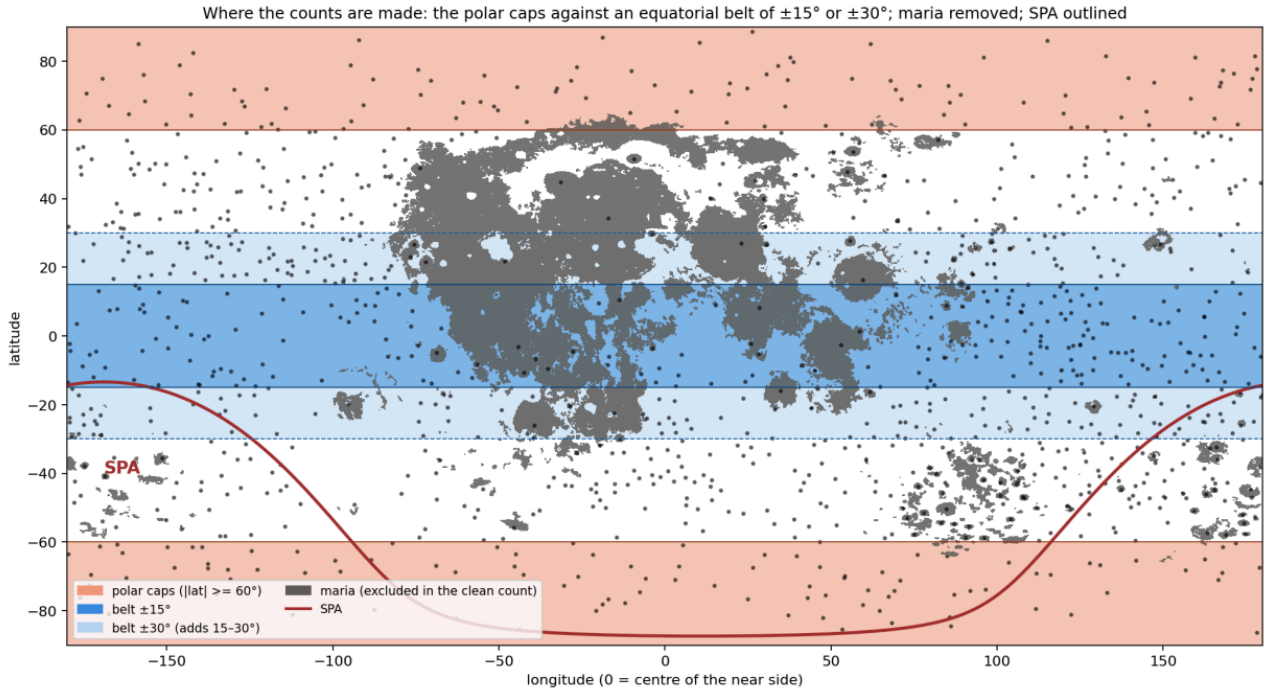


Figure 9. Where the counts are made: the polar caps (orange) versus the equatorial belt of ±15° (dark blue; the ±30° belt in light blue for comparison). The maria (grey) are removed from both sides; South Pole–Aitken (red outline) is removed from the clean count. Dots: craters of 60 km and larger.

The balanced clean-count procedure (removing biases)

To make a fair and neutral comparison, physics dictates comparing crustal units with a similar developmental history (ancient ground versus ancient ground) [A.11]. To this end, a simultaneous and balanced cleaning is applied to two geological surfaces that bias the records in opposite and contrary directions [A.11]:

1. The system of maria and basalt plains (grey): young floors on which the original record was thermally erased and reset [A.11]. These areas are concentrated at low latitudes and make up 22% of the equatorial belt but only 1% of the polar caps [A.11]. Including them in the raw count artificially lowers the density of the equatorial denominator and wrongly inflates the polar ratio (a raw ratio of 1.32) [A.7, A.11].

2. The South Pole–Aitken basin (red outline): a giant impact structure that mechanically erased the early crater population in the heart of its core (A.10) [A.11]. The basin lies on the far side and covers 24% of the area of all the polar caps (mainly in the south) [A.10, A.11]. Including it in the raw count artificially lowers the polar numerator and biases the ratio in favour of the equator [A.11].

3. The normalised balance: Subtracting these two elements from both sides of the equation at once neutralises the two opposite biases, and leaves pristine highland crust with a stable history (Highlands) both at the poles and at the equator [A.11]. The paper's conclusion (D.1–D.2) rests firmly on this clean, balanced count, which fixes the system ratio at 1.20 ± 0.05 [A.7, A.11].

The five limits of knowledge and the analytical constraints of the count

To maintain a strict, restrained academic standard, the study explicitly defines five basic rules that limit and delimit what a crater count can prove [A.11]:

- **The historical accumulation rule (Accumulative Record):** The crater record seen on the surface today is a cumulative function [A.11]. Every crater imprinted since the shell began to freeze and collect its record on the Earth has been preserved in it, unless a later surface (a mare or a basin) physically erased it [A.11]. The modern count reflects the sum of the populations from all periods together, and not a single chronological stage; the polar density excess is analysed as a mechanical pattern added on top of this overall background [A.11].
- **The diameter-threshold limit and the risk of saturation (Saturation Limit):** The completeness of identification and the quality of the data change drastically as a function of crater size [A.11]. Below the 40 km threshold, catalogue completeness is affected by the type of ground and by the intrusion of secondary craters (post-impact debris clusters) [A.11]. Therefore, the $\geq 20\ \mathrm{km}$ class is used to derive the numerical values, while the $\geq 40\ \mathrm{km}$ class serves as a control tool to confirm the direction of the sign [A.11]. Below the firm 20 km threshold no comparison is made, since there secondary craters dominate and the surface is in a state of saturation (an equilibrium in which new craters erase old ones at the same rate, and the count stops growing) [A.11].
- **Neutralising optical biases and solar illumination:** At the poles, sunlight strikes at a low angle to the horizon and casts long, giant shadows that bring out every furrow and small crater, producing a visual illusion of a far more cratered area than the equatorial belt (a sunset effect) [A.11]. To block this optical bias, the study rejects the use of images or photographs, and relies exclusively on topographic data and digital elevation maps that measure the actual ground and not its shadows [A.11].
- **The background presence of the contact basin:** The anomaly of Paper 11 (the near-side minimum in the contact zone) is present as a basic background beneath the entire reported array [A.11]. The polar excess is measured relative to it and does not replace it; the six-axis model (A.9) presents and balances both patterns simultaneously [A.11].
- **The requirement of a uniform database:** All the density data are taken from the global geometric catalogue of Robbins (2019), which is characterised by uniform resolution over the entire Moon [A.11]. The study rejects the use of classical geological maps, compiled from historical sources of unequal detail (several times denser on the near side than on the far side), in order to prevent a comparative distortion [A.11, D.3]. A valid scientific comparison exists only when both halves of the body are counted by the same method and with the same measuring tool [A.11].

Closing the descriptive stage and moving to the explanatory tests

With the completion of sections A.1–A.11, the descriptive part of the paper is fully closed [A.11]. The five spatial stations established the following database:

- Identification of a sharp density excess at high latitudes [A.11].
- Full separation and isolation of the polar pattern from the longitudinal (east/west) asymmetry [A.11].

- Confirmation of the robustness of the bipolar effect even under the constraint of a firm geological filtering mask [A.11].
- Decoding of the internal structure of the caps and proof that the cores converge to statistical equality [A.11].
- Integration of the trends into a coherent geometric order that confirms the existence of an ancient triaxial body [A.11].
- Isolation and quantification of the central minimum of the SPA basin and its transfer to the analysis of Paper 8 [A.11].

From here on, the scientific argument changes its face and moves from the question of empirical description (what is found on the surface and where) to the question of the physical mechanism (what can produce such a pattern) [A.11]. Every alternative model or hypothesis examined in the following sections of the paper will have to stand up to six firm boundary constraints set by the observation, with no room for exception [A.11]:

> **1.** An exceptionally high crater density in both polar regions simultaneously
>
> **2.** Coherent reproduction of the trend at both isolated diameter thresholds (20 km and 40 km)
>
> **3.** Survival and full robustness of the end excess after applying the mare and basin mask
>
> **4.** An explanation of the differential variance in the response of north versus south to the mask (the collapse of the SPA basin in the south)
>
> **5.** Morphological convergence to similar density values in the residual inner cores of the caps
>
> **6.** Preservation of the six-axis order, with the sub-Earth point fixed as the sparsest cap

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PART B

Part B — What we tried before the answer

Part A defined the density anomaly quantitatively and spatially: lunar craters are significantly denser in the regions of both poles than in the equatorial belt, with the effect organised in a global pattern around a complete ring of the surface and not as a single singular point [B]. Part C will develop and present the geodynamic equations of the mechanism of rounding and mechanical contraction proposed as the scientific solution [B].

This section (Part B) is the stage of preliminary mechanistic testing, and presents the six alternative hypotheses and interpretations that were examined and rejected during the study [B]. Their mechanical rejection is what led the study to the rounding solution, since the combined analysis of the six failures proves that the physics of the system leaves room for only one kind of generating cause [B].

The matrix of four structural questions:

Each alternative hypothesis is examined consistently and strictly against four firm questions, whose values are derived directly from the observational data of Part A [B]:

1. The sign test (Sign Test): Does the proposed physical mechanism actually create a density excess at the poles relative to the equator, or does it act the other way and deplete them [B]? (A mechanism that depletes the poles is rejected immediately at the outset) [B].

2. The magnitude test (Magnitude Test): If the dynamic sign is correct, can the mechanism produce the measured empirical ratio values – 1.32 in the raw count, 1.20 ± 0.05 in the clean count, together with the significant rising trend as crater diameter increases (D.1–D.2) [B]?

3. The shape test (Morphology Test): Does the mechanism place and organise the density excess in the exact geographic space where it is observed – held at both poles to a similar degree, and arranged according to the triaxial order of the six axes (A.9) [B]?

4. The mask test (Mask Test): Can the density excess produced by the mechanism survive the application of the geological mask (removal of the maria, the large basins and the SPA basin from both sides of the equation), as it actually survives in the residual counts of Part A [B]?

The tests are arranged in a methodological order from the coarsest test (sign) to the finest test (mask), so that most hypotheses are decided and rejected at early stages of the analysis [B].

B.1 Preservation on ancient highlands (Highlands Preservation Hypothesis)

Definition of the interpretation:

This hypothesis holds that the density excess at the poles is nothing but a reflection of the degree of surface preservation and of varying geological age [B.1]. The polar regions are ancient, pristine highland crust (Highlands) that did not experience the late volcanic flooding of the dark basalt plains and maria, which erased and covered the early record at the low latitudes of the near side [B.1]. Therefore, the poles are dense because they are older and were not affected by endogenic alterations, and not because some dynamic mechanism actively concentrated or collected craters there [B.1]. This is the initial explanation accepted in the planetary community, since crater density serves as the classical indicator for determining the relative age of surfaces [B.1].

The morphological prediction of the hypothesis (what it predicts):

The model dictates that the density curve must follow linearly the chronological age and the degree of preservation of the area cell [B.1]. The density is required to be completely uniform and homogeneous across all the unflooded highland crustal units, and to show low values (pits) only where a young mare or basin extends [B.1]. The hypothesis rules out the existence of an anomalous centre or apex, since an ancient highland of the same age is uniform in space [B.1]. Had thermal preservation been the only factor, the pole would show values identical to any other highland area of the same age on the body, and the global density map would consist of flat plateaus and local pits, without a concentrated peak curve [B.1].

Analysis of the empirical findings from the counts (what the surface shows):

The effect of geological preservation and the thermal impact of the maria is entirely real, and it does shape the lunar surface on the largest scale (as reflected in the exposure regions of Paper

11, in which the far side hosts a raw density of 219 craters per 10^6 km², the marginal belt 164, and the sub-Earth contact disc only 99) [B.1]. However, three firm observations from Part A rule out the interpretation and lead to its rejection [B.1]:

1. Robustness of the excess under the decisive mask test: The filtering mask of section A.3 omits from the calculation precisely the elements on which the hypothesis rests – the mare polygons and the giant basins [B.3, B.1]. This procedure normalises the denominator and ensures a clean comparison of highland versus highland on both sides of the equation [B.1]. Had preservation differences been the sole cause of the phenomenon, the residual density ratio would have fallen exactly to the uniform-distribution value (1.00) once the replaced crust was removed [B.1]. In practice, the raw ratio falls from 1.32 to 1.11 (removal of maria), and climbs back to a firm value of 1.20 ± 0.05 once the SPA basin is also neutralised [B.1]. The ratio does not drop to 1, and it continues to show a rising trend as a function of crater diameter up to a value of 1.53 in the large classes [B.1]. Any density difference that survives this like-for-like procedure cannot be attributed to the difference between flooded ground and pristine ground [B.1].

2. The bipolar peak stands above the oldest highland in the system: The far-side crown (around the coordinates of the anti-Earth point) represents in the dynamic model the oldest and safest crustal unit, which the mare surfaces hardly touched geographically [B.1]. After applying the mask, this sector carries a density of 221 craters per 10^6 km² [B.1]. By contrast, the two residual polar caps show considerably higher values: 233 in the north and 246 in the south [B.9, B.1]. The poles stand morphologically above the oldest highland, not level with it [B.1]. Had the chronological age of the crust been the deciding factor, the far-side crown should have led the global order; the findings on the surface rule this out [B.1].

3. The existence of a radial concentration (a centre) in the pristine northern core: Within the space of the north pole, where no mare polygon or Neumann basin penetrates within 20° of the apex, the density is not homogeneous and flat but shows a sharp concentration curve [B.1]. The density measure climbs continuously from 189 in the wide cap (60°) to a peak value of 300 craters per million km² in the inner 15° cap [B.6, B.1]. A uniform, static highland crust cannot produce a radial convergence of density toward a defined geographic point [B.1]. The northern core is the cleanest observation site for refuting the hypothesis, since it contains no foreign geological element requiring filtering by the mask [B.1].

Verdict:

The hypothesis of preservation on ancient highlands passes only the sign test, and fails completely in the mask test and the shape test [B.1]. The wide spatial variance on a global scale (such as the near-side versus far-side ordering) is explained through the geometry of the contact zones of the tip of the pear (Paper 3) and the distribution of the hardest ballistic landing vectors concentrated on the far side (Paper 6), and not as a function of static preservation [B.1].

The study keeps the preservation mechanism solely as the physical reason why the mare areas and young impact basins show low density values (resetting of the record), and not as a source or explanation of the polar density excess [B.1]. The presence of ancient highlands on the near side south of the crater Tycho, which show an exceptional density of 302 craters within 15°, is a further

mechanical reminder that the density of an ancient surface is not a function of chronological age alone [B.1]. What the preservation hypothesis leaves without a physical answer is precisely the core of the paper's study: a significant density excess preserved at the level of a highland-versus-highland comparison, and holding a geometric centre anchored on the body's rotation axis [B.1].

B.2 Rotational flattening during attachment (Rotational Flattening Hypothesis)

Definition of the interpretation:

This hypothesis is based on the mechanical principle that a viscous/soft body rotating rapidly about its axis undergoes a flattening deformation – it widens along the equatorial belt owing to centrifugal stresses, and shortens along the polar axis, like pizza dough that stretches and flattens when spun in the air [B.2]. Since the young Moon in the attached stage rotated in rapid synchronisation with the Earth (a period of about 5.5 hours), it underwent a severe global flattening (C.1) [B.2]. The initial idea holds that these compressive stresses physically pressed the polar regions inward, and thus mechanically brought the craters they carried closer together and raised their density [B.2].

The morphological prediction (what it predicts):

The physical geometry of a flattening deformation at constant volume (a state in which material is neither added to nor removed from the system, but only changes its configuration) shows a behaviour entirely opposite to the crowding prediction [B.2]:

- **In the polar space:** the upper and lower surfaces are squashed and force the material to spread sideways, which causes a two-directional stretching of the surface within the plane of the pole [B.2].
- **In the equatorial space:** the surface is stretched along the circumference of the equator, but shortens along the meridians [B.2].

As a consequence, the number of craters per unit area (ρ_{crater}) decreases and becomes depleted in the polar region, and the differential area-spreading equation is defined precisely by the axis ratio of the spheroid (a/c, where a is the equatorial radius and c is the polar radius) [B.2]:

$$\rho_{\mathrm{polar}} = \frac{\rho_{\mathrm{equatorial}}}{(\mathbf{a}/\mathbf{c})^{\mathbf{2}}}$$

This equation dictates that the more flattened a body is, the sparser and emptier of craters per unit area its poles become (the exact inverse of the rounding equation developed in C.3) [B.2]. For the axis-ratio values of the Maclaurin spheroid at peak flattening ($a/c = 1.69$, density 1,300 $\mathrm{kg/m^3}$, rotation of 5.5 hours) and in the late rounding stages ($a/c = 1.20$, density 2,600 $\mathrm{kg/m^3}$, period of 5.77 hours), the flattening mechanism would leave the poles at a theoretical disadvantage of only 0.59 and 0.83 relative to the equator (a structural deficit of 41% and 17% respectively) [B.2].

What the empirical counts say:

The physical data on the surface show a sharp density excess at the poles (a ratio of 1.32 in the raw count and 1.20 ± 0.05 in the clean count) [B.2]. Therefore, the mechanical sign of the hypothesis is fundamentally wrong; rotational flattening dilutes the craters at the apex, whereas the observations on the surface show them at maximum density [B.2].

Verdict:

The rotational-flattening hypothesis fails completely in the sign test [B.2]. If an elastic ball is pressed from above and below, the surface of the upper end does not contract – it spreads and widens, and points imprinted on it move apart [B.2].

However, the hypothesis understands the general geometry of the body correctly when it is read in the opposite direction: a crustal surface that was forcibly stretched during the flattening stage, and later released from the stress constraint, will contract sharply in the polar regions [B.2]. This reversed understanding is the mechanical heart of section C.3 [B.2]. This finding caused a fundamental methodological change during the study: instead of searching for a dynamic mechanism that produces flattening, the search began for the geodynamic process that undoes and releases the flattening (the process of rounding back to a sphere) [B.2].

B.3 The polar axis shortened by the tide — the Roche effect (Roche Tidal Stretching Hypothesis)

Definition of the interpretation:

During the attached stage, the young Moon was subject to the Roche effect – extreme tidal stresses that stretched its volume along the Earth line (the radial axis a) owing to the dynamic difference between the Earth's gravity, which pulled the near side inward, and the centrifugal pull of corotation, which drew the mass outward, while the interior of the body was held on the tip of the pear structure [B.3]. Since the volume of the body remained almost constant, the elongation of the radial axis came at the expense of a reduction of the two perpendicular axes [B.3]. Within the rotating system, the centrifugal term acted to stabilise and preserve the axis of orbital motion (b), and therefore the main burden of shortening and mechanical collapse fell on the polar axis (c) [B.3]. The poles were pressed and shortened toward each other, and the alternative idea holds that the craters on them were drawn and crowded together with them [B.3].

The morphological prediction (what it predicts):

The geometry of a free fluid body with a density of 2,600 $\mathrm{kg/m^3}$ under these tidal stresses is computed precisely as a triaxial axis ratio of $1 : 0.637 : 0.601$ (for axes $a : b : c$) [B.3]. This structure produces an ellipsoidal body very long toward the Earth and shortest at the poles (an egg lying along the tidal axis) [B.3]. When this tidal stretching deformation is applied to a crust carrying uniformly distributed craters, the density measure in each area cell changes in inverse proportion to the contraction factors (the exact inverse of the C.3 values) [B.3]:

Spatial sector examined	Change vector under Roche stretching	Change vector under rounding back (C.3)
The polar regions (the short axis c)	0.83× (dilution and moving apart)	1.21× (mechanical crowding)
The leading and trailing points (axis b)	0.88× (moderate dilution)	1.14× (moderate crowding)
The sub-Earth point (the long axis a)	1.38× (concentration and crowding)	0.73× (widening and dilution)

Reading the matrix across proves that a polar axis mechanically shortened under tidal stretching stresses spreads and widens the surface above it, exactly as in the rotational-flattening mechanism [B.3]. The pole is the physically most stretched sector of the shape, and therefore the craters there move apart and become diluted (0.83) [B.3]. By contrast, the ends of the Earth line (the sub-Earth point) are the regions where the surface converges and crowds to the system maximum (1.38) [B.3].

What the empirical counts say:

The actual observations on the surface present an entirely opposite picture: the poles are the densest regions in the system, while the sub-Earth point is the sparsest and emptiest of the six main caps (a raw value of 94, and a masked value of only 148) [A.9, B.3]. The tidal stretching mechanism predicts precisely the opposite in these two sectors [B.3].

Verdict:

The Roche-stretching hypothesis fails the sign test as a crowding-generating mechanism [B.3].

However, the physical effect itself is real and its magnitude is firmly preserved within the model equations: it defines the triaxial starting shape of section C.2, and the geodynamic rounding back from it toward a perfect sphere is the scientific answer of the study [B.3]. The Roche effect is responsible for setting the peak deformation and stretching of the attached body; the subsequent rounding (the release of the stretching, the return to a sphere) is what drives and physically brings the craters closer together [B.3]. Reading the right-hand column of the table (the rounding vector) reproduces in full agreement the order of the counts: maximal poles, intermediate sides, and the sub-Earth point at the bottom as the minimum [B.3].

B.4 Stronger gravity at the pole (Polar Gravity Excess Hypothesis)

Definition of the interpretation:

In a planetary body characterised by a flattened shape, the polar apex lies geometrically closer to the body's centre of mass than the equatorial belt does [B.4]. In addition, at the geographic pole (which sits directly on the rotation axis) the linear velocity is zero, and therefore the outward centrifugal component does not exist [B.4]. The combination of these two factors produces at the poles an effective gravity value (the self-attraction minus the centrifugal contribution) higher than that at the equator [B.4]. The alternative interpretation proposes that this enhanced gravity field at

the poles acted as a mechanical pump that drew loose elastic surface material along the crust toward the polar regions, and thus crowded the record around them [B.4].

The morphological prediction (what it predicts):

Physical law dictates that the gravity vector acts downward (radially toward the centre) and not sideways (tangentially along the surface) [B.4]. For crustal material to undergo active lateral motion along the surface toward the pole, the pole would have to function physically as the lowest point in terms of the body's effective potential (the floor of a potential valley, into which material slides and flows like water down a slope) [B.4].

In the dynamic system of the attached body rotating rapidly with the Earth, the physical situation is entirely the opposite [B.4]. In the rotating frame, the effective potential at the equatorial belt is lower than that at the pole by a firm factor of $\frac{1}{2}\Omega^2 R^2$ (where Ω is the rotation rate and R is the radius), a value representing the excess kinetic energy of motion that the equatorial belt has and that the polar axis lacks entirely [B.4].

In the late stages of rounding (density 2,600 $\mathrm{kg/m^3}$, period of 5.77 hours, mean radius $R \approx 1,890\ \mathrm{km}$), under a surface gravity value of $g \approx 1.37\ \mathrm{m/s^2}$ (C.7), the theoretical height equation is defined by [B.4]:

$$\mathbf{\Delta H = \frac{\frac{1}{2}\Omega^2 R^2}{g} \approx 120\ km}$$

This equation proves that on a flattened body that has not yet completed its collapse to its hydrostatic equilibrium shape, the polar regions stand as giant topographic hills about 120 km high "above" the equatorial belt in the energetic sense [B.4]. They are the points of highest potential (a hilltop), and therefore any loose material or elastic crust will experience an opposite push vector and slide away from them outward toward the equatorial potential valley [B.4]. This is the mechanism that produces flattening (B.2), not polar crowding [B.4].

What the empirical counts say:

The excess is located at the poles [B.4]. While the body was rotating, the combination of gravity and centrifugal acceleration acted to remove and dilute material from the poles, not to concentrate it there [B.4].

Verdict:

The enhanced-polar-gravity hypothesis fails completely as a lateral-displacement mechanism as long as the body is in the attached, rotating stage [B.4].

The effect is placed in its correct physical position only at the moment of the dynamic transition: it is the Moon's self-gravity that held it together as a whole body and prevented its disruption beyond the rigid Roche limit ($\approx 13,000\ \mathrm{km}$ at a density of 1,300) [B.4]. The moment the Moon's centre of mass crossed the corotation line and the mechanical compression on the base dropped to zero, the body was released from the constraint of excess synchronous rotation [B.4]. In this state, the flattening that the rotation had previously held becomes an anomalous load that the body cannot

bear [B.4]. Only at this moment does the pole of the flattened body collapse and turn from an energetic hilltop into the floor of a deep potential valley [B.4]. Now self-gravity acts as the main engine driving the flow and contraction of the surface inward toward the poles – this is the mechanism of rounding back (C.4) [B.4]. The geometric difference of 120 km represents the exact scale of the energy that the rotation held up, and its mechanical release during rounding is what crowds the craters [B.4].

B.5 A drain at the pole (Polar Drain/Vortex Hypothesis)

Definition of the interpretation:

This hypothesis is based on the intuitive mechanical analogy of "suction inward": the existence of a draining mechanism, crustal subsidence, or an internal dynamic vortex beneath each geographic pole [B.5]. According to this model, the magmatic or geodynamic system acts like water on the floor of a sink flowing and gathering toward the drain, and thus it actively draws, concentrates and collects the surface and the craters on it toward a single end point [B.5].

The morphological prediction (what it predicts):

The dynamics of a drain dictate a concentration directed exclusively toward a singular point (The Vertex Point) [B.5]. Therefore, the density map is required to show a steep and unbounded rising trend (a statistical singularity) as one approaches the pole point itself, with the geophysical topography obliged to show a depression or deep pit at the centre point [B.5]. The model requires the action of a lateral mechanical force vector directed straight at the coordinates of the pole [B.5].

What the empirical counts say:

The data measured on the surface rule out the structure of a point singularity [B.5]:

1. Arrest and stabilisation of the pristine northern core: Within the clean space of the north pole, the crater density ($\geq 20\ \mathrm{km}$) does indeed climb radially: from 189 in the 60° cap, to 227 (30°), and to 270 (20°) [A.6, B.5]. However, on entering the extreme core, the density curve stops completely and levels off: a value of 300 in the 15° cap versus a value of 298 in the 10° cap [A.6, B.5]. The data reveal a global concentration toward a wide polar region (Highlands Plateau) and not a point convergence into a drain [B.5].

2. Sensitivity of the residual southern sample: In the south, after applying the mask, the inner caps show values of 305 (at 15°) and 348 (at 10°) [A.7, B.5]. However, analysis of the system shows that within 20° of the south polar apex, the mask consists of an overlapping union with the SPA basin alone, and the 10° cap rests on a small sample of only 36 craters [B.5]. In such a small sample, a random fluctuation of a handful of craters shifts the numerical value by wide margins, and therefore the rise in the south carries the local geometry of the impact basin and the error intervals of the count, and is not evidence of a drain [B.5]. The scientific weight of the test rests on the clean northern sequence [B.5].

3. Absence of topographic evidence: The actual planetary topographic data show no pair of gravitational or structural depressions corresponding to the geographic poles [B.5].

Verdict:

The drain hypothesis passes the sign test (positive), but fails completely in the shape test [B.5].

Within a rotating planetary body, every physical force vector must preserve axial symmetry about the rotation axis, and such symmetry produces the geometric structure of a ring (Ring) and not of a single point; there is no physical force able to converge on the pole point itself [B.5]. The true density excess prevails along a complete ring – the ring standing at a distance of 90° from the sub-Earth point, passing continuously through both poles and both sides (A.9, C.3) – and the poles lead the top of the order only because the rotation axis sets their geometric centre of convergence [B.5]. The drain interpretation holds the correct sign (an excess) but a spatial pattern that distorts: it requires a point peak, whereas the observations show a constant plateau over a wide area cell and a global crowding band encircling the whole body [B.5].

B.6 Uniform change (Isotropic Volume/Scale Change Hypothesis)

Definition of the interpretation:

This hypothesis proposes to explain the density excess through global isotropic (uniform) processes, such as: a uniform volume expansion, a uniform thermal contraction of the whole body during its cooling stages, or the action of a strong self-gravity that draws all crustal material equally toward the centre of mass – that is, a mechanical change in the dimension and size of the body (R) without a change in its geometric shape (structure) [B.6]. Each of these events is a real, documented physical process in the early history of a young, hot planetary body, and each of them moves and changes the position of the entire surface [B.6].

The morphological prediction (what it predicts):

A uniform change in the dimension of the body changes every element and every crustal area unit by exactly the same factor and ratio [B.6]. The geographic spacing between the craters grows or shrinks homogeneously at every coordinate on the body, and therefore the density measure changes by a constant, uniform factor both at the poles and at the equatorial belt [B.6]:

$$\rho'_{\mathrm{polar}} = \rho'_{\mathrm{equatorial}}$$

For a uniformly cratered surface, a global change in size multiplies the numerator (the number of craters) and the denominator (the surface area) by exactly the same mathematical factor, and therefore the statistical ratio between them cannot change [B.6]. The mechanism works like the uniform inflation of a balloon: the operation makes all the points imprinted on it sparser, but it preserves equality and cannot produce a region denser or sparser relative to another [B.6].

What the empirical counts say:

The observations on the surface establish a sharp morphological difference: a ratio of 1.32 in the raw count, 1.20 ± 0.05 in the clean count, and a consistent climbing trend as a function of crater diameter up to a value of 1.53 in the large classes [B.1, B.6].

Verdict:

The uniform-change hypothesis fails absolutely in both the sign test and the magnitude test, since it cannot produce any density excess or spatial variance at all [B.6].

However, this is the methodological interpretation that charted the analytical path of the study [B.6]. Once a process of uniform change in size had been physically ruled out, physics dictated a firm conclusion: a change in size cannot produce a polar excess; only a change in the shape of the body (Anisotropic Distortion) can produce such an anomaly [B.6]. This change of shape must be non-uniform in space in a defined way: it must mechanically contract the surface in the polar regions at the maximum rate, and it must preserve full axial symmetry about the rotation axis [B.6].

Since hypotheses B.2 and B.3 represent real changes of shape but with the wrong mechanical sign (dilution of the poles), the scientific solution of the puzzle requires their exact geodynamic reversal [B.6]. This stage narrowed the scientific question: the search was directed exclusively at finding the specific change of shape that contracts the polar surface more than any other region, and only one geodynamic process in the history of the model satisfies this constraint [B.6].

B.7 The answer, and why it is the one that remains (The Flattening Release / Spherical Rebound)

A single geodynamic change of shape in the system gives a precise and complete answer to all the boundary constraints set by the observations, in which the previous interpretations failed: the mechanical transition of a highly flattened triaxial body that is released from its constraints and rounds back to the shape of a perfect sphere with a self-driven decrease in volume [B.7].

Closing the constraints and matching the physical setting:

1. The mechanical sign (positive): The mechanism is the exact geometric and mechanical reversal of hypotheses B.2 and B.3 [B.7]. The release of the stretching and the subsequent contraction act with the correct sign and crowd the crust [B.7].

2. The physical engine: driven by the self-gravity forces of section B.4 in their exact physical place – at the moment when the Moon's centre of mass crosses the corotation line, and the body's net compressive force on the base of the pear drops to zero [B.4, B.7]. The body is released from the excess centrifugal constraint while still attached and rotating in synchronisation with the Earth [B.7].

3. The spatial geometry: it produces precisely the structure of the symmetric ring around the rotation axis (B.5), with the polar regions leading the peak of mechanical crowding and contraction because the rotation axis passes through them and all the meridians retreat toward them (the poles are the only points that the rotation never moved) [B.5, B.7].

4. The magnitude test and chronological consistency: The physical dimensions are derived directly from the shape that the craters themselves record around the six axes (A.9) and from tests of agreement with the hydrostatic equilibrium shape of a Maclaurin spheroid (C.1) [B.7]. The rounding model does not predict a static value, but a dynamic chronological ordering that matches fully the empirical findings of the clean count [B.7]:

- Ancient giant craters ($60-100\ \mathrm{km}$): formed at peak flattening on the most deformed body ($1.8:1.4:1$, density $1,700\ \mathrm{kg/m^3}$), underwent the full rounding sequence and therefore show the peak crowding: predicted ~1.5 versus 1.53 ± 0.19 empirical [A.8.5, B.7].
- Intermediate/late craters ($20-30\ \mathrm{km}$): formed in late stages of the inward collapse on a body that had already begun to round (≈ 1.20, density $2,600\ \mathrm{kg/m^3}$), underwent only the end of the surface contraction and therefore show a moderate residual effect: predicted ~1.2 versus 1.09 ± 0.07 empirical [A.8.5, B.7].
- The whole population ($\geq 20\ \mathrm{km}$): the six-axis model reconstructs a combined shape of $1.37:1.15:1$, which independently predicts a ratio of 1.21, matching precisely the result of the residual clean count: 1.20 ± 0.05 [A.9, B.7].

5. The mask test: The effect is robust and fully survives the cleaning out of the maria, the basins and the SPA system, since it is embedded in the mechanics of the solid crust itself [B.7].

6. Defined timing and formation window: The process is placed chronologically in the early attached stage (before the final mechanical detachment, at which the Moon was released in a rigid round state with an axis ratio of ≤ 1.04) [B.7]. The interior of the body was soft and viscous enough to allow flow and rounding, while the outer shell was rigid enough to carry the craters as intact, isolated elements (the contraction closed the elastic spaces between them and built up the pressure of C.4) [B.7].

7. The state of the shell beneath the poles: if the polar surface contracted more than the equatorial surface, the material in it was pushed together and the crust there should have thickened [C.7]. Yet the polar crust is today thinner than the equatorial crust [D.6]. The explanation is that the polar shell was thinner still before the rounding: during the flattening the light material of the outer shell flowed from the poles to the equator (external rotational differentiation, C.8). The same answer therefore explains both the crater density at the poles and the crust beneath them [B.7, C.8].

The meaning of the rise with crater diameter:

The climbing trend of the density ratio as crater diameter increases is the clearest and most direct empirical evidence for the existence of a rounding process taking place along the timeline [B.7]. A crustal crater can preserve and record only the share of surface contraction that took place after its formation [B.7]. The largest (oldest) craters went through the whole transformation, while the small (later) craters received only the geometric closing chord [B.7]. No alternative hypothesis (from B.1 to B.6) can produce a structural dependence between the density excess and crater diameter; the rounding model requires it physically [B.7].

Summary matrix of the mechanistic test

The following is a quantitative and strict summary of the six rejected interpretations against the leading rounding hypothesis, defining the residual contribution of each rejected mechanism to the final scientific formulation [B.7]:

Section number	Hypothesis examined in the system	Sign test	Magnitude test	Shape test	Mask test	Mechanical component preserved within the final description
B.1	Preservation on ancient highlands	Yes	—	No	No	Defines the bias of the maria and low-lying basins (resetting of the record) [B.7].
B.2	Rotational flattening	No	—	—	—	Defines the initial stretching stress that the rounding releases [B.7].
B.3	Roche effect (tidal stretching)	No	—	—	—	Defines the triaxial starting geometry of section C.2 [B.7].
B.4	Enhanced polar gravity	No	—	—	—	Acts as the main engine driving the rounding collapse (C.4) [B.7].
B.5	Drain / vortex	Yes	—	No	—	Confirms the axial geometry of the global ring [B.7].
B.6	Uniform change (dimension/size)	No	No	—	—	Proves the exclusive physical need for a non-uniform change of shape [B.7].
C	The model of rounding and volume decrease	Yes (clean)	1.20 / 1.5–1.2	Polar ring	Yes	The confirmed scientific answer of the study [B.7]

Read the matrix from top to bottom: each rejected interpretation left behind a firm physical component, and the bottom row unites them into a single coherent structure [B.7]. The stretching of B.2 and B.3 provides the deformation measures to be undone, the self-gravity of B.4 provides the engine vector, the axial symmetry of B.5 defines the geometry, B.6 sets the rule of a necessary change of shape, and B.1 isolates the effect of the young surfaces [B.7].

We now have a single physical cause that successfully passes all four empirical tests, and six hypotheses that were rigorously rejected [B.7].

PART C

Part C — The geodynamic mechanism and its numbers

Part B of the study ruled out all six alternative explanations [B.7]. The analytical rejection left one possible mechanical cause: the geometric transition of a highly flattened triaxial body that is released from its stress constraints and rounds back to the shape of a perfect sphere, with a self-driven decrease in volume and a rise in density [B.7, C]. This part turns that answer into a

quantitative physical mechanism: it defines the shape dimensions of the young Moon, the stress vectors that changed it, the spatial effect on the crust, and the energy that drove the process [C].

The sequence of deforming forces and the morphological evolution:

The geometry of the young Moon was set under a dynamic equilibrium equation between three main force vectors [C]:

1. The pull of self-gravity: tended to press all parts of the body uniformly inward toward the centre, and on its own would have produced a perfect sphere [C].

2. The centrifugal acceleration of the fast rotation: in the stage of synchronous attachment to the Earth (a period of about 5.5 hours), the rotation pushed the equatorial belt outward and lowered the polar regions [C].

3. The tidal stretching stresses (the Roche effect): stretched the body extremely along the radial axis between the Earth and the Moon [B.3, C]. The Earth's gravity pulled the near side inward, while the centrifugal force of corotation drew the mass outward, with the base of the body physically held on the tip of the pear structure [C]. This stretching shortened the two perpendicular axes, and forced the polar axis to absorb the maximum mechanical shortening [C].

The chronology of the shape sequence unfolds along the mechanical stages of separation [C]:

- **The initial gathering stage:** a light sialic mass with a density of 1,300 $\mathrm{kg/m^3}$ consolidated at the tip of the Earth's pear structure and began to move outward on a base 6,000 km wide, with initial flattening under the Earth's constraint [C].
- **The stage of body stabilisation and maximum flattening:** as the mass moved away, the base of the pear narrowed to a width of 3,000 km [C]. When the pear reached corotation, the two sides of the body were pulled in opposite directions and the body reached its peak geometric deformation near the 3,000 km base [C]. This peak shape is analysed in sections C.1–C.2 [C].
- **The stage of attached rounding and volume decrease:** the moment the Moon's centre of mass crossed the corotation line, the net pull that pressed it onto the base of the pear dropped to zero [C, C.6]. Corotation allowed the body to detach from below and begin to round [C]. The release of the tidal and rotational stretching brought a sharp self-driven decrease in volume; the body became smaller, thicker and denser, while its outer shell progressively stiffened [C].
- **The final release stage (detachment):** at the moment of final mechanical detachment, the body was already completely round and its shell rigid [B.7, C]. After release, the slowing of the locked rotation removed only the small elastic bulge of the rigid body, and not the early flattening, which had disappeared earlier [C, C.5]. The rounding process is what the body's own gravity brought about the moment the stresses that held the deformation relaxed [C].

C.1 The initial shape at peak flattening: rotation and density

The question of this section quantifies mathematically how flattened and deformed the Moon was at the peak of the flattening stage, and how its shape changed through the stages of inward

collapse [C.1]. To compute the rotational component only two physical parameters are needed: the angular velocity of rotation (Ω) and the mean density of the material (ρ) [C.1].

For a self-gravitating homogeneous fluid (a body of uniform density held together by its gravity alone, with no initial mechanical strength), the exact hydrostatic equilibrium shape is defined as a Maclaurin spheroid (Maclaurin Spheroid) [C.1]. The degree to which the spheroid deviates from a perfect sphere is measured by its eccentricity, $e = \sqrt{1 - c^2/a^2}$ (where a is the equatorial radius and c is the polar radius; $e = 0$ gives a perfect sphere, and $e \to 1$ gives a flattened disc) [C.1]. The body remains stable at rest when the angular velocity Ω (expressed as $2\pi/T$, where T is the rotation period) satisfies the structural equation [C.1]:

$$\frac{\mathbf{\Omega^2}}{\pi \mathbf{G} \rho} = \frac{\mathbf{2}\sqrt{\mathbf{1 - e^2}}(\mathbf{3 - 2e^2})}{\mathbf{e^3}} \arcsin(\mathbf{e}) - \frac{\mathbf{6}(\mathbf{1 - e^2})}{\mathbf{e^2}}$$

- **The mechanics of the equation:** The left-hand side compares the outward centrifugal force (Ω^2) with the inward self-gravity of a uniform body ($G\rho$, where G is the universal gravitational constant) [C.1]. The right-hand side is a transcendental function of the geometric shape alone [C.1]. The right-hand side climbs from zero (for a sphere) to an absolute maximum of 0.4493, reached at an eccentricity of $e = 0.930$ (giving a sharp axis ratio of $a/c = 2.72$) [C.1]. Beyond this mechanical limit, no stable Maclaurin shape physically exists [C.1].
- **Parametric sensitivity:** The faster the rotation rate relative to the density, the more flattened the equilibrium shape [C.1]. A body of low density (light) is more susceptible to flattening than a dense body at the same rotation rate, because its weak gravity field offers less resistance to the centrifugal stress [C.1].

Solution matrix of the dynamic axis ratio (a/c):

Solving the Maclaurin equation for the parameter values of the detachment stages yields the following matrix of ratios of equatorial radius to polar radius [C.1]:

Spin period (T)	$\rho = 1,000$	$\rho = 1,300$	$\rho = 1,600$	$\rho = 1,700$	$\rho = 2,000$	$\rho = 2,600$	$\rho = 3,344$
5.50 hours (structural peak flattening)	No equilibrium	1.69	1.46	1.41	1.32	1.22	1.16
5.77 hours (late in crustal rounding)	2.22	1.56	1.39	1.35	1.28	1.20	1.14

- **Horizontal reading (the density vector):** The geometric shape undergoes continuous rounding (the value of a/c falling toward 1.00) as the density of the material rises, since a fixed rotation rate cannot hold a flattened deformation against a self-gravity field that strengthens and pulls inward [C.1].
- **Vertical reading (the rotation vector):** The shape rounds slightly as the rotation slows from 5.5 to 5.77 hours [C.1].

- **Calibrating the ends of crustal rounding (the highlighted values):** The matrix defines the two ends of the transformation of the pita part of the body: the stretched, inflated body at peak flattening (1.41 at a density of about 1,700 $\mathrm{kg/m^3}$), and the dense body in the late rounding stages (1.20 at a density of 2,600 $\mathrm{kg/m^3}$) [C.1].
- The pear reference value ($1.300\ \mathrm{kg/m^3}$): the value of 1.69 reflects the shape that the light mass gathered at the tip of the pear would have taken as a free fluid from rotation alone; this value serves as physical context only, and not as a measured peak value of the independent body [C.1].

The lower stability limit for the density of the material:

At a rotation period of 5.5 hours and a density of 1,000 $\mathrm{kg/m^3}$, the equation yields no equilibrium solution, because the centrifugal stress exceeds the maximum allowed point of the right-hand side [C.1]. In this hypothetical state, the body would break apart and shed mass from its equatorial margins [C.1]. This point defines a firm lower limit on the density of material required to keep the body intact at any rotation rate:

$$\rho_{\min} = \frac{\mathbf{\Omega^2}}{\mathbf{0.4493\pi G}}$$

For a rotation period of 5.5 hours, the minimum density limit is set mathematically at 1,070 $\mathrm{kg/m^3}$; a planetary body lighter than this value cannot hold itself together as a fluid at such a rotation rate [C.1].

Location of the peak deformation and the triaxial geometry

- **Timing of maximum flattening:** In the initial stages, the mass acted as an integral part of the tip of the pear (density 1,300, distance $\approx 8,991\ \mathrm{km}$), and its shape was dictated directly by the Earth [C.1]. Independent morphological flattening developed once the mass became an independent body able to stabilise itself, a condition met on crossing the rigid Roche limit [C.1]. Beyond this limit, self-gravity and material strength overcome the disruptive tidal forces of the Earth [C.1]. For light material ($1,300\ \mathrm{kg/m^3}$), the Roche limit lies farther out, at a distance of 12,994 km (the limit varies as a function of $\rho^{-1/3}$; for the release density of 2,600 $\mathrm{kg/m^3}$ the limit is closer, at 10,314 km) [C.1]. The body moved outward beyond 13,000 km while its base narrowed from 6,000 to 3,000 km, and reached maximum geometric deformation and flattening as it approached the 3,000 km base, very close to crossing the corotation line ($15,818\ \mathrm{km}$ at a period of 5.5 hours) [C.1].
- **The triaxial structure (an egg that is also a pita):** The shape at peak deformation was not a symmetric biaxial spheroid [C.1]. While the rotation flattened the body about its axis, the two-body tidal stresses of the Earth actively stretched it along the Earth-line axis [C.1]. The combination of stresses imposed a triaxial shape with three distinct axes: a long axis directed toward the Earth (a), an intermediate axis directed along the orbital motion vector (b), and a short, held axis connecting the poles (c) [C.1].

The empirical crater-excess record reconstructs and confirms these dimensions precisely (sections A.9, C.3): the original peak shape stood at a ratio of roughly $1.8 : 1.4 : 1$, within a morphometric scatter range of between $1.64 : 1.45 : 1$ and $1.95 : 1.33 : 1$ [C.1]. The whole population ($\geq 20\ \mathrm{km}$) converges to a combined average of $1.37 : 1.15 : 1$ [C.1]. This is a moderate and physically clear deformation: the long axis is 1.6 to 1.95 times the polar axis, and the intermediate axis is 1.33 to 1.45 times it [C.1].

Confirmation of the peak density measures ($\approx 1,700\ \mathrm{kg/m^3}$):

Determining the density of the body at peak flattening is based on cross-checking two independent dynamic tests, which yield full agreement [C.1]:

1. Confirmation of the pita part by the Maclaurin relation: The pita part of the shape ($b/c \approx 1.4$) is set exclusively by the rotational stresses [C.1]. To a first approximation, the difference between the intermediate and short axes of a tidally stretched body depends on the spin rate, and therefore it can be read directly from the Maclaurin equation [C.1]. At a rotation period of 5.5 hours, the Maclaurin relation gives an exact value of $b/c = 1.41$ for a body with a density of 1,700 $\mathrm{kg/m^3}$ (the highlighted value in the table) [C.1]. The empirical scatter range ($1.33-1.45$) bounds the density within limits of 1,615 to 1,960 $\mathrm{kg/m^3}$, with the core concentrated at 1,600–1,750 [C.1].

2. The geometric surface-closure limit: The transition from a stretched body to a perfect sphere involves a release of stretching and a global decrease in volume [C.1]. The geometric equations (C.3) prove that had the density of the body at the peak been higher than 1,800 $\mathrm{kg/m^3}$, the available stretching volume would have been too small; in that case, the ends of the tidal axis would have been forced to expand and grow during rounding instead of contracting and closing [C.1].

The two tests fix the value of the body at the peak as a stretched, inflated body with a density of about 1,700 $\mathrm{kg/m^3}$, yielding an equivalent-sphere radius (the radius of a sphere of the same volume) of about 2,180 km [C.1]. This value stands in contrast to a radius of 1,890 km at release, and a modern radius of 1,737 km [C.1]. The original sialic mass on the tip of the pear ($1,300\ \mathrm{kg/m^3}$, equivalent radius about 2,380 km) is the early starting base, before the dynamic flattening stage came into play [C.1].

The stages of mechanical rounding

When the centre of mass crossed the corotation line, the body began the process of rounding back to a sphere, which took place in two separate mechanical stages [C.1]:

1. Release of the egg part (the tidal axis a): released first and rapidly, because on the corotation line the net pull vector that pressed the body onto the base of the pear fell to zero [C.1, C.6]. This part of the stretching along the Earth line collapsed inward [C.1].

2. Release of the pita part (rotational flattening b/c): proceeded in parallel with the thickening and compaction of the material under an almost constant rotation period (C.4) [C.1]. This stage is defined along the rise of the density measures ($2,000 \to 2,300 \to 2,600\ \mathrm{kg/m^3}$) and a slight lengthening of the Earth's day (reaching 5.77 hours at the moment of release) [C.1]. Under these dynamics, the axis ratio fell and settled at a value of about 1.20 in the late rounding stages [C.1].

A fluid body completing a spin once every few hours must physically show a flattening deformation and cannot exist as a perfect sphere in a viscous state [C.1]. The final, rigid transition from a ratio of about 1.20 to a perfect spherical body at the moment of release is stabilised and reinforced by the stiffening of the outer shell [C.1]. This is a structural constraint of the model: an upper crust with a mechanical strength and rigidity of $\mu \geq 10\ \mathrm{GPa}$ can absorb the residual rotational stresses and stabilise a round body with an axis ratio of at most 1.04 [B.7, C.1]. At the moment of release the Moon is round, and its shell rigid [C.1]. The minor residual flattening that remained after release is only a small elastic bulge, which decays and disappears entirely as the body recedes and its orbital rotation slows (C.5) [C.1].

The Maclaurin equation is a basic physical law, and the solutions of the matrix depend on nothing other than the values of the angular velocity (Ω) and the density of the material (ρ) [C.1].

C.2 The tidal component: the free shape and the Roche effect

Whereas section C.1 computed the effect of the rotational stresses alone, this section brings in the second force vector that shaped the body morphologically: the Earth's tidal forces (the Roche effect) and the complex balance of potentials in the system [C.2]. The analysis serves as a geometric tool to illustrate the contribution of the radial stretching, and is not the basis for the statistical density test [C.2].

The two-body dynamics in the rotating system:

A planetary body in close proximity to the Earth and in synchronous rotational locking (the orbital period equal to the spin period Ω) is stretched extremely along the radial axis between it and the Earth [C.2]. In the attached stage, the mass is held at the tip of the pear structure, under a balance of opposing stresses: the Earth's gravity pulls the near flank inward, and the centrifugal force pulls it outward [C.2].

The classical physical model reproduces this structure by computing the shape of a free homogeneous fluid (with no contact constraint), which takes on the configuration of a Roche ellipsoid (Roche Ellipsoid) with three distinct axes ($a : b : c$): the tidal axis toward the Earth (a), the axis of orbital motion (b), and the short axis connecting the poles (c) [C.2]. Unlike the biaxial Maclaurin shape (which looks like a perfect circle when viewed from above the pole), the Roche ellipsoid is deformed and egg-like from every viewing angle [C.2].

The hydrostatic potential-balance equation:

The surface of a fluid body at rest must physically be an equipotential surface (Equipotential Surface), so that no material flows along it [C.2]. Within a uniform ellipsoid, the total potential is made up of the mathematical sum of three energy components [C.2]:

1. The self-gravity of the body: the energy rises as the square of the distance from the centre along the axes (x, y, z corresponding to the axes a, b, c), and the static coefficients A_1, A_2, A_3 (which satisfy the Chandrasekhar index equation: $A_1 + A_2 + A_3 = 2$) set the steepness of the rise [C.2]. The shortest axis (c) is assigned the maximum coefficient (A_3), since the mass is concentrated closer to it [C.2].

2. The tidal potential of the Earth: expressed as $-\mu(x^2 - \frac{1}{2}y^2 - \frac{1}{2}z^2)$, where the stretching factor is $\mu = \frac{GM}{r^3}$ (M is the mass of the Earth and r the orbital distance) [C.2]. This component lowers the potential along the tidal axis (x) and raises it across it (y, z), an action that applies radial stretching and tangential compression [C.2].

3. The centrifugal potential of the orbit: expressed as $-\frac{1}{2}\Omega^2(x^2 + y^2)$ (according to Kepler's third law in the system: $\Omega^2 = \frac{G(M+m)}{r^3}$) [C.2]. This component lowers the potential outward away from the rotation axis, and leaves the polar component (z) unchanged [C.2].

For the surface to balance in a closed shape, the net inward pull (gravity minus the rotation and tidal factors) multiplied by the square of the axis length must be exactly equal at all three ends [C.2]:

$$\mathbf{a^2}\left(\mathbf{A_1} - \frac{\mathbf{\Omega^2}}{\mathbf{\pi G\rho}} - \frac{\mathbf{2\mu}}{\mathbf{\pi G\rho}}\right) = \mathbf{b^2}\left(\mathbf{A_2} - \frac{\mathbf{\Omega^2}}{\mathbf{\pi G\rho}} + \frac{\mathbf{\mu}}{\mathbf{\pi G\rho}}\right) = \mathbf{c^2}\left(\mathbf{A_3} + \frac{\mathbf{\mu}}{\mathbf{\pi G\rho}}\right)$$

This combined equation yields two sets of conditions for the two independent axis ratios: b/a and c/a [C.2].

Solution matrix of the Roche ellipsoid equations

The following are the exact numerical values obtained by solving the system of equations, computed at a distance of 20,595 km (slightly beyond the fluid Roche limit of a body with a density of 2,600 $\mathrm{kg/m^3}$), serving as a morphological illustration tool [C.2]:

Orbital distance (r)	Density of the material (ρ)	Triaxial axis ratio ($a : b : c$)	Polar rise factor (R_s/c)	Radial subsidence factor (R_s/a)	Mechanical state and Roche-limit constraints
19,660 km	2,600	No equilibrium	—	—	Inside the fluid Roche limit; the body is tidally disrupted [C.2].
20,595 km	2,600	$1 : 0.637 : 0.601$	1.208	0.726	Convergence of the stretched Roche shape [C.2].
20,595 km	3,344	$1 : 0.763 : 0.726$	1.131	0.821	A denser body; the tidal deformation moderates [C.2].

- Decoding the dimensions of the stretched shape ($1 : 0.637 : 0.601$): an extreme asymmetry is established, in which the tidal axis facing the Earth is two-thirds longer ($\approx 66\%$) than the short polar axis [C.2].
- Normalisation to an equivalent-sphere volume (R_s): the mean radius of a sphere of the same volume is defined by $R_s = (abc)^{1/3}$ [C.2]. The last two columns quantify the degree of displacement of the surface during the geometric transition (the rounding) toward this equivalent sphere [C.2]:

- **The polar component:** the value of $R_s/c = 1.208$ dictates that during rounding, the polar apex must rise and leap outward by about a fifth of its height ($\approx 21\%$) [C.2].
- **The tidal-axis component:** the value of $R_s/a = 0.726$ dictates that the sub-Earth point must collapse inward and sink by about a quarter of its length ($\approx 27\%$) [C.2].

- **Independent mathematical confirmation:** a repeated solution of the boundary conditions from a free, analytical starting point reproduces the shapes precisely (values of $1 : 0.634 : 0.598$ at a density of 2,600, and $1 : 0.762 : 0.725$ at a density of 3,344), with a minimal consistency deviation of less than half a per cent ($\leq 0.5\%$), and a potential accuracy of two parts per thousand [C.2].

Geodynamic implications and the limits of the rigid Roche limit

The mathematical calculation establishes two basic physical rules that constrain the detachment stages [C.2]:

1. Ruling out the light fluid release: at the source density of the pear ($1,300\ \mathrm{kg/m^3}$), the fluid Roche limit lies very far out, at a distance of about 25,200 km [C.2]. This means that the body cannot physically reach its dynamic release stage while keeping this low density, since it would be tidally disrupted; this confirms the basic assumption of Paper 2, which established that the density must climb to 2,600 $\mathrm{kg/m^3}$ in order to pull the Roche limits inward [C.2].

2. Pinning down the exact fluid Roche limit: at a density of 2,600, solving the equations shows that at 19,660 km the shape breaks apart, while at 20,595 it is stable; the exact fluid Roche limit is fixed between them – at 19,971 km according to the classical coefficient, and at 20,100 km ($k = 0.09$) under the exact potential solution [C.2].

The rigid-crust limit in the dynamic sequence:

In the updated LEF model, the Moon is released and mechanically detaches much earlier and closer to the Earth – at a distance of about 16,400 km at the equator ($16,399\ \mathrm{km}$ at a period of 5.77 hours) and 17,287 km at the latitudes of 22.5° [C.2]. At these close distances, physics dictates that no stable free fluid shape exists or can exist at a density of 2,600 (the region lies deep within the fluid Roche limit) [C.2].

The physical meaning is firm: the young Moon held together and kept its structural integrity in this region thanks to its outer shell, which stiffened and froze early, and not by virtue of a fluid hydrostatic equilibrium [C.2]. Therefore, the triaxial model of this section serves solely as an indicator and illustration tool showing how the external tidal stresses imposed a sharp radial stretching along the a axis in addition to the rotational flattening of the b/c axis (C.1) [C.2]. The density excess and the statistical anomaly measured on the surface are exclusively the result of this combined release of stresses during the sequence of rounding and volume decrease (C.1, C.5, D.2) [C.2].

Whereas sections C.1 and C.2 defined the initial deformation dimensions, this section computes the transformation of the deformed surface during its mechanical collapse to a perfect spherical shape under a condition of volume decrease (compaction and stiffening of the core) [C.3]. The aim is to determine mathematically which crustal area cells close and contract at the maximum rate,

which at the minimum rate, and where the existing craters become crowded [C.3]. The output of the calculation is a measure of the differential density ratio between the poles and the equator, which is robust and independent of the body's overall isotropic contraction rate [C.3].

- **A chronological distinction in the series:** the geodynamic model defines three successive deformation stages:
  1. **Outward rounding:** the initial stage in which the sialic mass detached from the tip of the pear and became an isolated fluid sphere (Paper 3).
  2. **The flattening stage:** the stage of extreme two-body elongation and flattening near the corotation line.
  3. **Inward rounding (the rounding of this paper):** the stage of collapse of the stretched structure back to a sphere, which takes place late in the dynamic sequence, after the centre of mass has crossed the corotation line [C.3].

The squashed-balloon analogy:

To make the mechanics accessible, imagine a round balloon covered with dots (craters) distributed uniformly per unit area, which is pressed and firmly squashed between two flat palms (the two-body stresses) [B.5, C.3]. Owing to the pressure, its top and bottom become wide, flattened and flat, while the circumferential rim (the equator) is sharply stretched [B.5, C.3].

The moment the external stresses relax (crossing the corotation line), the balloon releases its residual elastic energy and springs back to its spherical shape [C.3]. In this geometric process, the wide, flat surfaces of the ends are rapidly gathered inward and converge into a narrow zone: the dots imprinted on them physically draw closer together at the maximum rate, while the middle rim undergoes the minimum change of area [B.5, C.3].

The geophysical mechanics of fixing the end points at the poles

Why is it precisely the geographic polar regions that function as the main centres of convergence and crowding of the crust [C.3]?

- **Refuting thickness or gravity hypotheses:** GRAIL satellite observations rule out the existence of a thickened crust or lateral gravity anomalies in the polar regions relative to the latitude belt in which they lie (B.4) [C.3].
- **The rotation-axis constraint (The Axial Geometric Anchor):** the poles are the structural end points through which the body's rotation axis physically passes [C.3]. Since the rotational flattening stresses act completely symmetrically about this axis, all the global meridians were deformed and stretched uniformly, while the two end points on the axis remained static (they have no "outward" momentum component of their own) [C.3].
- **The wheel-spoke mechanism:** the moment the rotational flattening is released, the crustal material that had previously been pushed out returns inward along exactly the same meridians [C.3]. Since all the meridians of a spherical body converge and meet at the poles, these end regions function as a structural hub that gathers into itself the surface of all the meridians [C.3]. This geometric property explains why there are exactly two polar crowding foci in the system,

located in exactly opposite positions facing each other [C.3]. The model fixes the ancient rounding poles as the geographic poles observed on the surface today [C.3].

Distribution of the differential tensile stresses over the surface

During the deformation stage, the crust in the polar regions was the flattest, most stretched and most widened in the system [C.3]. By contrast, the Earth's tidal stretching (the Roche effect) stretched the body extremely along the a axis (the radial line), with its near part held at the tip of the pear structure [C.3]. As a result, the surface at both ends of the Earth line (the sub-Earth and anti-Earth points) was compressed and less stretched than any other sector [C.3].

Therefore, at the moment the stresses were released and the body relaxed:

- **Maximum surface contraction:** takes place on the polar surface (the return of the maximum rotational stretching), and produces the highest crater-density excess [C.3].
- **Minimum surface contraction:** takes place at the ends of the tidal axis (the sub-Earth point), and leaves the crater record in its sparsest state (94 craters per 10^6 km²) [A.9, C.3].

The geometric balance equation: the combined density factor

The mechanical crowding is not limited to the polar ends alone; it affects a complete global ring standing at an angular distance of 90° from the Earth-line axis (the polar-tangential ring passing through both poles and both orbital sides) [C.3]. The reason is that the two ends of the tidal axis, as they collapse inward during rounding, push and send their share of the surface sideways, toward the middle ring [C.3].

The following is a mathematical calibration of the two-directional area-change factors, using the Roche ellipsoid (C.2) as an illustration model [C.2, C.3]:

Spatial location on the crust	Change factor from the Earth-line axis (a)	Change factor from the perpendicular axis (blc)	Combined density factor (ρ_{combined})	Geodynamic meaning in the crust
The polar regions (apex of the axis)	1.38× (tidal contraction)	0.88× (lateral stretching)	1.21×	Overall peak crowding [C.3].
The sides (leading and trailing)	1.38× (tidal contraction)	0.83× (polar stretching)	1.14×	Moderate crowding along the ring [C.3].
The sub-Earth point (tidal axis)	0.88× (tidal widening)	0.83× (polar widening)	0.73×	Maximum dilution (the density valley) [C.3].

- **Decoding the mathematical dynamics:** for each small crustal area cell, two perpendicular stress directions along the ground are defined [C.3]. The rounding shortens or lengthens the dimensions of the cell along these two axes, while the total crater count on it remains unchanged [C.3]. Therefore, the final crater density is computed as the product of the inverses of these two area-change factors ($\rho_{\text{combined}} = \text{Factor}_1 \times \text{Factor}_2$) [C.3].

- **System normalisation:** the values are expressed relative to the average global change of area [C.3]. A factor ≥ 1.00 indicates that the area cell closed and contracted in this direction more than the average (the craters drew closer together); a factor < 1.00 indicates a contraction below the average (the craters moved relatively apart) [C.3]. At the pole, the equation gives: $1.38 \times 0.88 = 1.21$ [C.3].
- **The order along the ring:** the poles and the sides undergo an identical contraction factor from the tidal axis (1.38), but the poles receive material from the sides at a rate exceeding the material they pass to them [C.3]. As a result, the poles settle at the top of the order (1.21) and the sides after them (1.14) [C.3]. The sub-Earth point widens along both its axes relative to the average, and therefore shows the minimum density value in the system (0.73) [C.3].

Confirmation of the order from the clean six-axis data

The following are the empirical data reconstructed from the Robbins catalogue in the 30° caps, after applying the "clean count" procedure (complete removal of the maria and the SPA basin from both sides of the equation), which confirm precisely the predictions of the geometric balance equation [A.9, C.3]:

End of the geometric axis	Crater density (≥ 20 km per 10^6 km²)	Crater density (≥ 40 km per 10^6 km²)	Mechanical status of the sector in the lunar crust
The sub-Earth point	94	32	The minimum base; the site of the contact disc of the pear [C.3].
The point opposite the Earth (anti-Earth)	204	60	Stands on the ring; acts as the summit of the free dome [C.3].
The leading side (90° W)	162	58	Stands on the ring; neutralised from the bias of the Orientale basin [C.3].
The south pole (90° S)	249	82	Stands on the ring; morphometrically leading (without the SPA basin) [C.3].
The north pole (90° N)	227	87	Stands on the ring; morphometrically leading [C.3].
The trailing side (90° E)	253	84	Stands on the ring; enriched by the geometry of returned material and rotational lag [C.3].

The empirical findings are firm: every point located along the polar ring shows a significant density excess compared with the density valley of the sub-Earth point, which contains less than half the number of craters of the poles [C.3]. The internal asymmetry between the trailing and leading sides is explained by the distribution of local ejection vents and the flow of momentum of returned material that lagged behind the rotational velocity and landed behind (A.9) [C.3]. The polar regions are the geometric site where the maximum crowding takes place and where the phenomenon is measured in its most pristine and clean form, and hence the name of the paper's sections [C.3].

C.3 The mathematical calculation of differential surface contraction and the morphometric predictions

This section is the firm analytical core of the paper [C.3]. It translates the geometric shape changes of the body into transformation equations of the surface, and computes precisely the differential crowding ratios between the different crustal regions [C.3]. The calculation is based on a complete physical separation between two independent components of collapse:

1. The change of shape (Anisotropic Deformation): computed under the firm constraint of constant volume toward an equivalent sphere of radius $R_s = (abc)^{1/3}$ [C.3].

2. The isotropic change of size (Isotropic Volume Contraction): the Moon underwent a global thermal contraction and release of stretching that reduced its volume by a factor of about 1.5 during rounding [C.3]. Since this shared contraction multiplies and shrinks every area cell on the body by the same constant factor, it cancels out mathematically and does not change any density ratio [B.6, C.3]. Therefore, the crowding measure of the poles versus the equator is entirely immune to the body's overall contraction rate [C.3].

The homogeneous ellipsoidal transformation equation:

The geometric transition is represented as a homogeneous elastic deformation that turns the ellipsoid shape into a sphere, as if the surface were drawn on a uniform rubber sheet [C.3]. Every coordinate along the long, intermediate and short axis is multiplied by the corresponding stretch factors: $R_s/a, R_s/b$, and R_s/c respectively [C.3]. A small area cell whose outward normal vector (perpendicular to the surface, of unit length) is defined as $\hat{n} = (n_x, n_y, n_z)$ changes its area by the combined geometric factor [C.3]:

$$\mathbf{Area_Factor}\mathbf{=J}^{\mathbf{-1}}\mathbf{=}\left[\mathbf{n}_{\mathbf{x}}^{\mathbf{2}}\left(\frac{\mathbf{R}_{\mathbf{s}}}{\mathbf{a}}\right)^{\mathbf{2}}\mathbf{+n}_{\mathbf{y}}^{\mathbf{2}}\left(\frac{\mathbf{R}_{\mathbf{s}}}{\mathbf{b}}\right)^{\mathbf{2}}\mathbf{+n}_{\mathbf{z}}^{\mathbf{2}}\left(\frac{\mathbf{R}_{\mathbf{s}}}{\mathbf{c}}\right)^{\mathbf{2}}\right]^{\mathbf{-1/2}}$$

Craters imprinted uniformly on the original surface change their densities in inverse proportion to this area factor (J), since the same number of craters compressed into a smaller area cell will show a higher density [C.3].

The meaning of the formula in the space of the crust:

The equation establishes that the surface area of a patch facing along one spatial axis is governed by the product of the stretches of the two axes perpendicular to it [C.3]. Since the product of the three stretches is exactly 1 (volume conservation in the shape stage), the change of area equals exactly the inverse of the stretch along the axis normal to the patch [C.3]:

- At the polar apex ($n_z = 1$): the factor shrinks to its minimum value, c/R_s [C.3]. The surface undergoes the maximum geometric contraction, equally in every tangential direction, and the residual crater density jumps at the maximum rate of R_s/c [C.3].
- At the sub-Earth point ($n_x = 1$): the factor takes its maximum value in the system, a/R_s [C.3]. The surface undergoes the minimum contraction, and the crater density there is the lowest,

standing at R_s/a [C.3].

- **The radial transition:** between the pole and the equator, the contraction factor weakens continuously with distance from the apex of the axis [C.3]. This mechanical property explains geometrically why the density profile of the pristine north pole (A.6) climbs and sharpens inward directly toward the geographic pole point [C.3].

Calibration of the biaxial model (the oblate spheroid)

For a symmetrically flattened body in which the long and intermediate axes are equal ($a = 4b$), the volume-conservation condition dictates $R_s^3 = a^2 c$ [C.3].

- The polar surface changes by a factor of $c/R_s = (c/a)^{2/3}$, representing a uniform biaxial shortening of $(c/a)^{1/3}$ along every ground direction [C.3].
- The equatorial surface changes by a factor of $a/R_s = (a/c)^{1/3}$, representing an east–west shortening of $(c/a)^{1/3}$ against a minimal contraction along the north–south axis – this is the area cell that contracts least of all on the body [C.3].

The statistical ratio between these two regions defines the absolute crowding equation of the pole versus the equator, in which the radius of the sphere (R_s) cancels out completely [C.3]:

$$\frac{\rho_{\text{polar}}}{\rho_{\text{equatorial}}} = \frac{\mathbf{R_s/c}}{\mathbf{R_s/a}} = \mathbf{\frac{a}{c}}$$

A firm rule is established: the maximum theoretical crowding at the pole point relative to the equator equals exactly the equatorial/polar axis ratio (a/c) of the original flattened shape [C.3]. A body that was 1.69 times as wide as it was tall at peak deformation will leave the craters imprinted on it at a density exactly 1.69 times higher at the polar apex relative to the equator [C.1, C.3]. However, when the averages over the final empirical caps are computed, the ratio value is smaller, because the contraction decays with angular distance from the pole [C.3].

Calibration of the triaxial model and agreement with the six axes

For a flattened and stretched triaxial body ($a : b : c$), the same differential equation sets a threefold density order: the density climbs by a factor R_s/c at the poles, by a factor R_s/b at the equatorial sides, and by a factor R_s/a at the base of the Earth line [C.3]. This order (poles $\rightarrow$ sides $\rightarrow$ Earth line) reproduces and fully matches the empirical six-axis order measured in section A.9 [C.3].

To test the validity of the model, the original shape of the body is reconstructed from the six-axis data, and this is carried over, unchanged, to predict the clean ratio of the 30° caps versus the equatorial belt ($|\phi| < 15^\circ$) on the pristine crust [A.9, C.3]:

Empirical database examined	Poles/sides ratio	Poles/ends ratio	Reconstructed triaxial shape (a : b : c)	Predicted caps/belt ratio	Measured caps/belt ratio (clean count)
All craters ($\geq 20\ \mathrm{km}$)	1.12	1.30	$1.37 : 1.15 : 1$	1.21	1.19 (interval: 1.20 ± 0.05 in D.1)
Craters $\geq 40\ \mathrm{km}$	1.11	1.53	$1.68 : 1.13 : 1$	1.28	1.39
Craters 60 – 100 km	1.19	1.78	$2.07 : 1.22 : 1$	1.42	1.53
Craters 60 – 300 km	1.28	1.73	$1.96 : 1.34 : 1$	1.48	1.61
Craters 100 – 300 km	1.58	1.61	$1.76 : 1.75 : 1$	1.64	1.84

- **Integration without tuning (Zero-Parameter Fit):** For the general population ($\geq 20\ \mathrm{km}$), the shape reconstructed from the six axes ($1.37 : 1.15 : 1$) predicts, completely independently, a ratio of 1.21 on the clean surface, a value matching with extreme precision the finding measured on the surface – 1.19 [C.3]. Two entirely disconnected geographic area cells agree with each other through the solution of a single geometric equation, without any need to fit arbitrary parameters [C.3]. The counting scatter of this shape is very small (a range of 1.08–1.22 for b/c, and 1.30–1.48 for a/c) [C.3].

Decoding the peak shape from the large craters

For the ancient size classes (60–100 km and 60–300 km), the six-axis uncertainty intervals are wider owing to small samples in each cap (only 13–68 craters, which scatter the values over wide ranges of 1.63–2.87 for a/c and 1.02–1.46 for b/c) [C.3]. Therefore, the physical peak shape is reconstructed through an analytical treatment based on a family of shapes defined by the dynamic deformation equation [C.3]:

$$a : b : c = (a/c) : [1 + f(a/c - 1)] : 1$$

where the measure $f = \frac{b-c}{a-c}$ represents the share of the long axis's excess that the intermediate axis also carries ($f = 0$ represents a pure egg, and $f = 1$ represents a pure pita) [C.3]. For each value of f, the axis-ratio value a/c is chosen that yields the exact clean ratio of the 60–100 km class (a value of 1.44, with the sub-Earth point neutralised from the belt on both sides owing to the depletion of the contact disc in A.9.2) [C.3]:

Shape factor (f)	Triaxial shape configuration (a : b : c)	Predicted values for the six caps (poles/sides , poles/ends)
0.19	$1 : 1.25 : 2.34$	1.95 / 1.22
0.35	$1 : 1.33 : 1.95$	1.72 / 1.28

Shape factor (f)	Triaxial shape configuration (a : b : c)	Predicted values for the six caps (poles/sides , poles/ends)
0.50	$1 : 1.39 : 1.78$	1.61 / 1.32
0.60	$1 : 1.42 : 1.70$	1.55 / 1.34
0.70	$1 : 1.45 : 1.64$	1.51 / 1.36
1.00	$1 : 1.52 : 1.52$	1.42 / 1.41

- **Convergence of the core around the peak shape:** The empirical values measured at the six axes yield end ratios of 1.19 / 1.78 for 60–100 km, and 1.28 / 1.73 for 60–300 km [C.3]. Cross-checking the data shows that the shapes located in the core range between $f = 0.35$ and $f = 0.60$ fall precisely within the counting scatter of the six axes and the clean belt simultaneously [C.3]. The centre of this family is firmly fixed around the system value: $1.8 : 1.4 : 1$ (a value of $a/c \approx 1.6{-}1.95$, and $b/c \approx 1.33{-}1.45$) [C.3]. On the full clean equatorial belt, these shapes predict excess ratios of 1.45–1.49, a value in full agreement with the finding on the surface – 1.53 ± 0.19 [C.3]. The local anomaly of the trailing cap (A.9) slightly raises the average of the sides, but does not change the validity of the reading [C.3].
- The physical implication of an enhanced f measure: linear tidal theory alone yields for a free body low f values of 0.22–0.25 near the line [C.2, C.3]. The excess of the value actually measured ($f \approx 0.35{-}0.60$) is direct mechanical proof that the elongation of the tidal axis (the egg part) was physically restrained and curbed relative to the rotational flattening, evidence for the existence of a rigid holding base that restrained the stretching (C.6) [C.3].

The mechanical model of the geometric chronological clock

The mathematical analysis of the transformation sequence is defined as a geometric chronological clock [C.3]. Each geometric stage along the path of inward collapse from peak flattening to a sphere represents a dynamic moment in history, and the residual crowding fixed in the crust depends exclusively on the remaining rounding that took place after the crater formed [C.3]:

Peak flattening (1.8:1.4:1) → predicted 1.49 mid-rounding (1.30) → predicted 1.26 late (1.20) → predicted 1.17

A crater formed at the peak absorbed the full sequence of surface contraction, while a crater formed late received only the closing chord [C.3]. Since ejections of material took place also during the growth, an intermediate shape was crossed twice (on the way up and on the way down); therefore, a large, crowded crater reflects formation near the peak point, and not necessarily earlier in chronological terms [C.3].

The calculation is carried out by direct integration on a 0.05° grid of the spatial density-factor function [C.3]:

$$\mathbf{J}(\lambda,\phi) = \mathbf{R_s}\left[\cos^{\mathbf{2}}\lambda\left(\frac{\cos^{\mathbf{2}}\phi}{\mathbf{a^2}} + \frac{\sin^{\mathbf{2}}\phi}{\mathbf{b^2}}\right) + \frac{\sin^{\mathbf{2}}\lambda}{\mathbf{c^2}}\right]^{\mathbf{1/2}}$$

where λ, ϕ are the angular coordinates on the surface; the precision achieved between the two analytical calculation methods is 10^{-7}, and the code is fully included in the reproduction package of the system [C.3].

The morphometric determination is firm: the craters themselves kept their original diameter and were carried as rigid elements (tiles) on a solid upper crust; what contracted and closed elastically was the surface and the seam spaces between them (C.7, D.2) [C.3]. The largest craters ($100-300\ \mathrm{km}$) show an empirical value of 1.84 ± 0.38, which suggests a shape reading beyond the theoretical peak, but falls within the wide counting error of the sample [C.3].

C.3.1 Mathematical calibration of the volume decrease and the absolute area factors

To complete the development of the mechanism, the mathematical calculation fuses together the two components of the crustal transformation: the geometric change of shape (at constant volume) together with the isotropic change of size (thermal contraction and compaction of the core) [C.3]. In the dynamic transition from peak flattening ($\rho \approx 1,700\ \mathrm{kg/m^3}$) to the spherical release ($\rho \approx 2,600\ \mathrm{kg/m^3}$), the global volume of the body falls by a firm factor of 1.53 [C.3.1].

This fall shrinks the equivalent radius of the young Moon by 13% (from 2,180 km at the peak to 1,890 km at release), and mathematically reduces the average global surface area by a quarter [C.3.1]. The following are the absolute area-change factors (combining the volume decrease and the rounding together) computed for the centre of the peak-shape family ($1.75 : 1.45 : 1$) at a density of 1,700 $\mathrm{kg/m^3}$ [C.3, C.3.1]:

Spatial sector of the crust examined	Absolute area-change vector	Dynamic and mechanical meaning for the surface
The average surface of the body	0.75×	Shared (isotropic) contraction arising from the overall volume decrease [C.3.1].
The surface in the polar regions (axis c)	0.55×	Maximum surface contraction in the system (dark cell) [C.3.1].
The surface at the sides (leading and trailing, axis b)	0.80×	Moderate surface contraction along the ring [C.3.1].
The surface at the ends of the Earth line (axis a)	0.97×	Minimum surface contraction in the system [C.3.1].
The surface of the polar caps versus the equatorial belt	0.67×	The combined average of the cap areas (0.58) versus the belt (0.86) [C.3.1].
Crater-density measure (caps versus belt)	1.49×	The predicted crowding that the statistical test reads on the surface [C.3.1].

- **Decoding the mechanical order:** Reading the table from top to bottom reproduces precisely the empirical six-axis order: the poles close and contract at the sharpest rate (0.55), the sides after them (0.80), and the ends of the Earth line show the minimum change (0.97) [C.3.1].

- **The motion vectors of the surface:** Under this dynamic geometry, the polar regions function as a rigid anchor and hardly move in space (the physical polar radius grows by only about 290 km as a result of the outward rounding, while the body as a whole contracts inward) [C.3.1]. All the other parts of the crust move actively along the body toward them: the areas of the sides retreat inward by about 430 km, and the surface at the ends of the Earth line collapses inward by about 900 km [C.3.1]. This is the squashed-balloon mechanism – the two ends of the axis are fixed, and all the rest of the material of the belly gathers and converges toward them [C.3, C.3.1].

Confirmation of the theoretical peak-density limit

The absolute calculation provides a second physical check for setting the peak-density limit ($\approx 1,700\ \mathrm{kg/m^3}$) [C.1, C.3.1]:

- For all area cells on the body to undergo positive closure and contraction (a factor value < 1.00), physics dictates that the density of the body at the peak must be fixed below an upper threshold of about 1,800 $\mathrm{kg/m^3}$ (exactly 1,790 $\mathrm{kg/m^3}$ for this shape configuration) [C.3.1].
- Had the body at the peak been denser than this value, the geometric energy would have been too small; in that hypothetical case, the surface at the ends of the Earth line would have been forced to expand and grow (factor ≥ 1.00) during rounding instead of closing, a finding that contradicts the crater record [C.1, C.3.1].

It is firmly emphasised that the scientific test of the paper does not rest on these absolute factor values, which are sensitive to the theoretical peak density [C.3.1]. The test is based exclusively on the statistical ratio between the last two rows ($\frac{1}{0.67} = 1.49$), which is robust and remains completely constant for any value of peak density, because the shared isotropic contraction term cancels out of it completely [C.3.1]. The scientific claim of the study rests on this ratio factor: the poles crowd to the system maximum and the equator to the minimum, without dependence on, or need for, an exact numerical determination of the contraction rate of the equator itself [C.3.1]. (The presence of mechanical strength in the crust would partially restrain the deformation, and yield more moderate contraction factors, a mechanism developed in section E.3) [C.3.1].

The evolution of the Moon's shape in three dynamic stages

Figure 10 brings together the entire geodynamic mechanism in a sequence of three successive stages of development, illustrating how the crater dots, imprinted uniformly on the flattened body, converge and crowd at the two poles as the body tends toward a spherical shape [C.3.1]:

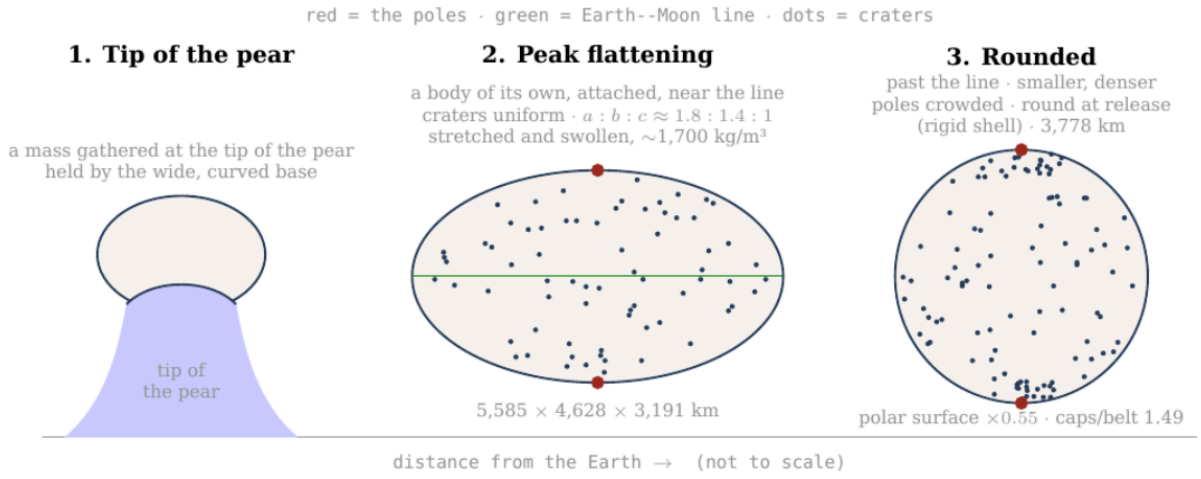


Figure 10. The Moon's shape in three stages: a mass gathered at the tip of the pear, held at its wide, curved base; at peak flattening, as it approached the 3,000 km base near the corotation line — a body of its own, still attached, stretched between the Earth's gravity on its near side and the centrifugal pull of corotation — triaxial, *a*:*b*:*c* ≈ 1.8:1.4:1, seen here from the side, with the Earth line horizontal — with the craters distributed uniformly over it; and round, smaller and denser, after its centre of mass has passed the line, with the craters formed on the flattened body crowded at the two poles. The dots are illustrative; the numbers are those of sections C.1 and C.3. The diameters of the body at the peak (along the Earth line, along the orbit and through the poles) are computed for the shape 1.75 : 1.45 : 1 and the Moon's mass at about 1,700 kg/m³ (equivalent radius about 2,180 km), and the round body at 2,600 kg/m³; the paper's claim rests on the ratio between the pole and the equator, which does not depend on the overall size. (In the diagram: horizontal axis — the distance from the Earth, not to scale; red — the poles; green — the line between the Earth and the Moon; dots — craters. Stage 1, "tip of the pear": a mass gathered at the tip of the pear, held at the wide, curved base. Stage 2, "peak flattening": a body of its own, attached, near the line; uniform craters, *a*:*b*:*c* ≈ 1.8:1.4:1; stretched and inflated, about 1,700 kg/m³; 5,585 × 4,628 × 3,191 km. Stage 3, "rounded": beyond the line, smaller and denser, dense poles, round at the moment of release (rigid shell), 3,778 km; the polar surface 0.55×, caps/belt 1.49.)

1. Stage 1 ("tip of the pear"): an initial sialic mass consolidates at the tip of the Earth's elongated structure, held at its wide, curved mechanical base (an initial base width of about 6,000 km at an orbital distance of $\approx 8,991\ \mathrm{km}$) [C.1, C.3.1].

2. Stage 2 ("structural peak flattening"): the mass separates from the pear material and becomes an independent attached body, moving beyond the rigid Roche limit ($\approx 13,000\ \mathrm{km}$) toward the corotation line [C.1, C.3.1]. The base narrows to a width of 3,000 km, and the body is stretched extremely between the Earth's gravity inward and the centrifugal force of corotation outward [C.1, C.3.1]. The body takes on a precise triaxial geometry of $a : b : c \approx 1.8 : 1.4 : 1$ (computed for the centre of the family $1.75 : 1.45 : 1$) [C.1, C.3.1]. In this stretched, inflated state, the density of the material stands at about 1,700 $\mathrm{kg/m^3}$ and its physical dimensions are set at $5,585 \times 4,628 \times 3,191\ \mathrm{km}$ [C.3.1]. The crater population is deposited and distributed over this flattened morphology in a completely uniform distribution per unit area [C.3.1].

3. Stage 3 ("the rounded body at release"): the Moon's centre of mass crosses the corotation line; the stretching is released, and the body collapses inward with a self-driven decrease in volume to a round, smaller and denser shape [C.1, C.3.1]. The density climbs to 2,600 $\mathrm{kg/m^3}$ and

the size of the body converges to a spherical diameter of 3,778 km at the moment of mechanical release, with its outer shell already rigid [C.1, C.3.1]. The geometric transition contracts the local area at the poles by a factor of 0.55×, and gives rise to the official bipolar density excess of the 30° caps versus the equatorial belt, fixed at a predicted value of 1.49× [C.3.1].

Summary of the combined mechanism:

It has been shown geodynamically that a flattened body rounding under a condition of ultimate volume decrease closes and reduces its polar surface at the maximum rate, and its equatorial belt at the minimum rate [C.3.1]. Crustal craters imprinted on the crust before the rounding stage end the sequence concentrated and crowded in the polar regions by an exact factor set exclusively by the original geometric shape (excess values of about 1.2 to 1.5 within the counted caps, depending on their time of formation relative to the peak point) [C.3.1]. The six-axis model reproduces and confirms the validity of the empirical phenomenon across all the size classes, without the need for free parameters [C.3.1].

C.4 Gravity as the physical engine, the slope profile, and the energy and pressure balance

This section defines the dynamic engine that drove the rounding process, quantifies the energy output released in the system, and decodes its effects on the interior of the lunar body [C.4]. The mechanism proves that there is no need to assume a new external force: on crossing the corotation line, the body is released from the compression on the base of the pear, and the Moon's self-gravity field, which strengthens as the volume shrinks and the material is compressed, is what forces the body to round [C.4]. Since the outer shell is rigid, this deformation builds a sharp internal pressure that finds release through the key geological openings already existing in the crust (the maria) [C.4].

1. The two-component release mechanism during rotational locking

The original deformation of the body was held by a temporary balance of forces: the stresses of fast synchronous rotation together with the two-body tidal stretching along the Earth line (the Roche effect), while the body was mechanically pressed onto the base of the pear [C.2, C.4]. The relaxation of the deformation and the inward collapse proceeded through the separate release of the two shape components [C.4]:

- Release of the egg part (the tidal axis a): this stretching component was held directly by the mechanical compressive force against the base of the pear [C.4]. The moment the Moon's centre of mass crossed the corotation line, the compression vector from below fell to zero and became an opposite pull directed outward [C.1, C.4]. The vertical pressure beneath the base dissipated, and the egg part collapsed rapidly inward [C.4].
- Release of the pita part (rotational flattening b/c): this component was released directly by the Moon's self-gravity [C.4]. On passing beyond the line, the spin rate remained almost constant (a moderate lengthening of the Earth's day from 5.5 to 5.77 hours), while the density of the material rose sharply from about 1,700 to about 2,600 $\mathrm{kg/m^3}$ owing to the combined volume

collapse [C.1, C.4]. As a consequence, the measure of the balance of rotation against self-gravity, expressed by the left-hand side of the Maclaurin equation:

$$\frac{\mathbf{\Omega^2}}{\pi \mathbf{G} \rho}$$

fell from 0.28 at the peak to 0.17 in the late stages [C.1, C.4]. This parametric fall forced the theoretical equilibrium shape to fall correspondingly from 1.41 to 1.20 [C.1, C.4]. Gravity is the sole engine of this stage: a given rotation rate can no longer hold a flattened configuration against the internal gravity field of a body that has become dense and pulls more strongly inward [C.1, C.4]. The dynamics are like a figure skater spinning with outstretched arms whose physical weight suddenly increases without her rotation speeding up – she can no longer hold her arms stretched out at the same radius [C.4].

2. The potential slope profile and the crustal flow

On a highly flattened body that has departed from its hydrostatic equilibrium, the effective potential vector changes and reverses its dynamic direction [C.4]. The structural flattening that the rotation previously held at the low density $(1,700)$ becomes an anomalous excess load that the enhanced gravity at the high density $(2,600)$ cannot bear [C.4].

As a result, the polar regions become geometrically the absolute minimum points of the effective potential (the floor of a potential valley), while the equatorial belt stands at the top of the order [C.4]. The elastic crustal materials and the whole surface experience a lateral push vector and flow down this energy slope – from the equatorial belt directly inward toward the poles, along all the global meridians [C.4]. The mechanical process works like a bowl of thick honey that has been tilted and returned to the horizontal: the viscous fluid will flow on its own, by its own weight alone, until it reaches a flat surface and a renewed equilibrium, without any need for additional external pressure [C.4].

3. The energy-output equations and the values of thermal compression

The process of rounding and inward collapse releases in the system two independent spatial energy terms, which turn into heat and thermally feed the interior of the body [C.4]:

a. The shape-change term (eccentricity energy):

The inward rounding lowers the global gravitational energy of the body from the deformation level of the ellipsoid to the minimum level of an equivalent sphere of the same volume [C.4]. For a flattened spheroid with eccentricity e (such as an axis ratio $a/c = 1.69$), the gravitational binding energy W (the work required for complete mechanical disassembly of the structure, defined as negative) is computed exactly by [C.4]:

$$\mathbf{W} = -\frac{\mathbf{3}}{\mathbf{5}} \frac{\mathbf{G M^2}}{\mathbf{R_s}} \frac{\arcsin\left(\mathbf{e}\right)}{\mathbf{e}}$$

where M is the mass of the Moon and R_s the radius of the equivalent sphere [C.4]. For a perfect sphere, the geometric factor stands at $3/5$ (0.60); the flattened shape carries an additional coefficient slightly smaller than 1 that depends exclusively on its eccentricity [C.4]. Since the sphere is the most physically bound shape, the geometric transition releases elastic energy, like a stone lowered into a pit [C.4]. For the system's triaxial peak shape $(1.75 : 1.45 : 1)$, the energy difference releases exactly $2 \times 10^{27}\ \mathrm{Joule}$, a value that is 2.1% of the total binding energy of the body (a value of about $3 \times 10^4\ \mathrm{J/kg}$) [C.4]. This energy, released as internal heat within a viscous core (which behaves mechanically like thick tar), raises the global temperature of the interior by about 30 kelvin [C.4]. This thermal factor supplies energy to keep the interior mobile and viscous, but cannot melt a solid crust [C.4].

b. The volume-decrease term (compression energy):

The second and dominant energy term in the system arises from the radial convergence of the mass [C.4]. The gravitational binding energy of a sphere, $-3/5(GM^2/R)$, deepens sharply as the radius of the body (R) decreases [C.4]. As the equivalent radius falls from about 2,180 km to about 1,890 km, the energy released in the system jumps to a value of $1.5 \times 10^{28}\ \mathrm{Joule}$ (a value of $2.1 \times 10^5\ \mathrm{J/kg}$) [C.4].

This compression term is seven times larger than the geometric shape-change term, and it releases a thermal output able to heat the interior of the body by about 200 degrees Celsius on global average (an exact value of 206 kelvin, computed under a fixed heat capacity of $1\ \mathrm{kJ/kg \cdot K}$) [C.4]. The energy acts like a cocked spring released from its constraints that returns the stretching work that held it; the work is converted directly into high values of hydrostatic pressure and internal heat, which keep the core in a mobile state and feed the crustal ejection vents (the maria) [C.4].

4. The resistance stresses of the shell and the chronological-clock mechanism of pressure

While the viscous interior of the body tends to round in accordance with the lines of gravity, the solid outer shell (the crust of the young Moon) froze early and has a high rigidity, required to preserve the integrity of the craters imprinted on it (C.7) [C.4].

- **Strain stresses in the crust:** forcing a solid shell to change its outer shape from the triaxial peak shape to a spherical structure forces it to undergo strain stresses (relative change of dimension along its length) of between 10% and 20% [C.4].
- **Collapse of mechanical strength:** under the elastic rigidity values of a solid rocky crust (tens of GPa), this strain produces an enormous internal stress of between 2 and 12 GPa [C.4]. Since the maximum shear strength of the shell materials is only about 0.1 GPa, the resulting stress is 20 to 120 times higher than the mechanical strength limit of the crust [C.4]. The shell cannot bear this load [C.4].

The rounding sequence and the change of pressure in the crust	Stage A: structural peak flattening	Stage B: intermediate rounding	Stage C: late rounding (release)
Assigned theoretical axis ratio $(a : b : c)$	1.8 : 1.4 : 1	1.40	1.20

The rounding sequence and the change of pressure in the crust	Stage A: structural peak flattening	Stage B: intermediate rounding	Stage C: late rounding (release)
Residual polar mismatch ($1-(c/a)^{2/3}$)	System maximum (1.00)	Measure decaying down the curve	Low residual value
Internal pressure profile in the crustal vents	Peak hydrostatic intensity	Intermediate, decaying pressure	Minimal pressure (tending to zero)
Morphology of the returned-material craters	Giant craters ($D \geq 60-100$ km)	Intermediate craters ($40-60$ km)	Small craters ($20-30$ km)
Degree of crater crowding at the poles	Maximum contraction (predicted ratio ~1.5)	Intermediate contraction (predicted ratio ~1.26)	Minimum contraction (predicted ratio ~1.17)

The shell does not need to break or fall apart completely to release these stresses [C.4]. The enormous hydrostatic pressure projected from the interior finds a rapid outlet through the system of conduits and key geological openings already existing and distributed in the crust – the array of mare vents (Paper 5) [C.4]. The mechanism works like a bottle containing a soft filling and a stiff skin pierced at several points: applying hand pressure will eject and spray the interior material out through the holes with high force, which will gradually decay as the pressure relaxes [C.4].

The magma and molten rock were sprayed and thrown forcefully through the mare vents, performed ballistic flights (Paper 6), and landed back on the solid crust, imprinting the ancient crater population (the returned material) [C.4].

The physical mechanics of the polar-mismatch equation

The dynamic profile of the internal pressure is computed precisely (Figure 12) as a function of the vector of geometric mismatch remaining between the polar surface of the rigid shell and the spherical shape toward which the interior tends:

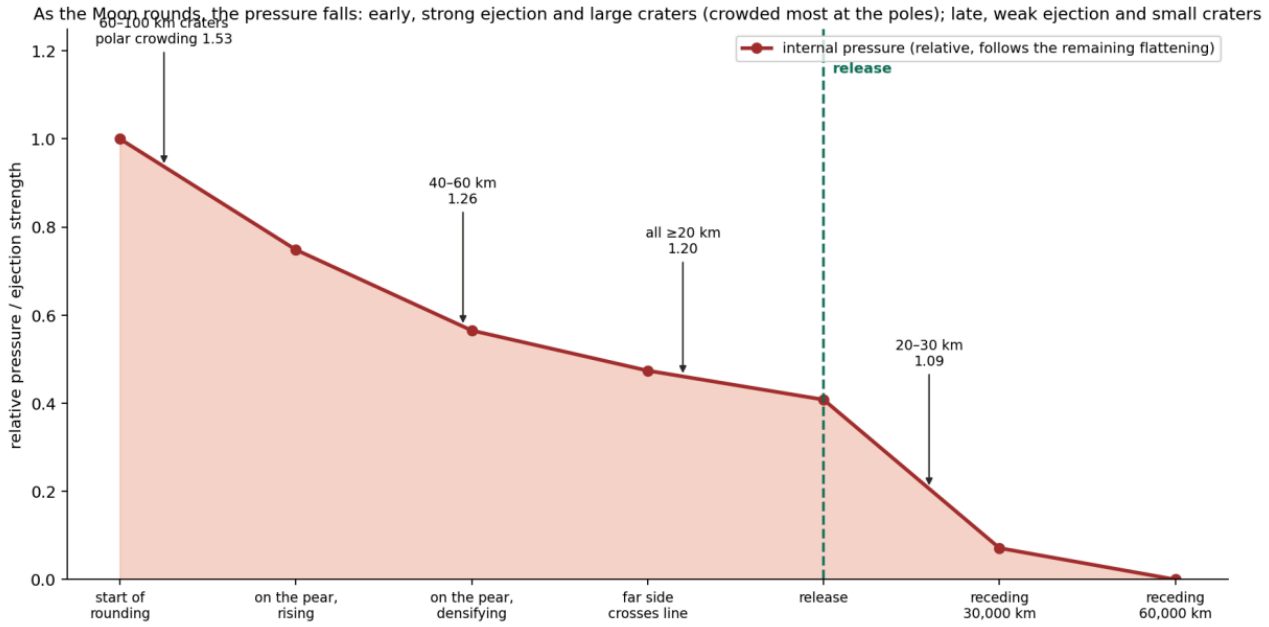


Figure 12. The internal pressure through the rounding stages, relative to its value at peak flattening (it follows the remaining polar mismatch, $1-(c/a)^{2/3}$), with the crater sizes whose clean polar crowding points to each stage (D.2). The axis labels in the figure keep the earlier names and the compressed scale of the stages: "*a*/*c*) "start of rounding 1.69) is peak flattening, about 1.8 : 1.4 : 1, near the corotation line; "on the pear, rising", "on the pear, densifying" and "far side crosses line" are the rounding stages, read as 1.40, 1.30 and 1.24.

Pressure_Profile}\mathbf{\propto 1-}\left(\frac{\mathbf{c}}{\mathbf{a}}\right)^{\mathbf{2/3}}

- **At the moment of peak flattening (stage A):** the mismatch is maximal, and therefore the hydrostatic pressure reaches its physical peak [C.4]. The ejection of material is at its most powerful, and produces the largest giant craters in the system [C.4]. These craters were imprinted on the most flattened body, and after their formation absorbed all the stages of the surface-contraction sequence, and therefore today show the maximum density excess at the poles (the geometric chronological clock) [C.4].
- **Through the rounding stages (stages B and C):** as the body collapses inward and rounds, the mismatch value steadily decreases, and with it the internal pressure profile in the vents falls [C.4]. The ejections gradually weaken, and produce craters of progressively smaller diameters [C.4]. These craters underwent less residual surface contraction, and therefore show a moderate density excess at the poles [C.4].

This is the firm physical basis that defines crater diameter as the chronological clock of the system (C.5, D.2) [C.4]. When a perfect spherical shape is reached, the surface becomes a hydrostatic equipotential, the slope vector drops to zero, the structural mismatch disappears entirely and with it the internal pressure in the vents collapses, and the residual crustal flow comes to a complete stop in the system [C.4].

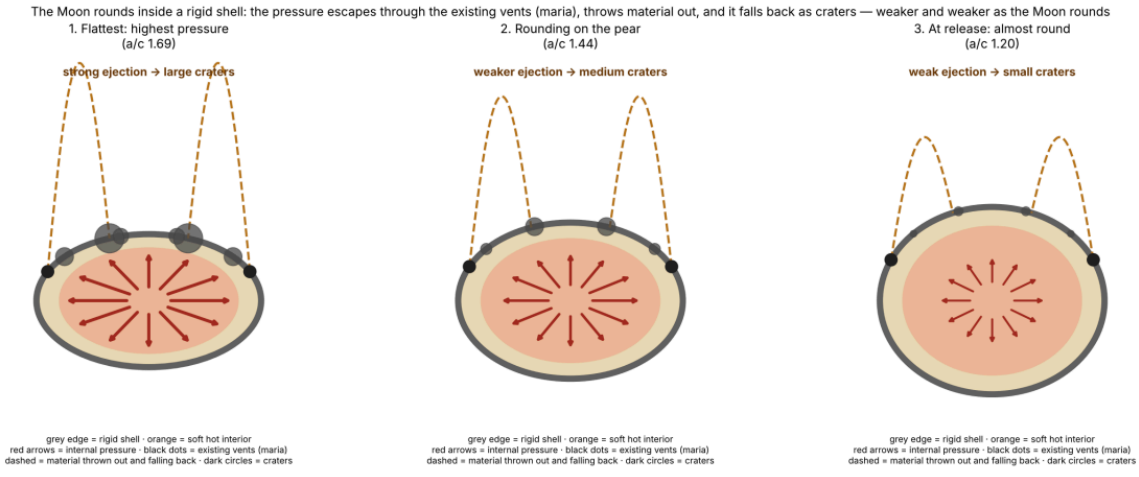


Figure 11. The Moon rounds inside a rigid shell. The pressure of the soft interior escapes through the existing openings (the maria), throwing material out, and the material falls back as craters — strong ejection and large craters at the most flattened stage, weaker as it rounds, weak late in rounding, near release. The panels carry the earlier labels "Rounding on the pear" and "At release: almost round" for these two rounding stages, and the compressed scale of an earlier version (*a*/*c* 1.69, 1.44, 1.20): they should be read as the peak shape (about 1.8 : 1.4 : 1), the rounding after the egg part was released (1.40), and late in rounding (1.20). Schematic.

C.5 Stretching in time: the dynamics of loads, the decay of flattening and the timeline of rounding

This section anchors the morphological evolution and the geodynamic forces of sections C.1–C.4 along a firm chronological timeline [C.5]. It defines, at each stage of the attached system, the strength of the Earth's stretching vectors, the mechanical mechanism that preserved the integrity of the body, and the exact level of triaxial deformation [C.5]. These values determine on which geometric shape each crater was imprinted, and from this is derived mathematically the level of residual polar excess density measured in it today (the geometric chronological clock) [C.5].

1. Dynamics of the tidal-load measure (k) against the stability threshold

The Earth's tidal forces act to stretch the volume of the Moon along the radial axis (axis a), while the Moon's self-gravity field tends to compact the material into a sphere [C.5]. The system stress-balance factor is computed along the sequence using the firm physical measure [C.5]:

$$\mathbf{k} = \frac{\mathbf{\Omega^2}}{\pi \mathbf{G} \rho} = \frac{\mathbf{M_E}}{\rho \mathbf{r^3}}$$

- **The physical mechanics of the measure:** the equation compares the Earth's external tidal pull at a given distance (M_E/r^3, represented as the square of the orbital angular velocity Ω^2) with the self-gravity field that a uniform body exerts on itself ($\pi G \rho$) [C.5]. The physical size of the body does not affect the value of the measure – it is set exclusively by the orbital distance (r) and the density of the material (ρ) [C.5]. The tide falls steeply as a function of $1/r^3$; a moderate recession reduces the load sharply, and the stiffening of the crust (the rise in density ρ) sharpens its decay [C.5].
- **The fluid stability threshold (the Roche limit):** a free fluid body (without mechanical strength and detached) can stabilise a closed shape only when the value of the measure firmly satisfies $k < 0.09$ [C.2, C.5]. Inside this Roche limit ($k \geq 0.09$), the tidal stresses tear the fluid apart [C.5]. By contrast, a rigid body that elastically resists deformation can hold together much closer to the source: the rigid Roche limit for light material ($1,300\ \mathrm{kg/m^3}$) is set at a distance of 12,994 km [C.1, C.5].

Matrix of the development of the k measure and the mechanical states through the attached stages:

The following is the exact numerical analysis of the mechanical stresses along the separation path, sorted chronologically from top to bottom [C.5]:

Dynamic stage in the evolutionary sequence	Orbital distance (r, km)	Computed value of the measure (k)	Density of material (ρ, $\mathrm{kg/m^3}$)	The preserving mechanical mechanism and the stability state of the crust
Stage 2: consolidation of the mass at the tip of the pear	8,991	2.01	1,300	No free equilibrium; the body is held and mechanically stabilised by the base of the pear [C.5].
Stage of the independent body (the rigid Roche limit)	12,994	0.67	1,300	No free equilibrium; the body is held by the base of the pear. This is the most extreme mechanical load since the body has held its own shape (k is 7 times higher than the fluid Roche threshold) [C.5].
Crossing of the corotation line by the far side	$\approx 14,100$	0.45	$\approx 1,500$	No free equilibrium; the body is in the midst of the dynamic flattening stage and is stabilised by the base [C.5].
Stage of approach to the line (3,000 km base)	$< 16,400$	< 0.25	$\approx 1,700$	No free equilibrium; the body reaches its structural peak flattening, held and stabilised at its base [C.5].
Stage of the centre of mass crossing (onset of separation)	16,400	0.17	2,600	No free equilibrium; the pressure beneath the base drops to zero, the body rounds (axis ratio ≤ 1.04) and is stabilised by its rigid shell [C.5].
Stage of orbital release at latitude 22.5°	17,287	0.14	2,600	No free equilibrium; final mechanical release from the Earth; the body is stabilised and held by its rigid shell [C.5].
The theoretical fluid stability threshold (Roche)	20,595	0.084	2,600	A free fluid equilibrium is reached ($k < 0.09$); the body can hold itself together for the first time without the help of a rigid crust (the fluid Roche limit lies at 19,971–20,100 km) [C.2, C.5].
The rejected critical branch of Paper 4	19,660	0.096	2,600	No free equilibrium; the distance lies within the fluid Roche limit for this density, and therefore the shape is mechanically rejected [C.2, C.5].

2. The geophysical mechanics of support by the base and the rigid crust

The numerical analysis reveals a sharp structural anomaly: through all the attached stages, the value of the tidal load (k ranging between 2.01 and 0.17) never reaches the free-fluid stability threshold (0.09) before the moment of release [C.5]. The young Moon moves outward and is released at a distance of about 16,400 km (beyond the equatorial corotation line), deep within the disruptive fluid Roche limit (which requires a minimum distance of about 20,000 km for this density) [C.2, C.5].

The survival of the structure is explained through a two-stage chain of mechanical support [C.5]:

1. The stage of support by the base of the pear (the flattening stages): the body is not torn or tidally disrupted at the rigid Roche limit ($12,994\ \mathrm{km}$), despite an extreme load seven times higher than the tearing threshold, because it sits and is fixed on a wide attachment base 6,000 km wide connected to the parent material [C.5]. The width of the base carries and balances the radial and tidal stress vectors that the self-gravity of the mass cannot bear on its own (Paper 3) [C.5]. As the body moves outward, the base narrows from 6,000 to 3,000 km, and the load decays gradually with arc distance [C.5].

2. The stage of support by the rigid crust (the release and rounding stage): when the body compacts and settles at a density of 2,600 $\mathrm{kg/m^3}$, the base of the pear narrows and the pressure on it drops to zero [C.5]. At this moment, the outer shell that has stiffened and frozen is the sole bearer of the residual stresses and stabilises the round body against tidal disruption [C.2, C.5]. The dynamic tidal calculation and the mechanical release equations of Paper 2 converge on the same physical constraint of a rigid crust at the moment of detachment [C.5].

3. Chronology of the four moments of geometric transition

The morphological evolution from the tip of the pear to the modern state is governed along four firm dynamic segments [C.5]:

- **Moment A (consolidation at the tip of the pear):** light sialic material ($1,300\ \mathrm{kg/m^3}$) concentrates at the tip of the pear structure, and begins its outward path and the development of initial flattening, while held on the 6,000 km base [C.5].
- **Moment B (structural peak flattening):** the mass takes shape as an independent attached body beyond the rigid Roche limit ($\approx 13,000\ \mathrm{km}$) [C.5]. The geometric flattening reaches its extreme peak as the base narrows to 3,000 km near the crossing of the corotation line (a predicted distance of $15,818\ \mathrm{km}$ at a period of 5.5 hours) [C.1, C.5]. The Moon is stretched between the two opposing pulls into a triaxial shape of $1.8 : 1.4 : 1$, in an inflated state with a material density of about 1,700 $\mathrm{kg/m^3}$ [C.1, C.5].
- **Moment C (attached inward rounding):** the centre of mass crosses the corotation line (the dynamic distance at which the synchronous rotational velocity equals exactly the free orbital velocity) [C.5]. The net compressive force drops to zero, the egg part (the tidal axis a) collapses inward and the body begins to round with a sharp decrease in volume [C.1, C.5]. While the spin period remains almost static (a moderate lengthening from 5.5 to 5.77 hours, which makes the corotation line migrate slightly from 15,818 to 16,399 km), the rise in core density to 2,600 $\mathrm{kg/m^3}$ increases the inward self-gravitational pull [C.5]. The same rotation can hold only a small and shrinking bulge; the axis ratio falls to 1.20, a figure indicating that about three quarters ($\approx 75\%$) of the original excess flattening disappeared and was absorbed into the crust while the body was attached [C.5]. Had the pear not been elongated and carried the mass up to this line, the lunar material would have collapsed back toward the Earth [C.5].
- **Moment D (spherical release and rigid recession):** at a distance of about 210 km beyond the corotation line, the Moon is released and finally detaches from the Earth in a round state (a maximum axis ratio of only 1.04) with a rigid shell [B.7, C.5]. After release, the sequence of

events changes its character and becomes small and carried out on a rigid body: the spin locks to the orbit and slows with recession (14 hours at a distance of 30,000 km, and 41 hours at a distance of 60,000 km) [C.5]. The residual elastic bulge raised on the rigid crust decays with it to values of 1.03 and 1.00 respectively, until the modern state is reached [C.5].

Analysis of the diameter curves and the chronological clock of pressure

Examining the measures derived from the diameter curves (Figure 13) yields the following scientific insights:

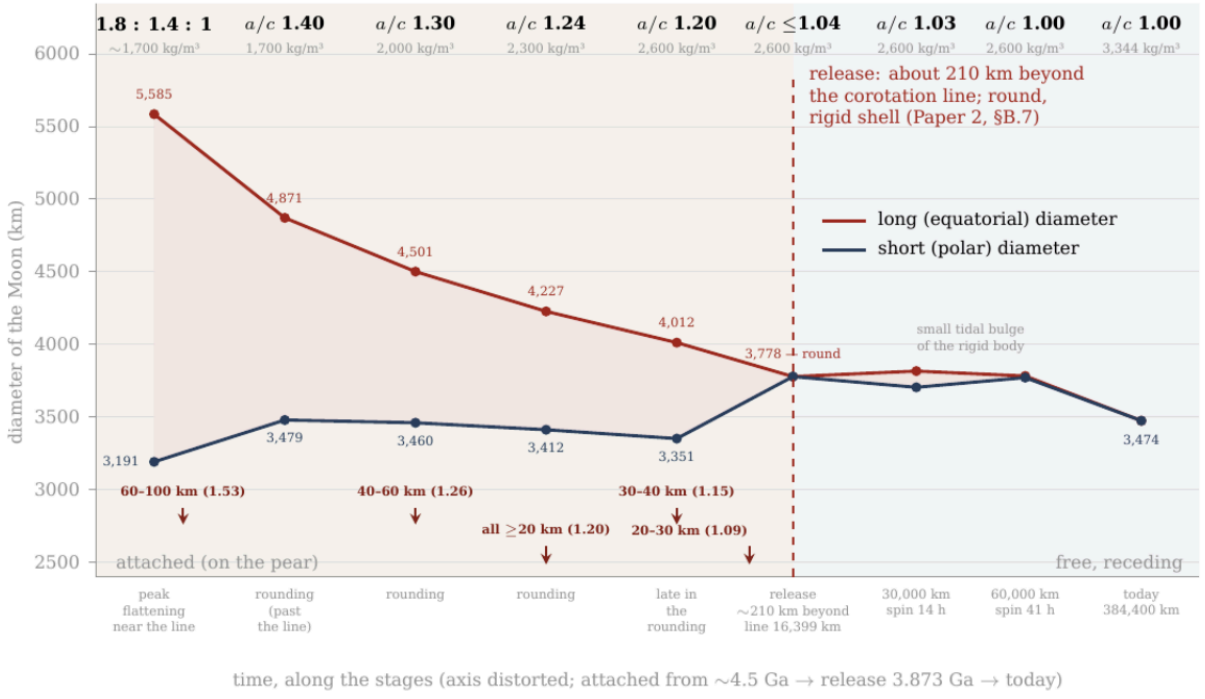


Figure 13. The long diameter and the short (polar) diameter of the Moon from peak flattening to today, against time (the axis is distorted, the early stages stretched). The flattening grew before the first point, as the body emerged on the tip of the pear and moved outward; it is not plotted. The gap between the two curves is the flattening: widest at its peak, as the body approached the 3,000 km base near the corotation line (1.8 : 1.4 : 1, at about 1,700 kg/m³; the long diameter is the Earth line, and the intermediate one, along the orbit, 4,628 km, is not plotted), closing during rounding after the centre of mass passed the line — first as the egg part is released (1.40), then with the thickening — down to about 1.20, and closed at the moment of release, about 210 km beyond the corotation line (the line at 16,399 km at the equator, 17,287 km at 22.5°), when the body is round within its rigid shell (*a*/*c* ≤ 1.04). After release only the small tidal bulge of the rigid body remains (1.03 at 30,000 km, a rotation of 14 hours; 1.00 at 60,000 km, a rotation of 41 hours). The arrows mark the crater sizes whose clean polar crowding (D.2) points to each stage: the large ones near the peak, the small ones late in rounding, before release. The diameters are computed from the volume of the Moon's mass at the sequence density assigned to each stage and from the shape of the stage (C.1; at the peak, 1.75 : 1.45 : 1); both curves descend because the volume decreases as the stretching is released — the equivalent diameter falls from about 4,350 km at the peak to 3,778 km at release; the ratio between the pole and the equator, which the paper tests, does not depend on the overall size. (An earlier version of the figure drew the body still flattened, *a*/*c* 1.20, at the moment of release.) (In the graph: vertical axis — the Moon's diameter in km; horizontal axis — the stages: peak flattening near the line, the rounding stages beyond the line, late in rounding, release about 210 km

beyond the line, recession and today; red — the long (equatorial) diameter, dark blue — the polar diameter; at the peak 5,585 and 3,191 km, at 1.40 4,871 and 3,479 km.)

1. Decoding the original gap between the curves: the distance and spatial gap between the red curve (the long equatorial diameter along the a axis) and the blue curve (the short polar diameter along the c axis) represents precisely the value of the geometric flattening [C.5]. This gap reaches its system maximum at the point of peak flattening (diameter values of $5,585\ \mathrm{km}$ lengthwise, versus $3,191\ \mathrm{km}$ across the pole; the intermediate axis b of $4,628\ \mathrm{km}$ is not plotted) [C.3.1, C.5]. The curves close and converge rapidly through the rounding stages: values of 4,871 km versus 3,479 km at the 1.40 stage, until equality and complete closure are reached at a uniform spherical diameter of 3,778 km at the moment of mechanical release [C.5].

2. Combined fall of the baseline (volume collapse): both curves show an overall downward trend along the sequence [C.5]. This trend stems from the fact that the volume of the body shrank drastically and the equivalent radius decreased from about 2,180 km at the stretched, voluminous peak to 1,890 km at the dense, rigid release [C.1, C.5]. The morphometric ratio between the pole and the equator is robust and unaffected by the overall size of the body [C.5].

3. The epiphany of the crater-size clock: the dynamic arrows linking the empirical diameter groups to the stages of the curve (D.2) represent the law of internal hydrostatic pressure in the vents [C.4, C.5]. The giant craters formed at the most flattened moment, when the pressure in the mare vents was at its extreme peak, and therefore they went through the mechanical maximum of surface contraction during rounding and today record the maximum density excess at the poles [C.4, C.5]. Small craters were imprinted at later stages when the pressure in the vents had already decayed, underwent less residual surface contraction, and therefore show a moderate excess on the surface [C.4, C.5]. The whole array falls precisely within the bounds of the predicted range of the balanced clean count (1.20 ± 0.05) [C.5].

1. The dynamics of differential crowding: why are the large craters the most crowded?

The degree of mechanical crowding of a crater on the crust is not set by its absolute chronological age, but is a geometric record of the body's shape and dynamic flattening at the moment of impact and crater formation [B.7]. The craters kept their original diameter and were carried as intact, rigid elements on the crust; what contracted and closed elastically was the surface and the seam spaces between them (C.7, D.5) [C.3]. The geometric mechanics are defined by the equation of residual rounding [C.3]:

- **Formation at peak flattening:** craters that formed when the Moon was at the peak of triaxial deformation ($1.8 : 1.4 : 1$) underwent the full sequence of inward rounding and surface reduction [B.7, C.3]. As a result, they went through the mechanical maximum of surface contraction, and today show a predicted density ratio of about 1.50 in favour of the polar caps.
- **Formation in middle and late rounding stages:** craters that formed when the Moon had already begun to collapse inward and had rounded to a shape of 1.20 (late in rounding) underwent less residual surface contraction, and therefore their predicted density ratio drops to about 1.17.

- **Post-release formation (during recession):** craters that formed on a body that had already completed the transition to a spherical shape and had left the corotation line undergo no surface contraction at all, and therefore their predicted ratio is 1.00 (a completely uniform distribution).

The vent-pressure mechanism and the crater as a clock of shape:

The causal link between crater diameter and its time of formation is governed by the internal pressure profile in the mare vents [C.4]:

1. The moment of peak flattening: represented the moment of maximum internal hydrostatic pressure, arising from the extreme morphological mismatch [C.4]. The pressure pushed and threw out enormous quantities of returned material at maximum force (Paper 6), and therefore most of the largest craters in the system formed at this moment [C.4].

2. The rounding stages: as the body converged inward, the mismatch decreased, and the pressure in the vents decayed continuously, thus producing craters of progressively smaller diameters [C.4].

Material was ejected and landed throughout all the stages of the transition (both on the way up to the peak and on the way down from it), and therefore the enhanced density ratio of the craters records precisely how flattened the body was at the moment of their imprinting [C.3]. Crater size acts as a firm geometric clock of the shape: the larger one formed closer to peak flattening, absorbed more rounding, and ends up more crowded at the poles [C.3, C.4]. This mechanical prediction is confirmed precisely in the clean count (D.2), in which the density ratio climbs from 1.09 in the smallest size class (20–30 km) to 1.53 in the class of giant craters of 60–100 km.

2. The three-stage evolution of the polar regions

The geographic polar regions are older than the formation of the flattening, and their dynamic evolution unfolds through three firm structural stages:

[Stage 1: outward rounding] ⟶ [Stage 2: the flattening stage] ⟶ [Stage 3: inward rounding]

(setting of the poles and the rotation axis) (growth of the deformation, creation of the large craters) (surface contraction, crowding of the craters)

- **Stage 1 — outward rounding (fixing the axis):** the original sialic mass, which sat on the tip of the Earth's pear, collapsed inward and became an independent fluid sphere (Paper 3 §B.4). At this stage the geographic rotation axis of the body was set; the craters existing on the shell then changed with the surface equally everywhere (without relative deformation), and new craters continued to accumulate [C.3].
- **Stage 2 — the flattening stage (growth of the deformation):** as the body moved outward toward the corotation line, the base narrowed from 6,000 to 3,000 km, and the centrifugal and tidal stresses stretched and flattened the structure into an extreme triaxial shape [C.1, C.5]. At this stage the differentiation of material progressed (the ejected material was light and full of

water), the pressure profile in the vents soared to its peak, and the largest giant craters were imprinted on the surface in a uniform distribution per unit area [C.4, C.5].

- **Stage 3 — inward rounding (convergence and crowding):** the centre of mass crosses the corotation line and the flattening relaxes [C.1, C.5]. The surface undergoes a differential collapse: it closes and contracts at the maximum rate at the poles (a factor of 0.55×) and at the minimum rate at the equator (a factor of 0.97×) [C.3.1]. The craters (which kept their original diameter) gather and crowd correspondingly toward the poles, since all the meridians retreat toward the two end points of the rotation axis – the only points that the rotation never moved in space (C.3) [C.3.1].

3. The chronological birth of the month and the orbit of the young Moon

At the moment of release and final mechanical detachment from the Earth, the Moon's spin period and orbital period coincide near a value of slightly more than 5.77 hours:

- **The birth coordinates:** the Moon begins the process of separation and rounding on the corotation line [C.5]. On this line, its orbital period and the Earth's spin take exactly the same time – a value of 6.25 hours at an orbital distance of 17,287 km (for the latitudes of 22.5°) [C.5].
- **The mechanical release:** takes place about 210 km beyond the equatorial corotation line ($16,400\ \mathrm{km}$), as the Moon is thrown into an independent orbit with a moderate initial eccentricity (ellipticity) of about 0.037 (Paper 4) [C.5].
- **The path of orbital lengthening (Kepler's law):** from the moment of detachment onward, the length of the month (the orbital period T) grows and lengthens as a function of the orbital distance (r) according to Kepler's third law alone ($T \propto r^{3/2}$), without the need for further assumptions or free parameters [C.5]:

$$\mathbf{T = 2\pi\sqrt{\frac{r^3}{G(M_{\oplus}+m)}}}$$

The dynamic curve (Figure 14) begins with a completely flat profile (a short month of a few hours near the Earth), and climbs sharply only far out: 14 hours at a distance of 30,000 km $\rightarrow$ 3.7 days at a distance of 16 Earth radii $\rightarrow$ until it settles at 27.3 days in its modern state today [C.5]. Once the mechanical release point has been set, the length of the month at every distance follows directly from Newton's law of gravitation [C.5].

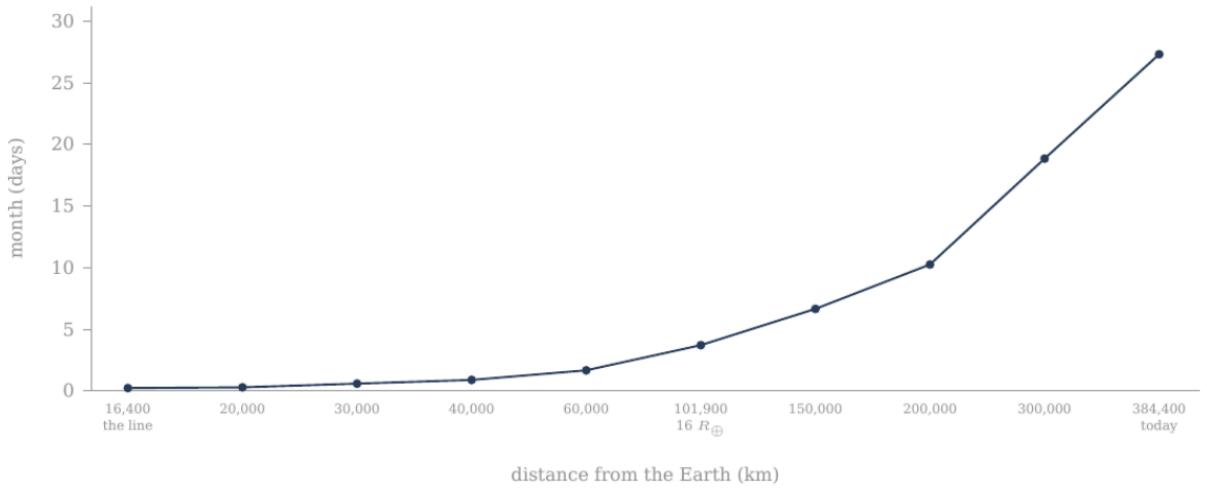


Figure 14. Where the month was born: the orbital period of the Moon — the month — against its distance from the Earth, from 5.77 hours at the moment of release (equal to the Earth's day at that time) to 27.3 days today, according to the law $\mu = G(M\oplus + m)$,$T = 2\pi\sqrt{r^3/\mu}$) Kepler, where r is the distance between the centres, $M\oplus$ the mass of the Earth and m the mass of the Moon). The curve starts almost flat — a month of hours — and rises steeply only far out. The curve starts at the corotation line, 16,400 km, and the release lies about 210 km beyond it. (In the graph: vertical axis — the month in days; horizontal axis — the distance from the Earth in km, from 16,400 (the line) through $R_\oplus$ 16) 101,900) to 384,400 (today).)

4. Chronology of the attached rounding stage and confirmation of the rigid sphere

The process of inward rounding proceeded as a long, graduated event through the attached stage, and is not pressed for time; the rate of inward collapse was dictated by the release of the mechanical compression on the base and by the thickening of the core, and not as a function of the relaxation of the crustal materials (C.7) [C.5].

The chronological timetable of the model (the 600-million-year window):

The geodynamic model aligns the overall time span of about 600 million years from the start of the bulge to the final release (dated to 3.873 billion years before present, that is about 4.5 billion years before today) [C.5]:

- **Segment A (bulge and pear):** about 400 million years for the uplift of the original mass on the pear material of the Earth [C.5].
- **Segment B (dynamic flattening):** about 100 million years of extreme triaxial deformation near the corotation line [C.5].
- **Segment C (inward rounding):** about 100 million years from the crossing of the corotation line (the start of the surface collapse) to the final mechanical detachment [C.5]. This is the exclusive morphogenetic window in which the craters of the paper were imprinted and preserved [C.5].

When was the spherical structure completed?

- **In the late rounding stages:** the axis ratio stood at about 1.20, a figure indicating that about 75% of the original excess flattening had already been absorbed and closed within the crust

while the body was attached [C.5]. Had the pear structure not lengthened so as to carry the mass beyond the line, the material would have collapsed back to the Earth [C.5].

The Moon detached from the Earth completely round (an axis ratio of at most 1.04), with its outer shell rigid enough to absorb the residual rotational stresses (Paper 2 §B.7) [C.5].

- **Post-release (decay of the rigid bulge):** everything that took place after detachment is a small residual effect raised elastically on the rigid body, and must not be confused with the original flattening of the attached stage [C.5]. The locked Moon followed its orbital period (14 hours at 30,000 km, and 41 hours at 60,000 km), and with it the residual dynamic bulge shrank from 1.03 to 1.00 [C.5].

At a distance of about 60,000 km (less than 10 Earth radii, and deep within the 16-radius limit at which the shape freezes according to Paper 4), the dynamic bulge disappeared completely and the Moon became a perfect sphere by every measure [C.5]. The tiny fossil bulge preserved in it today ($f \approx 0.001$, compared with a peak deformation of about 0.44) is unequivocal geophysical proof: the extreme flattening of the attached stage disappeared entirely, leaving only a small remnant, 400 times smaller than its original state [C.5].

The combined database at the end of Part C

The geodynamic model has completed the construction of the theoretical and mechanistic foundation, and fixes the following core measures [C.5]:

> $\mathbf{k_{max}} = \mathbf{0.67}$ — peak mechanical load at the rigid Roche limit (12,994 km), stabilised on the 6,000 km base
>
> $\mathbf{a : b : c \approx 1.8 : 1.4 : 1}$ — the geometry of peak flattening on the 3,000 km base, material density 1,700 kg/m³
>
> $\mathbf{\Delta}\mathrm{Volume} \approx \mathbf{1.53}$ — a global volume collapse that drives the potential slope profile inward toward the poles
>
> $\mathbf{T_{release}} \approx \mathbf{5.77}\ \mathrm{hours}$ — the birth of the month in a rigid rounded orbit (ratio ≤ 1.04), about 210 km beyond the line

C.6 The wide base: the mechanical support mechanism, the dynamics of accelerations and the point of detachment

This section decodes the mechanical and geodynamic balance of the lunar interior in the attached stage, and explains how the body was protected from tidal tearing [C.6]. As established in section C.5, the external tidal load on the body consistently exceeded the hydrostatic stability limit of a free fluid ($k \geq 0.09$) [C.5, C.6]. The following analysis quantifies what carried this energy difference along the separation path, and why the mechanical support lapsed at the moment of release [C.6].

a. The global balance: the structural role of the base width

During the stages of growth and flattening, the young Moon did not act as a free body, but was physically attached and anchored to the parent material [C.6]. This attachment took place on a

very wide crustal base, which was physically required to absorb and carry the excess tidal stresses that the self-gravity field of the mass could not balance on its own [C.6].

The mechanics are integrated along the sequence of detachment stages [C.6]:

- The sector of a body of its own ($12,994$ km): the moment the mass establishes an independent body configuration at the rigid Roche limit, the load measure climbs to the extreme value of $k = 0.67$ (seven times above the fluid Roche threshold of 0.09) [C.5, C.6]. The body was saved from complete tearing thanks to its anchoring on a rigid base 6,000 km wide [C.6]. The dynamic structure is like a drop of viscous fluid hanging from the tip of a syringe: the drop keeps its morphological integrity under the load of the external gravity field as long as the mechanical opening holds and stabilises it [C.6]. This wide base was required simultaneously both for the balance of radial pressures and resistances of Paper 3, and against the crust-tearing tidal stresses [C.6].
- **The stage of base narrowing and recession:** as the mass underwent rotational flattening and was pushed outward on the tip of the pear, the width of the holding base narrowed from 6,000 km to 3,000 km [C.1, C.6]. Despite the structural narrowing, the body remained stable because the tidal factor fell sharply with orbital distance as an extreme ratio of $1/r^3$ [C.5, C.6].

Matrix of the development of the tidal load (k) and the width of the holding base:

The following are the exact numerical values governing the detachment stages of the structure, incorporated in Figure 15 [C.6]:

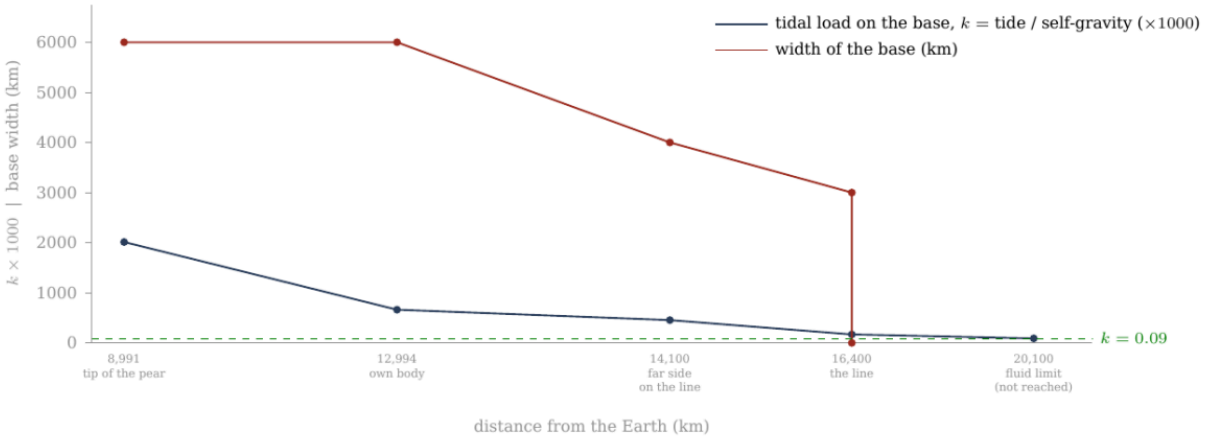


Figure 15. The wide base: the tidal load it carried (*k*, in dark blue, multiplied by 1000) and its width (in red) against distance from the Earth. The load is greatest when the body holds its own shape (12,994 km) and decreases with distance; the base narrows from 6,000 to 3,000 km as the body flattens and moves outward, and lets go at the moment of release: the separation begins when the Moon's centre reaches the corotation line (16,400 km), and the body is released about 210 km beyond it. The fluid threshold *k* = 0.09 (dashed) is reached only near 20,100 km (an earlier version gave 20,600), beyond release: the released body was held by its rigid shell. End points from Paper 3 (a base of 6,000 km on the tip of the pear, 3,000 km at the last contact) and from Paper 2 (density 2,600 at the moment of release); the points between them by interpolation. (In the graph: horizontal axis — the distance from the Earth in km: 8,991 tip of the pear, 12,994 body of its own, 14,100 far side on the line, 16,400 the line, 20,100 the fluid limit (not reached); vertical axis — *k*×1000 | base width in km.)

Orbital distance (r, km)	Dynamic stage in the evolutionary sequence	Density of the material (ρ, $\mathrm{kg/m^3}$)	Tidal-load measure (k) [C.5]	Physical width of the holding base (km)
8,991	Mass gathered at the tip of the pear	1,300	2.01	6,000 [A.11]
12,994	A body of its own, flattened (the rigid Roche limit)	1,300	System peak load — 0.67	6,000
$\approx 14,100$	The far side crosses the corotation line	$\approx 1,500$	0.45	$\approx 4,000$ (by interpolation)
$< 16,400$	Approaching the line (3,000 km base)	$\approx 1,700$	< 0.25	3,000
16,400	Centre of mass on the corotation line (rounding)	2,600	0.17	$0 \to 3,000$ (lets go at the moment of release)
$\approx 20,100$	The theoretical fluid limit after release	2,600	0.09	— (stabilised by a rigid crust)

The moment the centre of mass reaches the corotation line ($16,400\,\mathrm{km}$), the base lets go completely [C.6]. The detached body is released about 210 km beyond the line, deep within the fluid limit (which is reached only at 20,100 km), with the outer shell, stiffened to $\mu \geq 10\,\mathrm{GPa}$, carrying the load of the residual stresses alone from then on [C.5, C.6].

b. Dynamic analysis of the profile of accelerations and self-gravity

To decode the physics of the two-body balance in isolation, the static measures can be translated into a system of physical accelerations [C.6]. The equilibrium equation is governed by two opposite vectors that change their values along the timeline [C.6]:

1. The self-gravity vector at the surface (g): undergoes a consistent rise and climb [C.6]. Since the mass of the Moon is constant, the compaction of the core and the decrease in volume (the narrowing of the radius R) bring the surface materials closer to the centre, and sharply increase the inward pull: from 0.87 $\mathrm{m/s^2}$ in the initial pear stages, to 0.95 $\mathrm{m/s^2}$ in the middle stages, up to 1.37 $\mathrm{m/s^2}$ at the moment of spherical release ($1.62\ \mathrm{m/s^2}$ modern) [C.6].

2. The tidal acceleration vector across the body (a_{tide}): represents the pure mechanical difference between the Earth's pull on the nearest surface and its pull on the Moon's centre of mass ($a_{\mathrm{tide}} = \frac{2GM_E R}{r^3}$) [C.6]. This value undergoes a sharp fall and decay along the recession path [C.6].

Summary of the profile of accelerations and the direct ratio between them:

The following are the quantitative measures reflecting the balance of accelerations, confirmed graphically in Figure 16 [C.6]:

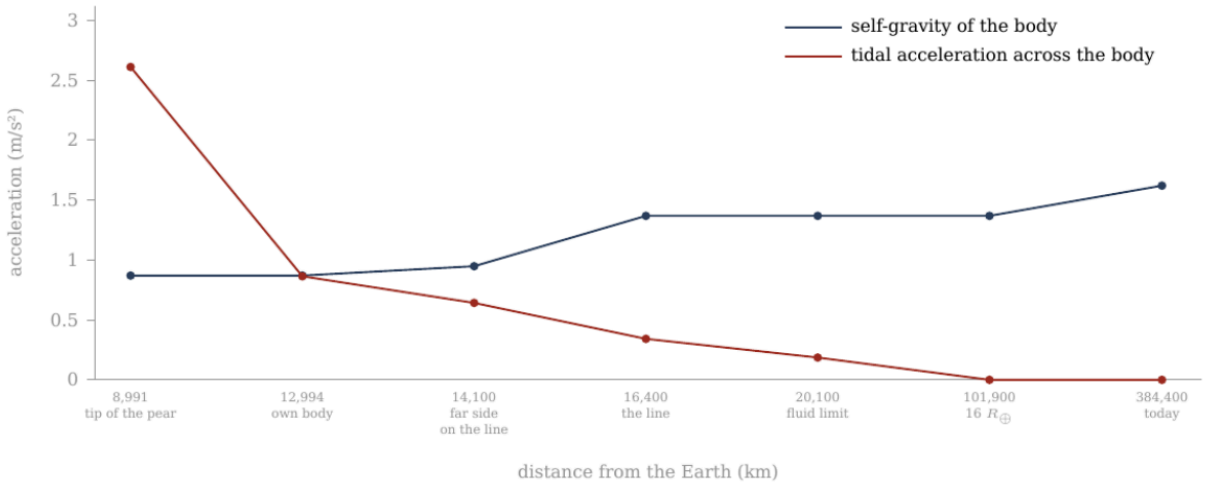


Figure 16. Self-gravity versus the tide, from the mass at the tip of the pear to today. The gravity of the body itself (in dark blue) rises as it compacts at constant mass — 0.87 m/s² as a foam, 1.37 at 2,600 kg/m³ at the moment of release, 1.62 today at 3,344; the tidal acceleration across it (in red) falls with distance, from three times the self-gravity at the tip of the pear to a quarter of it at the moment of release (about 210 km beyond the line at 16,400 km), and below the fluid threshold (0.135, the threshold *k* = 0.09 of C.5 expressed as this ratio) only near 20,100 km, beyond release (an earlier version gave 20,600). The two curves cross at the rigid Roche limit, where the body first holds its own shape: 1,300 kg/m³, 12,994 km (10,314 km at 2,600, Paper 2). The densities between 1,300 and 2,600 are stages in the sequence. (In the graph: vertical axis — acceleration in m/s²; horizontal axis — the distance from the Earth in km, from 8,991 (tip of the pear) through 12,994 (body of its own), 16,400 (the line) and 20,100 (the fluid limit) to 384,400 (today).)

Dynamic sector in the sequence	**Orbital distance (r, km)**	**Density (ρ)**	**Radius of the body (R, km)**	**Inward self-gravity (g, $\mathrm{m/s^2}$)**	**Tidal acceleration across the body (a_{tide}, $\mathrm{m/s^2}$)**	**Direct ratio of accelerations (a_{tide}/g)**
The early tip of the pear	8,991	1,300	2,381	0.87	2.61	3.0 (the tide leads by a factor of 3)
The rigid Roche limit	12,994	1,300	2,381	0.87	0.87	1.0 (the mechanical crossing point)
The far side on the line	14,100	1,500	2,269	0.95	0.64	0.67
The moment of release (detachment)	16,400	2,600	1,889	1.37	0.34	0.25 (gravity leads by a factor of 4)
Middle sector ($16R_E$)	101,900	2,600	1,889	1.37	0.001	0.001 (the tide is negligible)
The modern state today	384,400	3,344	1,738	1.62	0.00002	0

The two acceleration curves (Figure 16) cross and lock exactly at the rigid Roche limit ($12,994\ \mathrm{km}$), where the body first stabilises an independent shape and the tidal acceleration across it equals exactly its self-gravity (a ratio value of 1.0) [C.6]. Before the crossing point the external tide leads and tries to break the body apart; after the crossing, the Moon's self-gravity prevails and dominates the system, and from a distance of 16 Earth radii onward the tidal component becomes statistically negligible [C.6].

c. The methodological clarification: differences in the definition of the measures

To maintain firm mathematical consistency, the study clarifies the numerical difference between the two tidal measures used in sections C.5 and C.6, and proves that it is a difference of analytical definition and not a physical difference [C.6]:

1. The theoretical load measure (k): defined in section C.5 by $k = \frac{M_E}{\pi \rho r^3}$ [C.5, C.6]. The classical fluid Roche threshold, equal firmly to 0.09, is calibrated and formulated relative to this measure [C.5, C.6].

2. The direct ratio of accelerations: defined in the last column of this section as the direct division of the equation of tidal acceleration across the body ($\frac{2GM_ER}{r^3}$) by the equation of surface self-gravity ($\frac{4}{3}\pi G\rho R$) [C.6].

When the mathematical reduction is carried out, the radius of the body (R) cancels out completely, and the equation yields the reduced ratio value [C.6]:

$$\mathbf{Acceleration_Ratio}\mathbf{=}\frac{\mathbf{3}}{\mathbf{2}}\left[\frac{\mathbf{M}_{\mathbf{E}}}{\mathbf{\pi \rho r}^{\mathbf{3}}}\right]\mathbf{=1.5\times k}$$

A mathematical rule is established: the direct ratio of accelerations is a constant 1.5 times the load measure k of section C.5 [C.6].

This constant dependence dictates that the two systems move in full synchronisation through all the stations of the path [C.6]:

- A load value of $k = 0.67$ at the rigid Roche limit coincides exactly with an acceleration ratio of 1.0 [C.6].
- A load value of $k = 2.01$ at the tip of the pear coincides exactly with an acceleration ratio of 3.0 [C.6].
- A load value of $k = 0.17$ at the moment of release coincides exactly with an acceleration ratio of 0.25 [C.6].
- The theoretical fluid Roche threshold of $k = 0.09$ translates exactly into an acceleration ratio of about 0.135 [C.6].

This consistency (plotted in the graphs of Figures 15 and 16) proves full mathematical closure of the geodynamic system [C.6].

C.6.1 Chronology of the local effective gravity and the dynamic mechanics of detachment

Whereas the first part of section C.6 discussed the global balance of accelerations of the body as a whole, this part breaks down the data to local resolution [C.6]. The effective gravity – the actual force experienced by a free grain lying on the surface (the combination of inward self-gravity, tidal stretching, and outward centrifugal throw) – is not a uniform monolithic value, but varies drastically with the geographic coordinate at which the grain is located [C.6]:

1. In the polar regions (the rotation axis): expressed as $g + \omega^2 R$ (where ω is the angular velocity of rotation and R the radius of the body) [C.6]. Perpendicular to the tidal axis, the tidal forces physically compress and press the body inward, adding positively to the pull of self-gravity [C.6].

2. In the side regions (the axis of orbital motion): equal exactly to the value of pure self-gravity (g), since in this sector the centrifugal component fully offsets and neutralises the tidal factor [C.6].

3. At both ends of the tidal axis (the sub-Earth and anti-Earth points): expressed as $g - 3\omega^2 R$ [C.6]. At these coordinates, the external tidal stretching and the centrifugal throw of the rotation act simultaneously and jointly in the direction opposite to the body's self-gravitational pull [C.6].

Matrix of the spatial distribution of the local effective gravity:

The following are the exact numerical values (in $\mathrm{m/s^2}$) showing how the local stresses converged along the separation path [C.6]:

Dynamic stage in the evolutionary sequence	Orbital distance (r, km)	Polar apex ($g + \omega^2 R$)	Sides of the motion axis (g)	Ends of the tidal axis ($g - 3\omega^2 R$)
A body of its own (the rigid Roche limit)	12,994	1.30	0.87	0.43− (a firm negative value)
The far side crossing the line	14,100	1.27	0.95	0.01− (balance near zero)
The moment of release (mechanical detachment)	16,400	1.54	1.37	0.86
The modern state today	384,400	1.62	1.62	1.62 (homogeneous uniformity)

- **A cross-reading of the change in stresses:** the data show an extreme spread at the moment the body first stabilises an independent shape (12,994 km) [C.6]. The clearly negative value at the ends of the tidal axis ($0.43 - \ \mathrm{m/s^2}$) is the core of the analysis: were it not for the hold of the wide mechanical base 6,000 km wide, the crustal materials at the sub-Earth and anti-Earth points would have been torn off, lifted and thrown away from the body [C.6]. This is the physical reason why the wide support base developed and was needed precisely along the tidal line, while the polar regions were pressed inward with a force 1.5 times greater than the sides [C.6].

- **Convergence after crossing the line:** as the body recedes, the spread steadily narrows [C.6]. At the moment of release the distribution balances (1.54 at the poles versus 0.86 at the ends), every point is held firmly by its own self-gravity, and the negative values disappear [C.6]. In the modern state, the gravity field settles at full uniformity of 1.62 $\mathrm{m/s^2}$ at every point on the surface [C.6].

C.6.2 The mechanics of the collapse of the lower pressure on the corotation line

The dynamic transition from a wide base to complete detachment is governed by the simultaneous intersection of three physical constraints on the corotation line [C.6]:

1. Balance of velocities (inside the line): inside the corotation radius, a body rotating in synchronisation with the Earth moves at a linear velocity lower than that of a free Keplerian orbit [C.6]. As a result, the Earth's gravity pulls the mass inward with a force exceeding the outward centrifugal throw [C.6]. The body is pressed and firmly fixed onto its base, and the base presses back from below (this is the radial balance mechanism of Paper 3) [C.6].

2. The zero point (on the line): at the corotation point, the two force vectors (gravity inward and the throw outward) are exactly equal [C.6]. The vertical compressive force on the base falls mathematically to absolute zero [C.6].

3. The reversed pull vector (beyond the line): on crossing the line and receding beyond it, the velocity of the body is higher than the free orbital velocity, and the compression turns into an active pull directed outward, away from the base [C.6]. The tidal load decays to $k = 0.17$ – a load fully handled by the stiffened shell, while the contact area has narrowed to 3,000 km [C.5, C.6].

Since the base is no longer required to carry a compressive load, the Moon is thrown off and mechanically released about 210 km beyond the line, when about 58% of its mass has already crossed the boundary (Paper 4) [C.6]. With the detachment, the Earth's hydrostatic pressure flow that passed through the pear to the Moon stopped completely, thus closing the chapter of the powerful magmatic ejections [C.6].

C.6.3 Analysis of the compression equation against the residual tidal stretching

To quantify the mechanical engine that led to the release of the egg part (the tidal axis a), the following table tracks the net pull at the centre (inward gravity minus the outward centrifugal acceleration, measured at the Moon's centre of mass) along the last 1,000 kilometres of the path before the line and up to the moment of release [C.6]:

Base_Load\ (GPa)}=\frac{\text{Net_Attraction}\times \text{Mass}_{\text{Moon}}}{\text{Area}_{\text{Base}}}

For a fixed base 3,000 km wide (surface area of $7.07 \times 10^{12}\ \mathrm{m^2}$), the gradient of compression against the tidal stretching across the surface ($(2\mu + \Omega^2)R$) and the secondary load it exerts across the mid-plane are computed [C.6]:

Position of the body's centre relative to the line (km)	Net pull at the centre ($\mathrm{m/s^2}$)	Mechanical load on the base (GPa)	Stretching at the surface ($\mathrm{m/s^2}$)	Stretching load in the mid-plane (GPa)	Net pull (near / far end, $\mathrm{m/s^2}$)
1,000−	0.293	3.04 (positive compression)	0.585	1.14	0.23− / 0.97
750−	0.216	2.24	0.566	1.10	0.29− / 0.87
500−	0.142	1.47	0.548	1.07	0.35− / 0.77
250−	0.070	0.72	0.530	1.03	0.41− / 0.68
100−	0.028	0.29	0.521	1.01	0.44− / 0.62
The line (0)	0	0 (perfect balance)	0.514	1.00	0.47− / 0.59
100+	0.027−	0.28−	0.508	0.99	0.49− / 0.55
Release (+210)	0.057−	0.59− (outward pull)	0.502	0.98	0.51− / 0.52

- **Decoding the collapse of the compressive load (the blue/red cells):** the numerical analysis shows an extreme mechanical reversal [C.6]. The vertical compressive load on the base collapses sharply from $3.04\ \mathrm{GPa}$ at a distance of 1,000 km before the line, lands exactly at $0\ \mathrm{GPa}$ on the line, and reverses into an outward tensile stress of $0.59\ \mathrm{GPa}$ at the moment of release – a figure defining a sharp delta and mechanical transition of $3.63\ \mathrm{GPa}$ [C.6].
- **The static nature of the tidal stretching vector:** unlike the collapse of compression, the residual tidal stretching load across the mid-plane hardly changes along the whole segment, showing a negligible fluctuation of only $0.16\ \mathrm{GPa}$ (a factor of 22 in favour of the compression) [C.6]. The stretching remains stable (a change of 14% before the line and only 2.5% from the line to release) [C.6].

The scale for comparing the stresses in the system:

To understand the strength of the geodynamic forces acting in the crust, the data can be set against the mechanical limit values of the body [C.6]:

- **Self-cohesion energy:** the self-gravity field binds the two halves of the Moon with a strength of $2.7\ \mathrm{GPa}$ over the cross-sectional area of the base [C.6].
- **Strength of the crustal materials:** the mechanical strength limit of a solid rocky crust stands at about $0.1\ \mathrm{GPa}$ [C.4, C.6].
- **Peak-flattening load:** maintaining and sustaining the triaxial peak shape ($1.75 : 1.45 : 1$) away from a spherical structure requires a shear stress of the order of $\rho g(a - c)$, equal to $2.0\ \mathrm{GPa}$ at a density of 1,300, and rising to $3.7\ \mathrm{GPa}$ at the release density of 2,600 [C.6].

The geophysical conclusion:

The quantitative cross-check proves that within the bounds of the line, the vertical compression of the Earth produces exactly the order of magnitude of stress required ($\approx 3\ \mathrm{GPa}$) to maintain the integrity and deformation of the flattened shape [C.6]. At the moment of passing beyond the line, the collapse of the pressure from below and its reversal into an outward pull is what provides the full mechanical and energetic release for the inward collapse of the egg part of the shape, while the static stretching vector does not change and cannot bring about this release [C.6].

Since the compressive load is completely symmetric about the Earth line, it governed and preserved the egg part (the tidal axis a), while the release of the pita part (the rotation axis b/c) was driven exclusively by the rise in density and the compaction of the core (C.4) [C.6]. The tidal stretching that remained beyond the line exceeds the fluid stability limit up to a distance of 20,100 km, and the Moon keeps its integrity and stabilises a rigid spherical shape ($a/c \leq 1.04$) thanks to its strong elastic crust [C.2, C.6].

C.6.4 The plug-extraction mechanism and the end of the pear's activity

The geodynamic model establishes a direct causal link between the mechanical detachment of the Moon and the halt in the evolution of the terrestrial pear structure [C.6]:

- **The crustal plug effect:** throughout all the attached stages, the mass of the young Moon sat directly on the apex of the tip of the pear and functioned physically as a tight plug hermetically sealing a high-pressure bottle [C.6]. The presence of the plug blocked the main ejection vents of the pear, and forced the internal pressure to feed, stretch and lengthen the pear material further and further outward [C.6].
- **Release of the hydrostatic pressure:** at the moment of mechanical detachment and the residual recession of the Moon beyond the line, the plug was pulled out [C.6]. The key openings of the pear were exposed directly to space, and all the enormous internal hydrostatic pressure that had been stored within the pear was released and burst out in a sharp global ejection [C.6].

With the disappearance of the driving internal pressure, the vector that mechanically lengthened the pear structure stopped completely, and this explains geophysically three core puzzles:

> **1.** Why the pear structure retreated and shrank back immediately after detachment, instead of continuing to stretch
>
> **2.** Why the system was arrested and did not eject or release a second moon or further residual masses into orbit
>
> **3.** Why the ejecta were thrown and scattered in all directions, imprinting the global crater record

The Earth released its pressure at the moment the plug was extracted, and with that the mechanism was concluded [C.6].

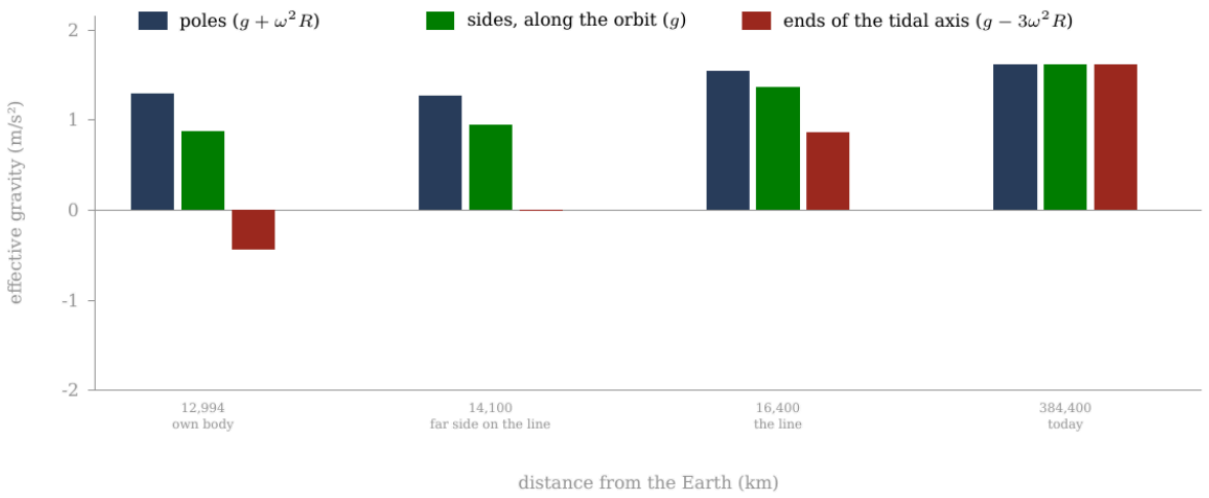


Figure 17. The effective gravity by location on the body rotating with the Earth — poles, sides and the two ends of the tidal axis — from the moment the body first holds its own shape to today. At 12,994 km the ends are negative (the red column hangs below the zero line): material there would have been pulled outward were it not for the base; the poles carry one and a half times as much. Where the far side crosses the line (about 14,100 km, at a stage density of about 1,500 kg/m³) the ends stand near zero. By release the spread has narrowed, and today, at 384,400 km, it has disappeared: 1.62 m/s² everywhere. (In the graph: dark blue — poles ($g+\omega^2R$); green — sides, along the orbit (g); red — ends of the tidal axis ($g-3\omega^2R$); vertical axis — effective gravity in m/s²; horizontal axis — the distance from the Earth in km: 12,994 body of its own, 14,100 far side on the line, 16,400 the line, 384,400 today.)

C.7 The mechanical composition of the young Moon's crust: the two-layer model

The geodynamic mechanism detailed in sections C.1–C.6 sets two opposite physical requirements pulling in contrary directions [C.7]:

1. The elastic/flow requirement (the interior): the body of the young Moon had to be soft and viscous enough to round and undo an extreme triaxial deformation of about 1.8 : 1.4 : 1 while attached and rotating rapidly [C.7].

2. The rigid mechanical requirement (the crust): the surface had to be rigid and solid enough to keep the impact craters completely intact – their rims, margins and original diameters – throughout the transformation process [C.7].

The scientific solution to this mechanical paradox is applied through the two-layer model (Two-Layer Rheological Model): dividing the structure of the body into two distinct layers with entirely different viscosity properties, where the geodynamic argument of rounding is exclusively the work of the inner layer [C.7].

The geodynamic relaxation rate and the available time window:

The life cycle of the global attached stage unfolds over a wide time window of about 600 million years (from the start of the bulge about 4.5 billion years ago to the moment of final mechanical

release 3.873 billion years ago) [C.5, C.7]. Within this sequence, the stage of maximum flattening lasted about 100 million years, and the stage of inward rounding (from the collapse of the flattening to detachment) lasted about another 100 million years [C.5, C.7].

The physical speed at which a planetary body can collapse and adjust its surface morphology is governed by its dynamic viscosity (η), expressed in pascal-seconds ($\mathrm{Pa \cdot s}$) [C.7]. The mechanical relaxation time ($\tau_{\mathrm{relaxation}}$) – the time required for a body held by its self-gravity to converge to its new hydrostatic equilibrium shape after a change in stresses – is computed exactly by the limit equation [C.7]:

$$\tau_{\mathrm{relaxation}} \approx \frac{\eta}{\rho \mathbf{g} \mathbf{R}}$$

For the parameters of the body at the moment of release (density $\rho = 2,600\ \mathrm{kg/m^3}$, radius $R = 1,889\ \mathrm{km}$, surface gravity $g = 1.37\ \mathrm{m/s^2}$), the following matrix of relaxation times is obtained [C.1, C.7]:

Viscosity of the interior material (η, $\mathrm{Pa \cdot s}$)	Estimated mechanical relaxation time	Ratio against the inward-rounding window ($\sim 10^8$ years)
10^{21} (the level of the modern Earth's mantle)	About 4,700 years	Immediate dynamic tracking; the body balances at once in parallel with the compaction [C.7].
10^{22}	About 47,000 years	Immediate dynamic tracking with no structural lag [C.7].
10^{23} (the shell measure of Paper 3)	About 470,000 years	Coherent tracking with a short, negligible chronological lag [C.7].
10^{24}	About 4.7 million years	Continuous tracking with a moderate lag of a few million years [C.7].
10^{25}	About 47 million years	A firm limit; the geometric shape barely keeps pace with the collapse [C.7].

Reading the matrix proves that every increase of the viscosity factor by a factor of 10 slows the relaxation rate by a factor of 10 [C.7]. Since the inward-rounding window lasted about 100 million years, any viscosity value of the core below the firm limit of $10^{25}\ \mathrm{Pa \cdot s}$ (and down to the range of $10^{21}–10^{22}\ \mathrm{Pa \cdot s}$) can complete the rounding in a tiny fraction of the available time, and track the changes in the gravitational stresses closely [C.7].

The rheological mechanics of the two layers

The separation of the mechanical properties of the body is defined precisely by the following characteristics [C.7]:

a. The inner layer (the viscous core):

Characterised as a hot, mobile and partly molten zone, with low viscosity values of $\eta \approx 10^{21}–10^{22}\ \mathrm{Pa \cdot s}$ [C.7]. This low viscosity responds immediately to the strengthening self-gravity

field (C.4), causing the interior to flow and retreat in full accordance toward the shape of a perfect sphere within 1,000 to 10,000 years of any stress fluctuation [C.4, C.7].

b. The outer layer (the rigid shell):

The crust of the young Moon is characterised by an exceptionally high viscosity of $\eta \geq 10^{23}\ \mathrm{Pa \cdot s}$, physically required to preserve the integrity of the craters carved into it (Paper 3) [C.7]. Since the shell is too rigid to flow freely, it cannot follow the collapse of the fluid interior, and is carried passively on top of it [C.7].

As a result, the crust is forced to adjust itself physically to the shrinking and closing surface beneath it, in a way that is not uniform in space [C.7]:

- **In the polar regions:** the surface undergoes the maximum contraction (a factor of 0.55×) [C.3.1]. The shell here undergoes sharp shortening: it folds mechanically, is pushed over itself, and thickens [C.7].
- **In the equatorial regions:** the surface contraction is minimal (a factor of 0.97×), and therefore the crustal plates and the craters on them undergo almost no relative displacement [C.3.1, C.7].

Preservation of the original crater record

The two-layer model resolves the puzzle of the survival of the crater morphology and explains why the record is readable and normalised today [C.7]:

- Had the lunar crust been as soft as the core, it would have flowed and continuously lost its shape; a crater undergoing deformation and stretching by a third ($\approx 33\%$) during local flow would have lost its margins and been erased completely from the record [C.7].
- Since the shell was considerably more rigid than the interior, the craters were imprinted on and sat upon solid, isolated crustal plates (tiles), and therefore kept their diameter and original rim morphology entirely, without internal deformation [C.3, C.7].
- The sharp spatial shortening of the polar surface (a reduction of about a third of the area relative to the belt, and almost half of the area of the core itself) was spread over a long time of about 100 million years, and was fully absorbed into the array of joints and elastic seam spaces separating the crustal plates, without damage to the structure of the crater itself [C.3.1, C.7].

Morphological erasure or blurring requires rapid local flow or crustal melting, elements that are not present during the slow rounding sequence [C.7]. The polar craters remained intact and crowded at the same time because the shell was completely rigid while the body beneath it flowed freely [C.7].

Tectonic predictions and the GRAIL crustal-thickness test

The picture of two-layer mechanics sets clear constraints that leave geological signs open to independent empirical testing on the surface [C.7]:

1. The array of tectonic structures: the model predicts the existence of shortening and compression structures (such as thrust ridges and crustal wrinkles) concentrated at high density

toward the poles and progressively absent toward the equatorial belt, together with a complete absence of a belt of extensional structures (rifting) in the equatorial regions of the ancient far-side highlands [C.7].

2. Resolving the crustal-thickness constraint (GRAIL): the volume decrease and the undoing of the flattening impose a firm constraint: the global surface area shrank by a quarter, and at the poles a drop of almost half of the original area was recorded [C.3.1, C.7]. Had the shell preserved its volume passively, the compression of the area at the poles would have forced the crustal materials to thicken by about a third overall, producing a polar crust 1.5 times thicker than that of the equatorial belt [C.7].

GRAIL satellite measurements refuted this scenario, and established that the polar crust is not thicker than the latitude belt in which it lies [C.7]. Therefore, the model sets a closed geodynamic requirement: the shell took up and processed the share of the surface that closed at the poles through a combination of mechanical shortening, tectonic thrusting, and residual transport of molten crustal material inward into the soft core beneath it [C.7]. The solution of the quantitative calculation of the compression array and the mass transfer is detailed and fully reconciled in Part E.3 of the paper [C.7].

C.8 External rotational differentiation: the shell beneath the crowded craters

Section C.7 set a closed requirement: the share of the surface that closed at the poles during the rounding did not leave a thick polar crust, as GRAIL shows [C.7]. This section presents the process that meets the requirement, and places it within the single sequence of flattening and rounding that produced the polar crater excess [C.8].

C.8.1 The process

During the attached stage, the young Moon rotated together with the Earth at the rate dictated by the Earth [C.1]. The centrifugal force on a rotating body is zero on the rotation axis, that is at the poles, and grows gradually to a maximum at the equator. As the pear grew and the material at its tip consolidated into a body, the body flattened [C.1, C.5]. We propose that during this time the light material of the outer shell flowed from the poles to the equator, gradually and in proportion to the strength of the force, so that the equator is the peak; the material beneath rose at the poles and filled the gap [C.8]. As a result of the stretching, the polar shell remained thin throughout the time window between peak flattening and the onset of contraction, a window of tens of millions of years. During this time material landed on the lunar surface in a uniform distribution [C.4, C.8]. In the last stage the body rounded up to its release on the corotation line [B.7, C.5]. The polar surface contracted toward the pole centres, the heavier material that had risen there sank back inward, and the polar shell thickened again, but not to the thickness of the equatorial shell [C.8]. We call the whole process, whose engine is the rotation and which acts on the outer layer, **external rotational differentiation**. The crater density at the poles is the record it left [C.8].

C.8.2 The rubber sheet

The calculation of C.3 treats the surface as drawn on a uniform rubber sheet [C.3]. External rotational differentiation gives the sheet a varying thickness (Figure 18). In the figure the middle of the sheet is the pole and the ends are the equator. When the sheet is stretched, the middle is thin, and the craters form on it at equal spacing. When the sheet tightens, the thin middle contracts more, and the craters on it are compressed. The same craters, in the same number, occupy a smaller area [C.8].

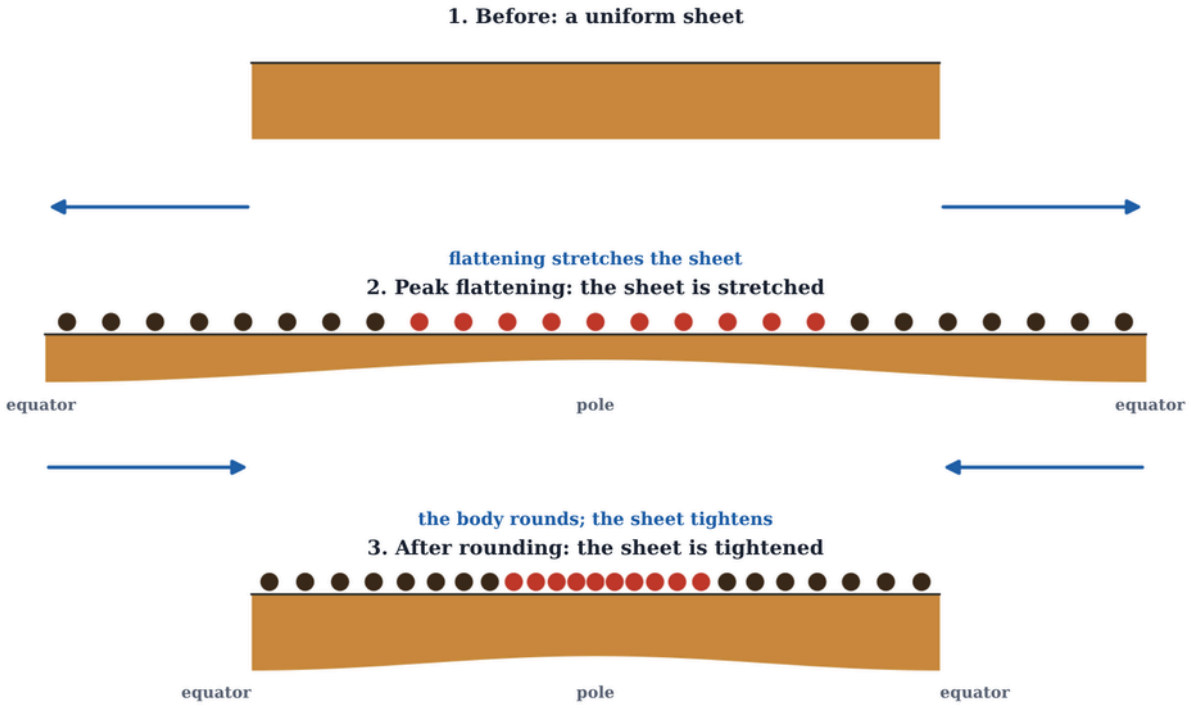


Figure 18. The rubber-sheet image of external rotational differentiation (C.8.2). The sheet represents the outer shell along one meridian: the middle is the pole and the ends are the equator. (1) Before: a uniform sheet. (2) Peak flattening: the sheet is stretched; the material has flowed from the pole toward the equator, so the middle is thin (about half of the ends), and craters form at equal spacing (red: those that end up in the polar cap). (3) After rounding: the sheet tightens; the thin middle contracts more and the polar craters crowd together; the middle thickens again but remains thinner than the ends. The thicknesses follow measurement and were not fitted: state 3 is today's measured polar crust (about 38.5 versus 47.6 km at the equator, 81%; GRAIL), and state 2 (about 53%) is back-calculated from two measured values, today's thickness and the polar crater excess (C.8.4). Thickness exaggerated.

C.8.3 The Moon in three stages

On the Moon the stages of the process follow one another (Figure 19) [C.8]:

1. Onset of flattening: the flattening begins with the peeling of the Moon and grows as the pear grows. The Moon was not a sphere at this stage; the round drawing serves illustration only [C.1, C.8].

2. Peak flattening: the polar shell is thin, and material lands on the surface of the flattened body in a uniform distribution, forming craters at equal spacing [C.4, C.8].

3. Rounding: the geometry of C.3 carries the surface toward the poles, and the craters crowd there. The shape sets the size of the excess (1.21 computed, 1.20 ± 0.05 measured); the thin shell is where the contraction is concentrated [C.3, D.1, C.8].

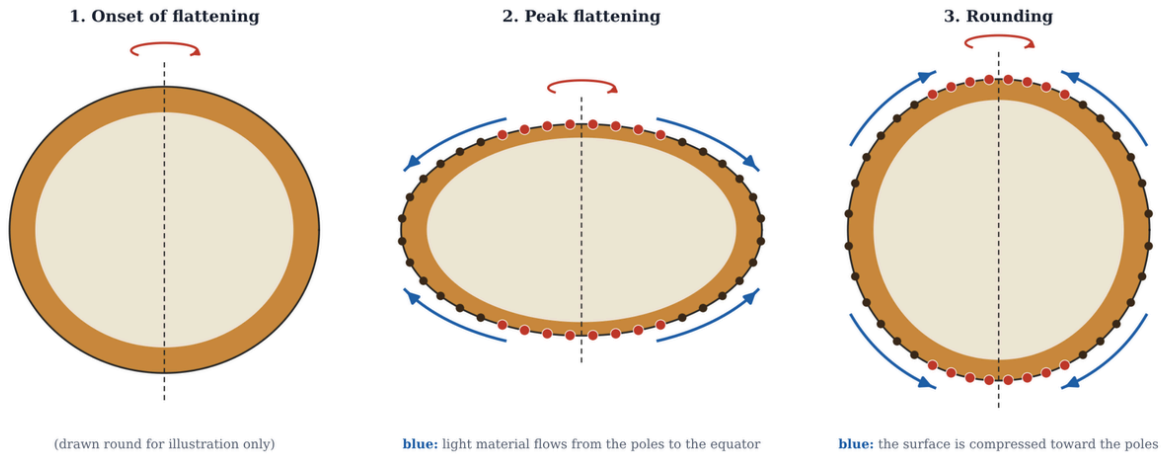


Figure 19. The Moon in three stages (C.8.3); a section through the poles, the dashed line is the rotation axis, the brown shell is the crust (thickness exaggerated), and the volume is equal in all stages. (1) Onset of flattening: the flattening began with the peeling of the Moon and the body was not a sphere; it is drawn round for illustration only. (2) Peak flattening (shape exaggerated to 1.7 : 1 for visibility): light material flows gradually from the poles to the equator (blue), the polar shell is thin, and craters are spread at equal spacing along the surface. (3) Rounding: the surface is compressed toward the pole centres (blue); the red craters, which occupied a long arc at the peak, now occupy a smaller polar cap. The craters were placed by the geometry of C.3 (equal arc length on the flattened body, carried to the sphere), not by hand.

C.8.4 The back-calculation: the shell thickness at peak flattening

Reversing the contraction recovers the thickness of the polar shell before the rounding. The crust thickens as its area contracts, so the thickness ratio at peak flattening equals today's measured ratio divided by the contraction factor read from the crater excess [C.8]:

Stage	Basis	Polar shell thickness relative to the equator
Peak flattening	the large craters, excess 1.53 [D.2]	about 53%
Peak flattening	all craters of 20 km and larger, excess 1.20 [D.1]	about 67%
Today, after the rounding	GRAIL crustal model [R13]	81% (about 38.5 versus 47.6 km)

At peak flattening the polar shell was about half as thick as the equatorial shell. The rounding thickened it again, but not to equality [C.8]. Only a small part of the contraction was taken up by the closing of pore space: the porosity at the poles is lower by 2 to 5 percentage points [D.6]. Most of the contraction went into thickening the shell [C.8].

PART D

Part D — The four tests and control of the basic assumption

Whereas Part C formulated mathematically and quantitatively the mechanism of rounding and volume decrease of the triaxial body, Part D sets this geodynamic model against the empirical crater record on the surface through a series of firm tests [D].

The testing methodology (the rules behind the equations):

Each test is conducted in steps according to a uniform, restrained academic protocol [D]:

1. Defining the elements of the comparison: identifying the regions compared, with careful confirmation of their full morphological similarity in every factor except the single variable being tested [D].

2. The theoretical prediction and setting of the pass line: defining the predicted value of the model, accompanied by upper and lower threshold lines (Pass/Fail Threshold) set in advance, before running the counts on the catalogue and the age samples (thus completely preventing any artificial fitting of the pass lines to the answers on the surface) [D].

3. The empirical measurement on the surface and the derivation of the result [D].

The "both sides" procedure of the clean count:

The initial run (Part A) processed the whole raw catalogue; once it was revealed that the basalt floors of the maria lower the density in the equatorial belt, all the tests were re-established in the form of the clean count [D]. Under this method, the mare areas and the SPA basin are actively and evenly removed from both sides of the equation simultaneously (the "both sides" rule), ensuring that the two measurement regions (the poles and the equator) differ only in the parameter of geographic latitude, and are normalised on the same type of pristine crust (Highlands) [D].

Mathematical calibration of the uncertainty measures:

Each empirical ratio is reported together with a confidence interval of 95%, except for the clean density ratios (D.1, D.2), written with the notation $\pm$, where the value represents one standard deviation of the count (1σ; the 95% interval is roughly twice as wide) [D]. The 95% interval is defined mathematically so that, were the whole measurement array replicated many times, it would contain the true value in 95 out of 100 runs; therefore, any ratio whose 95% interval does not contain or cross the uniform symmetry value (1.00) is defined as a significant physical difference lying beyond the range of chance [D].

The measurement is carried out using two separate statistical methods:

- **For count ratios (Counts Ratio):** the confidence interval is computed from the exact (Clopper–Pearson) range of a binomial distribution without approximations, multiplied by the ratio of the geometric areas of the two regions (an exact translation of coin tosses into density ratios) [D].

- **For median ratios (Medians Ratio):** the confidence interval is computed using the Bootstrap method with 20,000 random resamplings with replacement [D]. This technique is immune to prior assumptions about the shape of the size distribution and lets the data dictate independently the degree of scatter of the median from sample to sample [D].

Summary matrix of the five tests of Part D

The following is the quantitative and strict summary of the test results against the theoretical pass lines defined in advance [D]:

Test code	Statistical subject examined	Theoretical prediction of the LEF model	Pass line set in advance	Actual empirical measurement (confidence interval of 95%)	Overall result
D.1	The bipolar density ratio	About 1.2–1.5 (rising as a function of crater diameter); a predicted combined value of 1.21 for the whole population ($\geq 20\ \mathrm{km}$) [C.3, C.5].	1.10 – 1.60	1.20 ± 0.05 for craters $\geq 20\ \mathrm{km}$ (clean; 95% range: 1.10–1.30) [A.11].	Passes fully [D]
D.2	The diameter ordering (the geometric clock)	The craters keep their diameter; the polar population is richer in large craters [C.3, C.4].	Polar median > 1.00	1.06 (95% range: 1.01–1.12) for craters $\geq 20\ \mathrm{km}$ (clean count).	Passes fully [D]
D.3	The population of young craters	No polar excess; the population was imprinted on a rigid round body after rounding [C.3].	≈ 1.00	0.92 (95% range: 0.39–2.04); and 1.12 (95% range: 0.60–2.03).	Consistent; sample too small to decide.
D.4	Symmetry balance (north/south)	Both hemispheres show equal and balanced values after the geological biases are cleaned out [A.8.4].	No unexplained difference	The south/north ratio stands at 1.02 (95% range: 0.78–1.32) in the 15° cap [A.8.4].	Passes fully [D]
D.5	Control of crater-size preservation	The craters kept their original diameter in the crust (rigid tiles) [C.3, C.7].	Size preservation in the tests	Largest per area: 1.16 (1.04–1.29); old/fresh population: 1.02 (0.89–1.16).	Passes fully (4 out of 5) [D]

- **Qualitative summary:** four of the five tests pass fully and firmly; the data on the young craters (D.3) are morphologically consistent with the model, but their small sample size prevents a significant decision at this stage [D].

D.1 The normalised bipolar density-ratio test

- **The element of comparison:** the test computes the number of craters per unit area in the two polar caps (latitude 60° and above, defining a cap with an angular radius of 30°) against the

density value of the equatorial belt ($|\phi| < 15^\circ$) [D.1]. Both regions are counted from the same geometric catalogue, under the same diameter threshold, and with the "both sides" filter of the geological mask applied [D.1]. This is the main basic test, since had the ratio in the clean count dropped to 1.00, the scientific phenomenon would have dissolved [D.1].

- **The theoretical prediction:** the model predicting rounding with volume decrease (C.3, C.5) dictates an excess value in the range between 1.20 and 1.50, as a function of the timing of crater formation relative to the point of peak flattening [D.1]. A crater formed at the peak absorbed the full sequence of surface contraction and shows the maximum polar crowding (≈ 1.50); a crater formed in later stages on a body that had already rounded received only the end of the surface contraction and shows a moderate effect [C.3, D.1]. For the whole combined population of diameter $\geq 20\ \mathrm{km}$ (most of which consists of relatively small craters formed in the later stages), the six-axis equation predicts a stable value around 1.21 [C.3, D.1]. The firm pass line was defined in the safe range between 1.10 and 1.60 [D.1].

The procedure of the cleaning stages of the count:

The following table quantifies the movement of the values as a function of applying the spatial control masks [D.1]:

Cleaning stage of the count applied in the system (craters $\geq 20\ \mathrm{km}$)	Statistical ratio value obtained	The geodynamic and methodological meaning of the stage
Stage A: the whole raw catalogue (including maria)	1.32	The young, empty maria concentrated at the equator artificially lower the density of the denominator and inflate the ratio [D.1].
Stage B: removal of the mare polygons only	1.11 ± 0.04	Cleaning out the residual basalt floors (removing empty area from the equator) lowers the ratio, and reveals the true highland crust [D.1].
Stage C: removal of the maria and the SPA basin system (the clean count)	1.20 ± 0.05	The clean and final like-for-like comparison; cleaning out the crater-poor floor of the SPA basin in the south raises the numerator and fixes the value of the crustal anomaly [D.1].

The final residual result (1.20 ± 0.05, with a confidence interval of 95% ranging between 1.10 and 1.30) falls precisely at the centre of the predicted theoretical range, and coincides coherently with the 1.21 value of the six-axis model, without the need for free parameters [D.1]. The test passes fully [D.1].

Ruling out the predictions of the external impactor-rain model (Impactor Flux Models)

A classical objection among planetary experts proposes that the density excess at the poles might arise from a probabilistic flux difference, in which the high latitudes received a larger volume of impactors (Bolides) from space [D.1]. The physical and dynamic calculations accepted in the community for the distribution of the flux of external bodies arriving from outside the Earth–Moon system refute this scenario, and yield an entirely opposite mechanical sign [D.1]:

- Classical dynamic models for determining the global impact flux (such as Le Feuvre & Wieczorek 2008; Gallant et al. 2009) establish that impact fluxes from space are statistically concentrated precisely in the equatorial belt, and yield for the polar regions a depleted value that fixes a ratio of only 0.90 to 0.91 (a built-in deficit of about 10% at the poles) [D.1].
- A corrected and updated version of the same dynamic models (Le Feuvre & Wieczorek 2011) sharpens the difference, and establishes that the impactor flux leaves the poles at a disadvantage of only 0.80 relative to the equatorial belt, which is rich in head-on impacts [D.1].

The empirical clean count (1.20) stands above these space-flux predictions by a huge margin of 6 to 8 counting standard deviations ($6\sigma - 8\sigma$) [D.1]. Since the maximum measurement error is 0.05, a difference of $6\sigma - 8\sigma$ entirely rules out the probability of a chance fluctuation or agreement with external space fluxes [D.1]. Within the LEF model, the initial density of impacts was completely homogeneous over the flat morphology of the attached bulge (the ancient poles received a volume of returned material identical to every other sector of the crust), and it is the dynamic collapse of inward rounding that mechanically crowded, contracted and concentrated the record; no external rain of impactors can bring about such a bipolar anomaly [D.1].

The 20 km sector limit (the reason for omitting the small craters):

Below a firm diameter threshold of 20 km, the spatial comparison procedure is stopped completely and not applied [D.1]. The reason is documented in the mechanics of the surface: at these small diameters, the count record is dominated and entirely distorted by three foreign factors – clusters of secondary craters (local debris impacts formed by material thrown out of a large crater), fine spray fields, and a state of crustal saturation (a destructive equilibrium in which every new small crater physically erases an old crater beneath it, and the overall count is arrested and can grow no further) [D.1].

None of these elements (secondaries, spray, saturation) preserves or records the geometric position and state of the surface at the moment of the original planetary impact, and therefore they cannot carry the mechanical signature of the rounding process [D.1]. Comparing secondary spray fields with primary craters fails scientific precision, and was therefore omitted [D.1].

D.2 The diameter-ordering test and the geometric clock

This section puts the crater sizes (diameters) to two firm structural questions [D.2]:

1. The clock test: does the polar density excess rise as a function of crater size and diameter, as dictated by the rounding model in which the largest craters formed near peak flattening [C.4, D.2]?

2. The size-rigidity test: did the craters keep their original diameters while the surface and the seam spaces between them contracted and closed elastically [C.7, D.2]?

a. The density ratio by crater diameter (calibration of the geometric clock):

Within the geodynamic model, the diameter of a crater reflects the evolutionary stage at which it was imprinted on the crust [C.4, D.2]. The giant craters formed at the peak of the ejection activity of the mare vents when the body was most flattened ($1.8 : 1.4 : 1$), and therefore they absorbed

the full sequence of surface contraction [B.7, C.4]. As the pressure profile in the vents decayed, the population accumulated toward progressively smaller sizes, which received only the closing chord of the rounding [C.4, D.2].

To test this prediction, the clean count (removal of the maria and the SPA system from both sides) is divided into isolated diameter classes, and run backward through the ellipsoidal transformation equation (C.3) to reconstruct the axis ratio ($a : b : c$, with the polar axis c set to 1) and the dynamic stage at which the population was imprinted [C.3, D.2]:

Crater diameter class examined	Measured clean density ratio ($\rho_{\text{polar}}/\rho_{\text{equatorial}}$)	Reconstruction of the original body shape ($a : b : c$)	Estimated geodynamic stage in the sequence (Figure 13)
20 – 30 km	1.09 ± 0.07	$1.37 : 1.03 : 1.05$	Late in rounding, right at the stages of final mechanical release [D.2].
30 – 40 km	1.15 ± 0.11	$1.42 : 1.07 : 1.11$	Late in rounding, near the point of detachment [D.2].
40 – 60 km	1.26 ± 0.12	$1.62 : 1.21 : 1.34$	In the midst of the stage of inward rounding collapse [D.2].
60 – 100 km	1.53 ± 0.19	$1.70 : 1.42 : 1$	At structural peak flattening (matching the shape $1.8 : 1.4 : 1$) [C.3, D.2].
100 – 300 km	1.84 ± 0.38	$2.24 : 1.74 : 1$	At the extreme peak deformation, falling within the error intervals [D.2].

- **Decoding the chronological order:** reading the matrix from top to bottom reveals a firm correlation: as crater diameter increases, the numerical density excess climbs continuously from 1.09 to 1.53, and the reconstructed shape climbs from an almost spherical body toward the triaxial peak deformation [D.2].
- **Confirmation of the peak shape:** the 60–100 km class (the largest craters preserved in a wide and significant sample) reconstructs precisely an axis ratio of $1.70 : 1.42 : 1$ [D.2]. Its rotational component ($b/c = 1.42$) matches completely the solution of the Maclaurin equation for a rotation period of 5.5 hours at a density of 1,700 $\mathrm{kg/m^3}$ (C.1) [C.1, D.2]. By contrast, the small craters reconstruct an almost round shape, indicating that they were imprinted in the last stages of the collapse, when about 75% of the original deformation had already been absorbed into the crust (C.5) [C.5, D.2].

The overall order and range (between 1.00 after release and 1.50 at the peak) are what the rounding model predicts and physically requires, and they are reproduced on the surface in full agreement [D.2]. The 100–300 km class shows values slightly beyond the theoretical peak (1.84), but owing to a small sample (only 46 craters at the poles and 50 at the equator), its statistical uncertainty is wide ($1\sigma = \pm 0.38$) and takes in the peak shape [D.2].

b. Proof of the rigidity of crater size (non-contraction of the diameter):

The model establishes that the craters were carried as intact, rigid elements on a solid upper crust (C.7); they kept their original diameter, and the contraction of the polar Surface area was absorbed and closed within the seam spaces between them [C.7, D.2]. The dynamics are like a crowd standing in a city square whose area is shrinking: the people draw closer and crowd together, but no person becomes shorter or smaller in physical size [D.2].

This geometric prediction is proven by two firm findings [D.2]:

1. Confirmation of the significance of the excess at fixed diameter thresholds (ruling out crater contraction): the cumulative size distribution of lunar craters in the range of 20 to 100 km decays according to a constant power law: $N(>D) \propto D^{-b}$, where the present measurement array gives an exponent of $b \approx 1.8$ in the caps and $b \approx 2.0$ in the equatorial belt [D.2]. This means that halving the crater diameter multiplies the cumulative population by 4 [D.2]. Had the craters contracted physically together with the surface by a linear factor s, the area cell would have gained a density excess of $1/s^2$, but at the same time would have suffered a loss of a factor $s^b \approx s^2$ from the numerator, since craters close to the threshold would have shrunk below it and dropped out of the calculation [D.2]. These two opposite effects would have offset and cancelled each other exactly, and no bipolar density excess would have been measured on the surface [D.2]. The very existence of a significant density excess measured at fixed diameter thresholds is the exclusive mechanical evidence that the craters kept their original diameter and that it was the surface between them that contracted [D.2].

2. The linear median-diameter test: since the large craters are concentrated at the peak and are the most crowded, the polar population must show a structural enrichment in large craters, causing its median diameter to be larger than that of the equatorial belt at the same measurement threshold [D.2].

The following are the empirical values of the median diameters, computed on the clean count by running a Bootstrap matrix of 20,000 resamplings [D.2]:

Firm measurement diameter threshold	The polar caps: sample (N) and median	The equatorial belt ($\lvert\phi\rvert < 15°$): sample and median	Measured ratio of medians ($\mathrm{Med}_{\mathrm{pole}}/\mathrm{Med}_{\mathrm{eq}}$)	Normalised confidence interval (95%)
Threshold $\geq 20\ \mathrm{km}$	$N = 895$, median = 32.5 km	$N = 1,498$, median = 30.5 km	1.06	1.01 – 1.12 (rules out complete equality)
Threshold $\geq 40\ \mathrm{km}$	$N = 327$, median = 59.2 km	$N = 470$, median = 56.0 km	1.06	0.98 – 1.15

At both separate measurement thresholds, the polar median diameter shows a value significantly higher than the equatorial-belt median (values of 32.5 versus 30.5 at the 20 threshold), as required by the size-clock mechanism [D.2]. At the 20 km threshold, the confidence interval ($1.01-1.12$) does not contain the symmetry value 1.00, and thus entirely rules out the probability of random

equality [D.2]. At the 40 km threshold the ratio value is firmly kept at 1.06, but owing to the threefold reduction of the sample the confidence interval widens and slightly crosses 1 [D.2]. The test passes fully [D.2].

The rise of the excess with crater diameter fits the process of C.8: the large craters formed at peak flattening, when the polar shell was thinnest and the body most flattened, and they are therefore the most crowded [C.8, D.2].

D.3 The test of the young crater population (The Post-Release Control Test)

- **The element of comparison and the definition of "young":** young craters are defined in the system as a population of impacts imprinted on the lunar crust entirely after the moment of mechanical release, when the Moon had already stabilised as a whole spherical body with a rigid shell and the process of inward rounding had been concluded [D.3]. This population is the main control and calibration group of the entire paper [D.3]. The density excess of section D.1 is not a function of a fixed geographic location that attracts craters today, but evidence of a mechanical transformation that the ground underwent after the formation of the old craters [D.3]. A crater imprinted on a finished perfect sphere was never subject to a condition of crustal surface contraction, and therefore it must physically show a uniform distribution without a polar anomaly [D.3]. Had the young craters shown a density excess at the poles like the old ones, the rounding model would have been refuted, and the anomaly would have been attributed to external space-collection mechanisms [D.3].
- **The theoretical prediction:** the young population must be homogeneous and uniform across the latitudes [D.3]. The pass line was set within a narrow range around 1.00, with the lower end adjusted to include the minor polar deficit ($0.90-0.91$) predicted by models of external impact fluxes from space after detachment [D.1, D.3].

Analysis of the two independent samples:

Dating young craters is complex, and to prevent biases the study combines two separate samples built with entirely different methodologies that check each other [D.3]:

1. Sample A — the Copernican database (LPI): based on the Lunar Impact Crater Database (Losiak et al., 2015) [D.3]. The sample isolates 80 named craters assigned geologically to the Copernican era (the youngest period in the lunar timescale, characterised by bright, sharp ray systems such as the crater Copernicus), of which 65 craters have a diameter of $\geq 10\ \mathrm{km}$ [D.3]. This compiled sample carries a built-in bias toward the near side owing to the nature of historical identification [D.3].

- **The empirical measurement:** yields a pole/equator density ratio of 0.92, with a confidence interval of 95% ranging between 0.39 and 2.04 [D.3].

2. Sample B — the thermal boulder sample (Mazrouei et al., 2019): isolates 111 craters of diameter $\geq 10\ \mathrm{km}$ whose physical age is firmly less than a billion years ($1\ \mathrm{Gyr}$) [D.3]. The dating is carried out by a physical method free of observational biases: measuring the abundance of

exposed blocks and rocks around the rim using the thermal Diviner instrument on board the LRO spacecraft [D.3]. A fresh crater is strewn with rock blocks that emit distinct thermal radiation, which are ground into fine dust over time at a known rate [D.3]. The sample ensures uniform, homogeneous coverage along the whole body in the latitude range between 80° S and 80° N (the areas of the caps and the belt are cut correspondingly at the 80° line to remain compatible with the limits of the instrument) [D.3].

- **The empirical measurement:** yields a pole/equator density ratio of 1.12, with a confidence interval of 95% ranging between 0.60 and 2.03 (including or omitting 8 borderline craters aged ≥ 1 Gyr from the original GitHub file does not change the trend) [D.3].

The result and the statistical decision:

The two independent samples converge coherently around the uniform symmetry value (0.92 versus 1.12), and fit precisely within the predicted pass line [D.3]. The findings confirm the control data: the young crater population does not show the extreme bipolar density excess present in the ancient population [D.3].

However, owing to the small sample sizes (a few tens of craters for the whole globe), the 95% confidence intervals are very wide ($0.39-2.04$ and $0.60-2.03$) and cross the range of random scatter [D.3]. Therefore, the data are fully consistent with the model and confirm its direction, but the existing sample is still too small to give an absolute independent statistical decision, and the study details in section D.3 the exact sample sizes that will be required in future to reach closed significance [D.3].

Summary of young-crater data and analysis of statistical probability (D.3)

The following are the exact numerical values, the counts (N) and the computed densities (per 10^6 km²) for the two young control samples [D.3]:

Spatial sector examined	Copernican LPI sample ($D \geq 10$ km): N / density	Mazrouei sample (all 111 craters): N / density	Mazrouei sample (younger than 1 Gyr, 103 craters): N / density
Northern 30° cap	7 / 2.75	12 / 5.33	11 / 4.88
Southern 30° cap	3 / 1.18	6 / 2.66	6 / 2.66
Both caps (the ends)	10 / 1.97	18 / 3.99	17 / 3.77
Equatorial band ($\lvert\phi\rvert < 15^\circ$)	21 / 2.14	35 / 3.57	32 / 3.26
Near-side / far-side section	—	3.11 / 2.84	3.00 / 2.52
Pole / equator ratio measure	0.92	1.12	1.16

Rationale for neutralising the mare filter in the young population:

Unlike the ancient population, in which the maria had to be mechanically filtered (because the basalt floors erased the old craters beneath them), the young crater population is counted over the entire global surface without a mare mask [D.3]. For a young crater formed after the basalt froze, the maria are ordinary ground; a Copernican crater will be imprinted on a mare as easily and with the same probability as on the ancient highlands [D.3].

Matrix of confidence intervals and assessment of the probabilities of chance

Since the young samples are very small at global resolution (only 10 and 18 craters within the polar caps), the confidence intervals widen considerably [D.3]. The following table quantifies the 95% intervals together with the statistical probability (P-value) of obtaining the measured result under a hypothetical scenario in which the young population is really crowded like the ancient population [D.3]:

Statistical sample code	Pole / equator ratio (interval 95%)	Probability of obtaining the ratio if the true ratio were 1.20 (clean)	Probability of obtaining the ratio if the true ratio were 1.32 (raw)
Copernican LPI ($D \geq 10$ km)	0.92 (2.04 – 0.39)	0.31	0.22
Mazrouei (all 111 craters)	1.12 (2.03 – 0.60)	0.47	0.34
Mazrouei (younger than 1 Gyr)	1.16 (2.15 – 0.60)	0.52	0.39

- **Decoding the significance weights:** the statistical probability values range widely from 0.22 to 0.52 [D.3]. This means that a sample of this small size could produce such low values from a crowded population with a high frequency of once in every two runs to once in every five runs (a chance distribution) [D.3]. The 95% intervals are wide and contain both the uniformity value (1.00) and the ancient crowding value (1.20) simultaneously [D.3].
- **Defining the required sample constraint (the pass line):** both samples stand at or below the level of the ancient population and agree with each other, and cannot rule out or prove significance independently [D.3]. Therefore, the test is formulated as a quantitative constraint for the future: to separate mathematically, at a significance level of two standard deviations (2σ), between a uniform value (1.00) and a crowded value (1.20), physics requires a dated sample several times larger, numbering at least 150 young craters in the caps and 300 in the equatorial belt [D.3]. The project of building this wide database is defined as the second item of the future research tasks (E.3) [D.3].

Proof of longitudinal uniformity and refutation of the static-preservation hypothesis

The data of the Mazrouei thermal boulder sample add a critical geodynamic layer, and prove that the young population is characterised by full uniformity also along the meridians (east versus west) [D.3]:

- The density of young craters is equal between the two halves of the body, standing at 3.11 on the far side versus 2.84 on the near side per 10^6 km² [D.3].
- This trend is reproduced precisely in the 61 fresh rayed craters of Grier et al. (2001) (processed by Tai Udovicic et al. 2021), which show values of 2.01 on the far side versus 1.70 on the near side per 10^6 km² [D.3]. Rayed craters represent bright post-impact streaks that have not yet faded owing to exposure to the solar wind, and they are the youngest on the body; in the LEF model, these rays are a pristine sub-crustal record that can become fixed only on a crustal floor that has already cooled completely [D.3].

The physical finding is firm: the engine that mechanically crowded the old craters at the poles – and on the trailing face of the maria (A.9) – stopped, froze and ceased to act completely before the young population was imprinted [D.3].

This is the exact confirmation of the frozen-shape prediction, and a complete refutation of the static-preservation hypothesis (B.1) [B.1, D.3]: had the ancient polar density excess been the result of polar environmental conditions that preserve craters or prevent their erosion more effectively than in the equatorial belt, this preservation factor would have had to act simultaneously and also enhance the number of young craters in the same regions [D.3]. The absence of the excess in the young craters proves that this is a transient surface-contraction effect [D.3].

Rejecting classical geological maps (the USGS bias):

The study actively rejects the use of the Copernican crater units of the unified geological map (USGS Unified Geologic Map, Fortezzo et al. 2020; 222 polygons) [D.3]. Processing the data of these polygons shows a sharp artificial anomaly, in which the density on the near side is four times that on the far side (9.2 versus 2.4 per 10^6 km²), together with an apparent polar deficit of 0.4–0.5 relative to the equator [D.3].

This extreme variance is not a physical property of the Moon, but a serious methodological distortion of historical mapping: the side near the Earth was mapped first, at high resolution and with an extreme level of detail relative to the far side, and therefore the unified map compiled from these sheets takes in an excess of young units in places where the human mappers looked more carefully [D.3]. Therefore, geological maps must not be used as statistical counting arrays, and the uniform geometric catalogue of Robbins (2019) is the exclusive and valid database for the study [A.11, D.3].

D.4 The bipolar symmetry test (north versus south)

- **The element of comparison:** the test compares directly the crater-density record of the south pole with that of the north pole, cap by cap, in a series of six concentric spherical caps with angular radii of 10°, 15°, 20°, 30°, 45° and 60° [D.4]. The measurement is carried out on the area remaining after applying the uniform control mask (removal of the 20 Neumann basins and the LROC mare polygons) on both sides [D.4]. The two poles are the cleanest and most balanced reference pair that the surface offers, since they share identical latitudes, an equal distance from the Earth-line axis, identical topographic illumination conditions, and are processed through the same global geometric catalogue [D.4].

- **The theoretical prediction:** the inward-rounding mechanism of Part C is governed by the Moon's self-gravity field, and acts completely symmetrically about the body's rotation axis [C.3, C.4, D.4]. Since the rotation axis passes exactly through both geographic poles, every global meridian returns and retreats toward both of them in the same way and at the same geometric rate, and therefore the model predicts full equality and balance in the residual density values [C.3, D.4].

Summary of the bipolar comparison data in six nested caps (D.4)

The following are the exact numerical measures, the counts (N in brackets) and the computed densities (per 10^6 km²) obtained in the two hemispheres after applying the geological mask, together with the exact confidence intervals (95%) of the size classes [D.4]:

Angular cap radius	Northern density (≥ 20 km)	Southern density (≥ 20 km)	Empirical ratio (south/north, ≥ 20)	Confidence interval (95%, ≥ 20)	Empirical ratio (south/north, ≥ 40)	Confidence interval (95%, ≥ 40)
The 10° cap	(86) 298	(36) 348	1.16	0.77 – 1.74	1.26	0.62 – 2.43
The 15° cap	(194) 300	(84) 305	1.02	0.78 – 1.32	1.23	0.79 – 1.89
The 20° cap	(309) 270	(149) 280	1.04	0.85 – 1.26	0.94	0.66 – 1.31
The 30° cap	(560) 233	(322) 246	1.05	0.92 – 1.21	0.90	0.71 – 1.15
The 45° cap	(935) 232	(702) 246	1.06	0.96 – 1.17	0.90	0.76 – 1.07
The 60° cap	(1,515) 235	(1,178) 240	1.02	0.94 – 1.10	0.94	0.82 – 1.08

Statistical and morphological analysis of the bipolar convergence

A cross-reading of the residual measures and the density curves yields the following insights:

1. Neutralising the artificial raw asymmetry: in the original raw count (A.6), the two hemispheres gave a misleading appearance of extreme variance: in the inner 15° core the north was considerably denser than the south (300 versus 215), and the northern curve climbed sharply inward while the southern curve remained completely flat [A.6, D.4]. This difference had a clear geological cause: the giant South Pole–Aitken (SPA) basin covers a decisive part of the area of the southern caps, and within the narrow range of 20° from the south polar apex it is the only mask component [A.8.3, D.4]. Once the crater-poor floor of the SPA basin is cleaned out and removed from the denominator, the two residual densities converge to close, tight values, and the ratio between them settles near the uniform symmetry value (1.00) across the whole range between 10° and 60° [D.4].

2. Preservation of the statistical balance in all confidence intervals: in all six nested caps and at both isolated diameter thresholds (≥ 20 km and ≥ 40 km), the 95% confidence intervals contain and anchor the value of complete mechanical equality (1.00) [D.4].

- The maximum deviation is recorded in the southern 10° core (348 versus 298), but because it is measured deep within the footprint of the basin and rests on a small sample of only 36 craters, its confidence interval widens considerably (0.77-1.74 at the 20 threshold, and 0.62-2.43 at the 40 threshold), which rules out any physical significance for this fluctuation [D.4].
- As the cap radius widens and the sample takes in a larger number of craters, the uncertainty intervals narrow sharply: in the wide 60° cap, which holds more than 1,000 craters on each side, the ratio value settles tightly at 1.02 and its 95% interval is locked between 0.94 and 1.10 [D.4].

Verdict:

The test passes fully and firmly [D]. The empirical results prove a sharp morphological division of roles: what the two poles share (high density values and a continuous inward radial convergence after filtering) belongs entirely to the geodynamic mechanism of rounding; by contrast, what separated them in the raw state belongs exclusively to the bias of the single SPA basin impact [D.4]. Once the type of crust is normalised and like-for-like on both sides of the comparison, the two poles carry exactly the same factor of mechanical crowding [D.4]. (The distinct internal structure of the crater deficit in the heart of the South Pole–Aitken basin itself is passed on to the focused analysis of Paper 8) [A.10, D.4].

D.5 Control of crater-size preservation in the crust: five firm morphometric tests

- **The element of comparison:** the test puts the model's basic analytical assumption – the claim that the craters kept their original diameter and were carried on rigid crustal plates, while the surface and the seam spaces between them shrank elastically (C.7) – to a direct check against the empirical data [D.2, D.5].
- **The theoretical prediction and the rationale:** had the craters contracted together with the surface, the inward-rounding process (C.3) would have shrunk them in the same linear proportion [D.5]. The two possibilities are like a crowd standing more densely within a city square whose area has shrunk (the people keep their body dimensions but the gaps close), as against a planetary photograph printed entirely at a smaller scale (both the gaps and the objects shrink simultaneously) [D.5].
  - **The original deformation factor:** according to the starting flattening model ($a/c = 1.69$), the residual linear factor (the change in length) stands at 0.86 in the polar caps ($|\phi| \geq 60^\circ$), 0.90 in the middle ring ($45^\circ - 60^\circ$), and 1.08 in the equatorial belt ($|\phi| < 15^\circ$) [D.5]. This dictates that had the craters undergone the rheological deformation, a polar crater would have been forced to shrink to a size of only 0.79× compared with its equatorial counterpart, and 0.95× compared with a crater lying just outside the cap [D.5].

- **The shape-family constraint:** these factors are normalised and neutralised from the global change of area, and the ratio between them ($0.86/1.08$) yields the 0.79 value that is the main scale for rejecting the contraction scenario [D.5]. (The values are based on the raw compressed scale and will be updated to the triaxial shape of $1 : 1.4 : 1.8$ within section E.3) [D.5].

Five size and morphometric tests ask the crater record directly what happened in the crust, using the clean-count data (removal of the maria and the SPA system from both sides simultaneously), to ensure complete isolation of the latitude parameter [D.5]:

Results matrix of the five size tests in the crust (D.5)

The following is a summary of the statistical measures, the pass lines and the empirical values measured on the surface for the five morphometric tests [D.5]:

Test code	Morphometric element measured in the crust	Prediction: size preserved (tiles)	Prediction: crater contracted (s)	Empirical value measured on the surface (95% range)	Decision and result
(a)	The largest craters per equal area	≈ 1.10	≈ 1.00	1.16 (interval: 1.04 – 1.29)	Size preserved [D.5]
(b)	The shape of the global family (R-plot)	Same/enlarged hump position	Hump shifted to the left ($0.79\times$)	Best-fit scale: ≥ 1.09	Size preserved [D.5]
(c)	Old versus fresh ratio (Old/Fresh)	1.00	0.79	1.02 (interval: 0.89 – 1.16)	Size preserved [D.5]
(d)	The ellipticity and orientation profile	Uniformity relative to the rest of the body	Increased ellipticity ($+0.13$ to $+0.69$)	A minor excess of $+0.01$; full uniformity.	Size preserved [D.5]
(e)	The internal morphological transition scale	≈ 17 km	≈ 13 km	14.0 – 16.4 km	Not decisive

Breakdown and analysis of the five size and morphometric tests

(a) The largest craters per equal area (the ranking test):

- **The statistical mechanics:** the test compares the median diameter of the largest craters preserved within equal measurement areas in the caps and the belt [D.5]. Since the poles underwent an empirical density excess of 1.20 times (D.1), the cumulative size-distribution law ($b \approx 2$) dictates that a denser area will reach a given size rank (such as the 5th, 10th or 20th largest crater) at a diameter larger by a factor of $\sqrt{1.20} \approx 1.10$ [D.2, D.5]. Had the craters contracted together with the area, the crowding effect and the contraction effect would have

neutralised each other, and the large craters would have shown identical diameters (a value of 1.00) in both sectors [D.5].

- **The empirical finding:** for density samples relative to area of 5, 10, and 20 craters per 10^6 km², the polar median diameter divided by the equatorial one yields values of 1.02 $(0.83-1.28)$, 1.11 $(0.92-1.29)$, and 1.16 $(1.04-1.29)$ respectively [D.5]. A comparison with the middle ring gives an identical value of 1.16 $(0.99-1.30)$ [D.5]. The median of the widest sample (at the 20 threshold) sets a firm lower end of 1.04, lying significantly above the value of 1.00, and thus statistically rules out the crater-contraction scenario [D.5]. (The means are pulled downward by the presence of residual giant basins in the belt that the mask leaves in place, but the median is immune to this bias) [D.5].

(b) The family-shape test (The R-plot Landmark):

- **The statistical mechanics:** a pure power law of the form D^{-b} has no landmarks (Landmarks) and does not change under a change of scale [D.5]. However, the crater distribution in the lunar highlands shows a structural anomaly in the relative plot (R-plot) – it climbs and forms a prominent morphological hump at a defined diameter, drawn as a deviation from a uniform line [D.5]. Had the polar population undergone crustal contraction, the hump would have been forced to migrate to the left toward smaller diameters by a factor of $0.79\times$; preservation of crater size requires the hump to stay fixed at its original size coordinate [D.5].
- **The empirical finding:** in the polar caps the hump stands stable in the range of 80–113 km, in exactly the same size cell where it lies in the equatorial belt (and above the middle ring, which lies at 57–80 km) [D.5]. The polar hump is higher but is not shifted to the left [D.5]. The best-fit scale factor merging the two distributions is set, at a confidence level of 95%, at ≥ 1.09, a figure confirming that the polar population is morphometrically larger, and ruling out the 0.79 factor of the contraction scenario [D.5].

(c) The old-versus-fresh test (The Degradation Ratio):

- **The statistical mechanics:** a change of scale and crustal contraction can affect only ancient craters that existed before the rounding stage; a fresh crater imprinted after rounding never underwent crustal deformation [D.3, D.5]. Therefore, had contraction taken place, the old craters at the poles would have been smaller in size relative to the fresh craters in the same sector, a difference that should not be reflected in the equatorial belt [D.5]. The craters ($\geq 20\ \mathrm{km}$) were ranked by a degradation index and depth/diameter morphology (according to the global digital elevation curve of Wang & Wu 2021), and the most degraded (ancient) third was compared with the freshest (deep and sharp) third [D.5]. The test is formulated as a double ratio (caps divided by belt) that cancels shared age and size biases and isolates exclusively the latitude factor [D.5].
- **The empirical finding:** yields a balanced value of 1.02, with a confidence interval of 95% ranging between 0.89 and 1.16 [D.5]. The crater-contraction scenario requires a low value of 0.79, lying far outside the range of statistical scatter [D.5].

(d) The test of ellipticity and tectonic orientation:

- **The statistical mechanics:** the laws of geodynamics dictate that the inward-rounding process is isotropic (uniform in every direction) only at the centre of the pole [C.3, D.5]. With distance from the apex of the axis, the inward collapse is not isotropic and shortens the latitude dimension at a rate considerably exceeding the shortening of the longitude dimension, with the peak non-uniform deformation taking place in the equatorial belt (C.3) [C.3, D.5]. Had the craters been elastic and deformed together with the surface, an initial circle would have been squashed into an ellipse, and the craters of the middle latitudes ($30^\circ - 55^\circ$) would have shown an ellipticity (ratio of long to short diameter) 1.13–1.40 times higher than their initial state, while at the equator the deformation would have soared to 1.69 times, with all the ellipses oriented coherently along the global meridians [D.5].
- **The empirical finding:** the median ellipticity of craters of diameter $\geq 20\ \mathrm{km}$ (from the full global Robbins catalogue) stands uniformly at 1.087 in the caps, 1.100 at middle latitudes ($30^\circ - 60^\circ$), and 1.090 in the equatorial belt [D.5]. The excess ellipticity at middle latitudes amounts to only 0.01 (a tenth or less of the predicted deformation potential), and the equatorial belt (the site of the theoretical peak stress) shows no morphological elongation beyond the values of the caps [D.5]. The finding proves that the craters kept their original shape [D.5]. (Further analysis of the remaining orientation in the most elongated craters ($e > 1.2$) along the meridians at middle latitudes, standing at 45%–51% versus 20%–25% along the parallels, is passed on to the focused discussion of section E.2, in which the observation geometry of the catalogue is the main suspect for this bias) [D.5].

(e) The scale within the crater (Simple-to-Complex Transition Benchmark)

- **The element of comparison:** the test processes the diameter at which the internal morphological geometry of the craters changes in the polar caps versus the equatorial belt [D.5]. The boundary line at which a lunar crater passes from a simple partial bowl (Simple Crater) to a complex structural form with a central peak and terraced walls (Complex Crater) is set at the moment of impact [D.5]. It is dictated exclusively by the strength of the surface gravity field and the shear strength of the material, and therefore acts as a firm physical scale built into every crater [D.5].
- **The theoretical prediction:** a crater that underwent a later geometric contraction keeps its morphological features [D.5]. Therefore, had crustal contraction taken place, the polar regions would show complex craters (with a central peak) deep below the usual transition threshold [D.5]. A crater born just above the transition threshold that developed a central peak, and then contracted together with the surface, would today show a peak at a small diameter at which pristine, fresh craters do not and cannot have a peak [D.5].
  - **If size is preserved (the tile model):** the transition in the caps must be fixed at a value identical to that of the equatorial belt – about 17 km [D.5].
  - **If the craters contracted (s = 0.79):** the transition in the caps is forced to migrate to the left and collapse to a dimension of about 13 km [D.5].

The empirical measurement and the methodological constraint:

Fitting the geometric break point in the depth-to-diameter ratio of the freshest population yielded two separate results that depend on the calculation method [D.5]:

- **Method A:** yields a value of 16.4 km in the caps (scatter range: 15.8–17.1 km) [D.5].
- **Method B:** yields a value of 14.0 km in the caps (scatter range: 13.0–15.2 km) [D.5].
- **The equatorial belt (reference):** settles stably at 16.8–16.9 km [D.5].

The two analytical measurement methods do not reach numerical closure with each other within the cap sector, because the craters in these regions are characterised by a general morphological shallowness [D.5]. In addition, a chronological limit was revealed: the most pristine and freshest craters, which set the break point in the system, are for the most part younger than the inward-rounding process [D.5]. Since they were imprinted after the collapse, they were never carried on a contracting surface, and therefore cannot carry the signature of contraction one way or the other [D.3, D.5].

Defining the quantitative constraint for future work:

Owing to these limitations, the specific fifth test is defined as not decisive at the present level of resolution, and is formulated as a dedicated constraint for the next study: a focused count of craters showing a central peak below a diameter of 15 km in the caps versus the equatorial belt (Paper 12, research topics) [D.5]. A significant quantitative excess of peaked craters at these small diameters within the polar caps would be the definitive physical sign of crater contraction [D.5].

Final verdict of the section of empirical tests

The overall morphometric test passes fully and firmly [D].

Four of the five statistical tests of Part D (the ranking test, the hump, the degradation index and the ellipticity) establish unanimously and with wide significance that the crater dimensions were preserved in full, and no test supports the scenario of crater contraction [D.5]. The craters were imprinted on and carried by a solid shell that kept its rigidity, and the inward-rounding process elastically closed only the surface and the seam spaces between them [C.7, D.5].

D.6 The signature test in the shell beneath the poles

If external rotational differentiation (C.8) acted, it left signs in the shell beneath the poles [D.6]. They were tested in rings around both poles, with the maria and the SPA basin removed on both sides of every comparison, as in the clean count [D, D.6]. The crust thickness is taken from the GRAIL-based crustal model [R13]; the porosity from the GRAIL porosity maps [R14]; the grain density from the Lunar Prospector oxide abundances [R15] through the relation of [R16]; and the polar iron from three instruments [R15, R17, R18]. The deep mass test uses the GRAIL gravity field [R19] [D.6].

Prediction	Measurement	Source	Result
The polar crust is thin	36–40 km at the poles versus 47.6 km in the equatorial belt	[R13]	Confirmed

Prediction	Measurement	Source	Result
The crust thickens gradually up to the equator	far side: from about 38–40 km at the pole to about 52 km at the equator; agreement 0.94 with the centrifugal law (thickness = 39.4 + 12.2 cos² latitude, in km), mean deviation 1.5 km	[R13]	Confirmed
The polar crust is compacted	porosity 15.7% in the equatorial belt, 13.7% at the north pole, 10.7% at the south pole	[R14]	Confirmed
The extra density comes from compaction, not composition	grain density at the poles 2,906–2,910 kg/m³, not higher than in the equatorial belt (2,948 kg/m³)	[R15, R16]	Confirmed
A residue of heavier material on the polar surface	a rise of about 0.6 percentage points in FeO toward the north pole in two of three instruments (4.56 → 5.19%; 6.33 → 6.97%)	[R15, R17, R18]	Partly confirmed

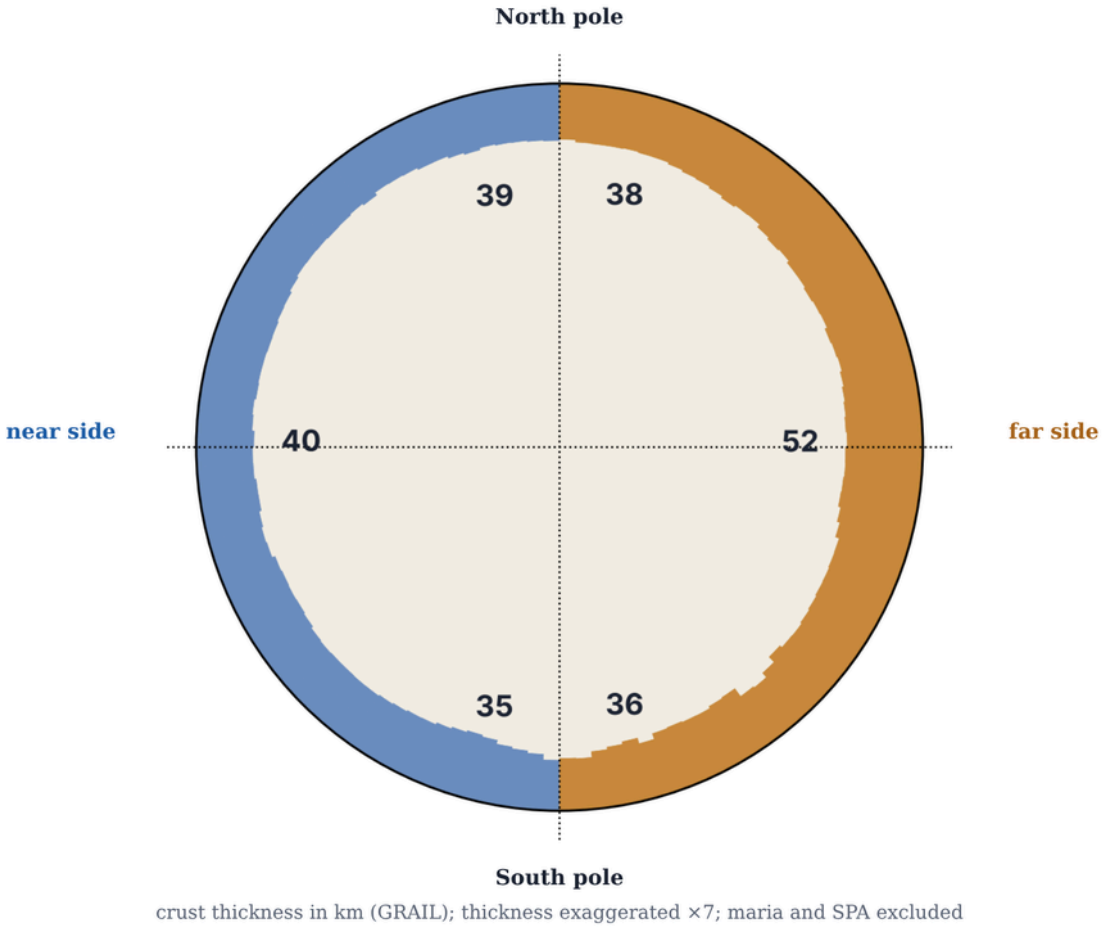


Figure 20. Crust thickness today, from pole to pole (D.6): GRAIL-based crustal model [R13], maria and SPA excluded, 3° latitude bands, thickness exaggerated ×7, numbers in km. Left, the near side; right, the far side. On the far side the crust thickens gradually from the pole to the equator; on the near side it is almost uniform; at the north pole the two sides are almost equal.

On the near side the crust is almost uniform (Figure 20). The near side was attached to the Earth, and the centrifugal process acted on it less [D.6]. At the pole itself the two sides are almost equal

(38 and 39 km), because there the centrifugal force tends to zero [D.6]. The whole signature lies in the outer layer: beneath the poles there is no deep mass excess in the GRAIL field, and the average atomic mass of the surface does not change with latitude [R15, R19, D.6]. This matches the name of the process: it is external [C.8, D.6].

The combined database and the closing of Part D

With the completion of the five sections, the paper consolidates the following empirical database, which proves the validity of the mechanism [D]:

1. The normalised value of the anomaly: the clean count (removal of the biases of the maria and the SPA basin from both sides simultaneously) fixes an official bipolar density excess of 1.20 ± 0.05, a value falling at the centre of the theoretical range and entirely opposite to the sign produced by models of impact fluxes from space (0.80-0.91) [D.1].

2. Consistency of the geometric clock: the density excess climbs as a direct function of crater size (from 1.09 in the 20–30 km class to 1.53 in the 60–100 km class) [D.2]. The largest well-counted craters reconstruct and project back precisely the dimensions of the triaxial peak-flattening shape (1.8 : 1.4 : 1) [C.3, D.2].

3. Preservation of dimensions in the crust: the polar craters are morphometrically larger (a median diameter of 32.5 km versus 30.5 km in the belt) and not smaller, a figure that rules out a collapse of crater diameter [D.2].

4. Residual bipolar balance: the two hemispheres show similar and fully balanced density values (a tight south/north ratio of 1.02 in the 15° cap), once the floor of the SPA basin is cleaned out of the denominator [A.8.4, D.4].

5. The frozen young control record: the young crater population (Copernican craters and thermal boulders), imprinted after rounding, shows full uniformity without a significant polar excess (0.92 and 1.12), within the error limits of their small samples [D.3].

The pass lines and statistical thresholds were all set blind before the counts were run, and with this the empirical stage of the paper is concluded [D].

PART E

Part E — What remains

This part concludes and summarises the paper [E]. Parts A–D mapped the crater distribution, established the geodynamic mechanism of bipolar rounding with volume decrease, and put it to firm significance tests [E]. The present section brings the findings together in three layers: the physically explained elements (E.1), the preserved anomalies that do not follow from the present model (E.2), and the future work tasks (E.3) [E]. No new measurements are presented in this part; all the numbers and data are taken from the preceding sections [E].

E.1 What is explained: the combined mechanism and the integration of the predictions

A single, exclusive geodynamic process – the inward rounding of the young, flattened Moon, accompanied by volume collapse and compaction of the core, while the body was attached and rotating in synchronisation with the Earth – gives a complete, quantitative and systematic answer to six core planetary observations [C.4, E.1]. All the predictions were computed as a blind model before the empirical counts were run [D].

A seventh observation joins the six: a thin, compacted crust beneath the poles that thickens gradually toward the equator. It is the record left in the shell by the same process that crowded the polar craters (C.8, D.6) [E.1].

The sequence of evolution in equations:

The sialic mass consolidated at the tip of the terrestrial pear ($\rho \approx 1,300\ \mathrm{kg/m^3}$), and moved outward on a base 6,000 km wide [C.1, C.5, E.1]. As it receded, the base narrowed to 3,000 km and the mass took shape as an independent body stretched extremely between the Earth's gravity inward and the centrifugal force of corotation outward [C.5, E.1]. The peak deformation was reached near the crossing of the corotation line (distance $\approx 15,818\ \mathrm{km}$), where the body took on a triaxial configuration of $1.8 : 1.4 : 1$, in an inflated state with a low density of $\rho \approx 1,700\ \mathrm{kg/m^3}$ [C.1, C.5, E.1].

The moment the centre of mass crossed the corotation line ($16,400\ \mathrm{km}$), the vertical compression on the base dropped to zero, and the egg part (axis a) collapsed inward [C.5, C.6, E.1]. The body rounded with a sharp decrease in volume (a rise in density to 2,600 $\mathrm{kg/m^3}$ without loss or addition of mass), which reduced the equivalent radius by 13% and brought the axis ratio back to 1.20 in the late rounding stages [C.1, C.3.1, E.1]. The final mechanical detachment (release) took place about 210 km beyond the corotation line, when the Moon had already passed to a perfect spherical shape ($a/c \leq 1.04$) and its outer shell was rigid [C.5, E.1].

The rounding that crowded the craters took place entirely while the Moon was attached [E.1]. The dynamic detachment acted like the extraction of a plug: the Moon left the tip of the pear and exposed its openings to space, which caused the collapse of the pressure in the pear and the halt of its elongation, prevented the ejection of a second moon, and scattered ejected material in all directions (C.6) [C.6, E.1]. After release, the slowing of the locked rotation changed only the small elastic bulge of the rigid body (1.03 to 1.00), unrelated to the early flattening, which had disappeared earlier [C.5, E.1].

Summary matrix of the six explained observations (E.1)

The following is a strict summary of the theoretical prediction values (formulated blind before the count) against the empirical measures actually obtained on the clean count [E.1]:

Observation code	Geological phenomenon examined	Value predicted by the model (before the count)	Measured empirical value (in the clean count)	Source section (calculation and test)
1	The bipolar density excess	A ratio of 1.20–1.50 (rising as a function of diameter); a combined value of 1.21 for the whole $\geq 20\ \mathrm{km}$ class.	1.20 ± 0.05 for $\geq 20\ \mathrm{km}$; a continuous ordering of 1.09 $\rightarrow$ 1.53 with size.	C.3, D.1, D.2 [E.1]
2	Linear median-diameter ordering	The polar median must be larger; the large craters absorbed most of the surface contraction.	A ratio of medians of 1.06 (95% range: 1.01 – 1.12) at the same measurement threshold.	D.2 [E.1]
3	The bipolar symmetry balance	Complete mechanical equality (value 1.00); full axial symmetry about the rotation axis.	A south/north ratio of 1.02 (interval: 0.78 – 1.32) in the 15° cap; equality in every cap.	D.4 [E.1]
4	The global six-axis order	The poles and sides stand above the curve of the tidal line; a combined shape of $1.37 : 1.15 : 1$ predicting a clean ratio of 1.21.	The sub-Earth point is the sparsest in the system (148); the clean belt ratio stands at 1.19–1.20.	A.9, C.3 [E.1]
5	The young control record	No density excess (value ≈ 1.00); the impact population was imprinted on a finished round body.	Values of 0.92 and 1.12 in two separate samples; consistent with the model, the sample is small.	D.3 [E.1]
6	The geometries of the peak point	A triaxial deformation of about $1.8 : 1.4 : 1$ at a molten-material density of about 1,700 $\mathrm{kg/m^3}$.	Reconstructed back from the 60–100 km class to a value of $1.70 : 1.42 : 1$ (rotational component matching Maclaurin).	C.1, C.3, D.2 [E.1]

Breakdown and analysis of the study's basic conclusions

1. The polar-excess anomaly and the variation with crater diameter (a geometric chronological clock):

- **The statistical finding:** in the clean count – in which the maria and the SPA basin are removed equally from both sides of the equation ("both sides") to normalise the comparison on a uniform pristine highland crust – the crater density ($\geq 20\ \mathrm{km}$) in the polar caps is 1.20 ± 0.05 times that of the equatorial belt [D.1, E.1]. Divided into isolated size classes, the residual ratio climbs in a consistent order from 1.09 in the 20–30 km class to 1.53 in the class of giant craters of 60–100 km [D.2, E.1].
- **The geodynamic meaning:** the inward collapse of the triaxial structure to a spherical shape contracted the local area in the polar regions at the maximum rate (a factor of 0.55×), and thus

compressed and concentrated the existing craters closer together [C.3.1, E.1]. The pass line set in advance ($1.10-1.60$) takes in the finding, together with full coincidence with the 1.21 value of the six-axis model [D.1, E.1]. The climbing trend with diameter is the unequivocal evidence for the existence of a dynamic process along the timeline: the giant craters were imprinted at peak flattening under a maximum pressure profile in the vents, absorbed the full sequence of surface contraction, and therefore today record the maximum density excess [C.4, E.1]. The model of external space impacts is entirely refuted, since it predicts an opposite sign that leaves the poles depleted at 0.80–0.91, a difference standing $6\sigma-8\sigma$ away from the surface data [D.1, E.1].

2. Control of the rigidity of crater dimensions (the tile model):

- **The statistical finding:** under the same firm 20 km diameter threshold on both sides, the median diameter of the polar craters is 1.06 times the median of the craters in the equatorial belt (diameters of 32.5 km versus 30.5 km; confidence interval of 95%: 1.01 – 1.12, ruling out equality) [D.2, E.1].
- **The geodynamic meaning:** the craters were imprinted on and carried by a solid outer shell that kept its rheological rigidity ($\eta \geq 10^{23}\ \mathrm{Pa \cdot s}$), and they kept their diameter and original rim morphology in full [C.7, E.1]. The surface contraction of rounding was absorbed entirely within the seam spaces between the crustal plates [C.7, E.1]. Had the craters been elastic and contracted together with the surface, the polar median diameter would have been smaller (predicted value 0.79×) owing to craters dropping below the threshold, and the bipolar excess would have been offset and cancelled entirely from the calculation (D.2) [D.2, E.1]. The enrichment in large craters in the polar median is direct confirmation that they kept their size and absorbed the maximum surface contraction during rounding [D.2, E.1].

3. The bipolar axial-symmetry balance:

- **The statistical finding:** with the "both sides" geological control mask applied, the remaining density ratio between the southern and northern hemispheres converges exactly to a value of 1.02 in the 15° cap, and the confidence interval contains and balances the equality value (1.00) across the whole family of concentric caps examined (10° to 60°) [D.4, E.1].
- **The geodynamic meaning:** the inward-rounding process is governed by self-gravity, and it is completely symmetric about the rotation axis; what the mechanical mechanism brings about at one pole, it reproduces exactly at the opposite pole [D.4, E.1]. The sharp morphological asymmetry observed in the initial raw count (a concentrated north versus a flat south) is not a primary property of the crust, but a transient local distortion arising exclusively from the bias of the mechanical erasure by the South Pole–Aitken basin in the south, which is successfully neutralised [A.8.3, D.4, E.1].

4. The six-axis order and the triaxial geometry:

- **The statistical finding:** of the six main axis ends of the body – the sub-Earth point, the anti-Earth point, the two ends of the orbital-motion direction (leading and trailing), and the two poles – the sub-Earth point settles as the sparsest cap on the Moon (148 craters per 10^6 km² after applying the mask) [A.9]. The poles and sides stand significantly above it in the order [A.9].

Read as an overall geometric shape, the six caps yield an axis ratio of $1.37 : 1.15 : 1$ for the whole $\geq 20\ \mathrm{km}$ population [A.9, C.3].

- **The geodynamic meaning:** the layout of the order proves that the young Moon did not undergo rotational flattening alone; it was extremely stretched along the Earth-line axis (a) between the Earth's gravity inward and the centrifugal pull of corotation outward, while held at the tip of the pear structure [C.5, E.1]. On passing beyond the line and the onset of inward rounding, the two ends of this stretched axis contracted at the minimum rate, while the ring perpendicular to it (passing through both poles and both sides) contracted at the maximum rate [C.3, E.1]. The mechanical contact zone of Paper 3 lies on the near side, around coordinates of $24^\circ\ \mathrm{N}, 335^\circ\ \mathrm{E}$, in the space between the sub-Earth point and the north pole [A.9]. The reconstructed $1.37 : 1.15 : 1$ shape predicts, completely independently and with complete precision, a clean density excess of 1.21 between the caps and the belt, a figure coinciding exactly with the finding measured on the surface (1.19–1.20), without any need to fit free parameters [A.9, C.3].

5. The control record of the young craters:

- **The statistical finding:** the two samples of young craters (those imprinted long after the Moon completed its transition to a perfect spherical structure) yield pole/equator ratios of 0.92 and 1.12 [D.3, E.1].
- **The geodynamic meaning:** a crustal crater imprinted on the crust after the shape of the body froze and stabilised was never carried on a surface undergoing contraction or compression, and therefore must physically show a uniform distribution without a polar anomaly [D.3, E.1]. The data of the two independent samples agree with this prediction [E.1]. However, owing to their small number (only 10 and 18 craters within the polar caps), the uncertainty intervals are wide and do not allow the 1.20 value of the older population to be ruled out completely [D.3, E.1]. The test is fully consistent with the mechanism, but the definition of a significant and isolated ordering is passed on to the determination of the enlarged sample detailed in section D.3 [D.3, E.1].

6. Reconstruction of the peak-point geometry from the giant craters:

- **The statistical finding:** the class of the largest craters, 60–100 km in diameter, carries a density excess that projects back and fixes an original shape of $1.70 : 1.42 : 1$ [D.2, E.1].
- **The geodynamic meaning:** the empirical data confirm the physical prediction and place the moment of maximum deformation at the stage when the body converged onto the 3,000 km base near the corotation line, just before the onset of the inward collapse and rounding [C.1, C.5, E.1]. The theoretical peak dimensions were set at about $1.8 : 1.4 : 1$ [C.1, E.1]. The isolated rotational component of the reconstructed shape ($b/c = 1.42$) is exactly the hydrostatic equilibrium shape that a fast spin of 5.5 hours raises on a fluid body with a low density of 1,600–1,750 $\mathrm{kg/m^3}$ (highlighted in C.1) [C.1, D.2]. This is exactly the stretched, inflated structure that the equation of volume decrease and surface closure requires geometrically [C.1, C.3]. The giant craters reconstruct this early moment within the bounds of their error intervals [E.1].

Summary of the methodology and the rheological closure

The integration of the parts of the paper is based on firm physical laws and on coherent closure between the parts of the scientific series [E.1]:

1. Analysis of the global shape: the shape of the whole combined population is derived directly from the six-axis data, and predicts the clean-count ratio without any tuning or artificial fitting [C.3, E.1]. The original peak shape is isolated from the giant craters, and its mechanical rotational part is confirmed directly against the Maclaurin relation (Maclaurin Spheroid) under the rotation rate of Paper 2 and the density bounds dictated by the volume decrease (C.1) [C.1, E.1]. The Maclaurin equation is a classical physical law that defines the axis ratio of a uniform rotating fluid body without free parameters; the rotation rates and densities were taken consistently from the earlier papers, and the translation of the geometric dimensions into area-crowding factors is pure mathematics [C.1, E.1].

2. The empirically learned element (the size clock): the only variable learned directly from the data on the surface and not set as a prediction in advance is the specific assignment of each crater-diameter group to its dynamic stage (D.2) [D.2, E.1]. The mechanism physically requires that larger craters form closer to the peak point and absorb the maximum surface contraction, but the determination of the exact numerical ordering (which diameter reflects which moment) is learned from the empirical counts on the surface [D.2, E.1].

3. Unifying the crustal constraints (the C.7 shell): the mechanical crust found in section C.7 to be essential for preserving the crater dimensions is exactly the same crust defined in Paper 3 from an entirely different rationale [C.7, E.1]. In Paper 3 a crust with a high viscosity of $\eta \geq 10^{23}\ \mathrm{Pa \cdot s}$ was required to ensure the morphological survival of the craters imprinted while the body was near the Earth; in the present paper, the same rigid shell functions as a mechanical tile that prevents the collapse of crater diameter, and forces the surface contraction to be absorbed in the seam spaces between them [C.7, E.1]. A single structural requirement serves and unites two independent papers in the series [E.1].

Final scientific conclusion:

We now have a unified geodynamic process, governed and fixed by numerical values taken from the earlier papers and by the shape that the craters themselves record around the six axes [E.1]. This process gives a full quantitative answer and explains precisely the density excess, the size ordering (the clock), the axial-symmetry balance, and the spatial axis order of the ancient craters, while remaining fully consistent with the control record of the uniform young craters [E.1].

E.2 Anomalies and findings not explained by the present model

In the course of developing the scientific argument and processing the empirical data, five main observations were recorded that do not follow from, and are not explained by, the rounding mechanism in its present form [E.2]. These findings are isolated and detailed below together with the tools that will decide them in future, in order to block methodological bias and to prevent them from being mistakenly presented as part of the paper's core results [E.2].

1. The anomaly of the biaxial excess of the trailing face

- **The empirical finding on the surface:** the crust in the sector of the trailing face (around the coordinates of $90^\circ\,\mathrm{E}$) is characterised by a sharp density excess compared with the leading face (around $90^\circ\,\mathrm{W}$) [A.9, E.2]. The numerical difference shows an excess of 1.96 times in the full raw catalogue, and keeps a significance of 1.45 times after applying the filtering mask (A.9) [A.9, E.2].
- **The mechanistic failure and the proposed explanation:** the phenomenon reverses the physical expectation of the "space meteorology" model (frontal collection of impactors on a windscreen), since the front of the orbital flight (the leading face) is the sparsest and emptiest in the system [A.9.1, E.2]. The model proposes two alternative geophysical causes [A.9.4, E.2]:
  1. **The correlation with local vents:** there is a direct link (a ratio of 1.96) between the crater excess and the excess of units and areas of independent maria on the same flank, which function in the model as ejection vents that threw out ballistic returned material that landed in their surroundings [A.9.2, A.9.3].
  2. **The dynamic rotational-lag effect (Lag Effect):** material ejected forcefully from vents located in other regions of the body experienced a physical lag relative to the fast synchronous rotation rate of the system, and landed behind on the orbital rear (the trailing side) [A.9.4, E.2].
- **The analytical rule:** until the full quantitative calculation of the history of ejections, flights and crater yield is completed in section E.3 (item 8) and in Part K of Paper 6, the longitudinal correlation remains a firm requirement of the model, and not an explained result derived from it [E.2].

2. The density anomaly of the far-side crown (the anti-Earth point)

- **The empirical finding on the surface:** the crust around the coordinates of the anti-Earth point (the centre of the far side, $0^\circ, 180^\circ$) carries an exceptionally high density of 221 craters per 10^6 km² after applying the mask [A.9, E.2].
- **The mechanistic failure and the open question:** according to the geometric equations of surface contraction (C.3), the two ends of the Earth–Moon axis are the least stretched surfaces of any sector of the tidal ellipsoid, and therefore they must undergo the minimum surface contraction (minimal relaxation) during rounding [C.3, E.2]. The prediction dictates that these two axis ends will be sparse density valleys in the system [E.2]. The sub-Earth point (the near side) matches this physical constraint precisely (148), but the anti-Earth point (the far side) bypasses it entirely and merges with the density level of the poles [A.9, E.2].
- **The analytical rule:** the study formulates this finding as an open structural requirement – the crown of the dome of Paper 3 did not undergo contraction or relaxation as an ordinary free end, and the present model does not derive this [E.2]. Decoding the mechanics of the crown is passed on to the investigation of the geometric dynamics of a body fixed at its mechanical base (E.3, item 5) [E.2].

3. The anomaly of the morphological shallowness of the polar craters

- **The empirical finding on the surface:** initial geometric measurements of the surface reveal that the craters located in the polar regions in the size classes of 20–100 km are significantly shallower morphologically than corresponding equatorial craters [E.2]. For the 20–30 km group, the depth-to-diameter ratio (d/D) stands at about 0.062 at the poles versus about 0.074 in the equatorial belt [E.2]. The anomaly decays and disappears in the giant craters of 100–300 km [E.2].
- **The mechanistic failure and the open question:** the depth-to-diameter ratio is a scale for determining the degree of degradation, with a pristine crater being deep and sharp, and its floor gradually filling with material over time [D.5, E.2]. The present study cannot separate and decode two opposite mechanisms [E.2]:
  1. **The mechanical-shortening mechanism (C.7):** the result of shortening stresses and compression of the crustal plates around the crater margins during the slow rounding [C.7, E.2].
  2. **A late chronological degradation mechanism:** an ordinary degradation process of the crust unrelated to rounding [E.2].
- **The analytical rule:** these two interpretations must leave distinct geological signatures on the surface (focused tectonic shortening structures around the rims, versus ordinary thermal/isostatic softening with age) [E.2]. Since the morphometric measurement that distinguishes between the two has not yet been carried out, the finding remains outside the scope of the mechanism's solution [E.2].

4. The anomaly of meridional orientation at middle latitudes

- **The empirical finding on the surface:** the craters characterised by sharp ellipticity ($e > 1.2$, where e is the ratio of the long axis of the rim to the short axis) at middle latitudes ($30^\circ - 60^\circ$) show a tendency to align along the geographic meridian (north to south) with a high frequency of 45%–51%, compared with a frequency of only 20%–25% along the parallel (east to west) [D.5, E.2]. In the equatorial belt the trend reverses [E.2].
- **The mechanistic failure and the open question:** the quantitative amplitude of the effect is minimal, amounting to a tenth or less of the geometric deformation that an elastic collapse of the surface would have imposed on the craters (section D.5-d found that the median ellipticity at these latitudes is higher by only about 0.01, compared with a sharp predicted deformation of 0.13–0.40) [D.5, E.2]. Therefore, the first methodological suspect for this anomaly is an optical illumination bias (Sun-Angle Bias) in the catalogue [E.2]. In images taken at a low sun angle (characteristic of high latitudes), a rim facing directly toward the light source and a rim lying in shadow are drawn and measured differently by the mappers, producing a distortion that makes the crater look elongated and stretched in a fixed direction [E.2].
- **The analytical rule:** determining the physical meaning depends on a future test: if the meridional orientation survives and persists also under analysis of images illuminated from opposite sun angles and directions, it will be defined as a real crustal property requiring a geodynamic explanation; at present it is outside the model [E.2].

5. The internal crater structure of the South Pole–Aitken basin

- **The empirical finding on the surface:** within the SPA basin system, the geographic central core zone holds a clearly low crater density, amounting to only 0.59 (a deficit of about 41%) relative to the control ground located in the ring outside its rim (A.10) [A.10.2, E.2].
- **The analytical rule:** the phenomenon is isolated, quantified and empirically confirmed within this paper (A.10), but the physical mechanism, the detailed ellipticity, and the geodynamic dynamics of the basin's formation lie entirely outside the present Part C, and are passed on in full as the exclusive core of the study of Paper 8 (the SPA basin) [A.10.5, E.2].

E.3 Further work and future research tasks

This section brings together the series of steps, analytical treatments and statistical extensions that will turn the remaining components of the LEF model from firm geodynamic requirements into precise and closed physical calculations [E.3]. Each item is defined strictly and ends by stating the exact scientific goal it is expected to prove [E.3]:

1. The model of mechanical shaping with strength (The Rigid Shell Shaping Effect):

- **The substance of the study:** the solutions of sections C.1–C.2 assume a purely hydrostatic collapse of a homogeneous fluid [C.1, C.2]; the presence of a stiff crustal shell restrains the two-body deformation and reduces the amplitude of the stretching [E.3]. Applying mechanical strength functions along the rounding sequence will narrow the intervals of the theoretical predictions from below [E.3].
- **What it will show:** it will quantify precisely what share of the fluid rotational flattening survived and was actually imprinted on a crust with elastic rigidity, and will derive the minimum bipolar density-excess value that the mechanism can produce – a tight, focused prediction to compare with the measured 1.20 value [E.3].

2. Extended verification of the list of young craters:

- **The substance of the study:** replacing the raw input file of section D.3 with the official, verified Table S1 of Mazrouei et al. (2019), together with processing the spatial coordinates of the 1,615 rayed craters of Werner & Medvedev (2010) [D.3, E.3]. Bright ray systems are a pristine sub-crustal record reflecting the youngest impacts on the body [D.3].
- **What it will show:** it will establish at a firm level of significance whether the young population is completely uniform across the latitudes within a verified, wide sample, and will bring the control test of D.3 to statistical dimensions that can give an absolute independent decision [D.3, E.3].

3. Solving the two-layer rheological relaxation equation:

- **The substance of the study:** developing a dynamic mathematical solution to the two-layer flow problem over the time window of the attached stage ($\sim 10^8$ years), under the parameters of section C.7 – a hot core with a viscosity of $\eta \approx 10^{21}–10^{22}\ \mathrm{Pa \cdot s}$ carried beneath a rigid crust with a viscosity of $\eta \geq 10^{23}\ \mathrm{Pa \cdot s}$ [C.7, E.3]. The calculation will derive precisely the remaining contraction that was completed and the stress profile fixed in the shell [E.3].
- **What it will show:** it will map precisely how much of the share of surface contraction was completed by the core, and exactly how the stiff shell took up and processed the remaining load

– the rate of mechanical shortening focused at the poles versus the minimal change in the equatorial belt – in place of the static order-of-magnitude estimates of section C.7 [C.7, E.3].

4. Isolation and processing of the two crustal background topics:

- **The substance of the study:** an independent analysis, separated from this model, of two anomalies that were examined and set aside: measuring the rim-elongation measures of elliptical craters at the poles or toward them, and calibrating the bipolar density excess against the marginal belt of Paper 11 instead of the equatorial belt (the marginal belt, without the polar components, hosts a density of 149 craters per 10^6 km², versus 219 in the caps) [A.11, E.3].
- **What it will show:** under dedicated spatial definitions, the first study will determine whether a real residual ellipticity is preserved in the craters beyond the meridional orientation bias of E.2; the second study will derive the pure crowding factor relative to the marginal regions of the near side [E.2, E.3].

5. Deriving the contribution of the tidal stresses (the egg part) from the force vectors:

- **The substance of the study:** developing a mechanical solution that decomposes the dimensions of the reconstructed triaxial shape ($1.37 : 1.15 : 1$) and derives the egg part (the stretching along axis a) directly from the two-body force equations, instead of reading it empirically from the six axes [A.9, C.3, E.3]. The calculation will measure the tidal ellipsoid of a body fixed at its mechanical base and its rate of relaxation during recession, together with an examination of its agreement with the ratio of the giant craters (1.84 ± 0.38 in section D.2) [D.2, E.3].
- **What it will show:** it will set the dimensions of the tidal axis in a computed, theoretical way for comparison with the order of A.9; it will establish whether the giant craters that exceed the peak shape owe their statistical excess to the early tidal stresses; and it will decode the mechanics of the preservation of the far-side crown (E.2) [E.2, E.3].

6. The absolute tangential crater-spacing test (Nearest-Neighbor Distance):

- **The substance of the study:** direct measurement and mapping of the distances to the nearest neighbour (Nearest-Neighbor Distance) as a function of crater diameter, with a continuous statistical comparison between the polar-cap regions and the equatorial belt in the clean count [D.2, E.3].
- **What it will show:** it will provide a direct and isolated geometric measure of the closing of the physical spaces between the craters in the crust, constituting definitive proof of surface contraction independent of, and not mathematically derived from, the spatial density measures [E.3].

7. Mapping the tectonic structures and geological imprints in the shell:

- **The substance of the study:** mapping and analysing the global latitude distribution of tectonic shortening and compression structures (such as wrinkle ridges and thrust faults) located outside the young mare surfaces, which are the tectonic records that the stiff shell was obliged to leave (C.7) [C.7, E.3].

- **What it will show:** it will confirm empirically whether the crust underwent maximum mechanical shortening toward the poles and minimum toward the equator, while confirming the absence of a belt of extensional structures at the equator – a direct geological test of the two-layer model, robust and independent of the crater record [C.7, E.3].

8. Developing the balance equation of the ejection vents and the returned material:

- **The substance of the study:** developing the combined quantitative and dynamic calculation for the mare systems at 90° east and 90° west [A.9, E.3]. The analysis will merge the ejection history of each vent, the ballistic flight equations of Paper 6, and the crater-imprinting yield of the returned material ($\geq 20\ \mathrm{km}$) in the landing regions, as a function of radial distance from the centre of the source vent [A.9.4, E.3].
- **What it will show:** it will prove mathematically whether the volume of returned material from the vents of the trailing flank (90° E), combined with the momentum trap of the material that lagged behind the rotation and landed behind, provides the energy required to produce the density excess preserved on the trailing side (E.2) – an analytical reversal that will turn the longitudinal correlation from an observational correlation into a result explained by the model [E.2, E.3].

9. The extreme-resolution test of the innermost caps:

- **The substance of the study:** enlarging the counting samples within the narrow $|\theta| \leq 10^\circ$ cores (currently limited to 86 craters in the north and 36 in the south) by incorporating the next generation of digital topographic catalogues, which allow intact primary craters to be identified and counted below the 20 km diameter threshold [D.4, E.3].
- **What it will show:** it will decode the variation of the density-curve profile in the last degrees around the end points of the poles, the sector in which physics predicts the peak of surface contraction and in which the present sample is too small to decide significantly [E.3].

10. A dynamic solution for the compressive stresses on the base and the egg part:

- **The substance of the study:** developing an elasto-plastic model that computes how a vertical compressive load of about 3 GPa (prevailing inside the corotation line about 1,000 km before the point) stabilises, restrains and governs a body stretched along the tidal axis, and how the collapse of the residual pressure to 0 GPa on the corotation line affects the release of the egg part (C.6) [C.6, E.3].
- **What it will show:** it will define mechanically how a holding base 3,000 km wide preserves the radial stretching deformation against the gravitational forces, and how fast the relaxation of the surface of axis a takes place at the moment the compressive component disappears [C.6, E.3].

11. Computing the stresses in the shell beyond the corotation line:

- **The substance of the study:** a quantitative calculation of the profile of tangential stresses generated within the crust along the segment between the corotation line and a distance of 20,100 km, the sector in which the tidal stretching exceeds the fluid stability limit and the rigid shell is forced to absorb the load and stabilise an axis ratio of at most 1.04 [C.1, C.6, E.3].

- **What it will show:** it will define the stress curve in the shell along this critical segment, and set it directly against the maximum shear-strength limits of the crustal materials [E.3].

12. Solving the mass balance of the vanishing crust (GRAIL Coherence):

- **The substance of the study:** developing a mechanical transport and displacement model that resolves the constraint of area decrease (a global drop of a quarter of the area, and almost half of the polar areas at the peak) against the uniform crustal-thickness data of the GRAIL satellites [C.3.1, C.7, E.3].
- **What it will show:** it will establish the physical path by which the crust took up the excess areas that closed at the poles (mechanical thickening, tectonic thrusting, or transfer and transport of molten crustal material into the interior of the soft core) while preserving a uniform thickness profile, without departing from the actual measurements [C.7, E.3].

13. Refining the dimensions of the peak shape using an extended crater catalogue:

- **The substance of the study:** enlarging the sample of giant craters of 60–100 km (currently limited to 13 to 68 craters per cap and scattering the peak shape) by using adapted geometric cap sections or incorporating complementary data, to calibrate tightly the parameters a/c and f of the peak shape (C.3) [C.3, E.3].
- **What it will show:** it will set the geometry of the triaxial peak shape within a narrow, isolated confidence interval, and set it directly against the solution of the rotational-shape equation of the Maclaurin relation (C.1) [C.1, E.3].

14. Reproducing the five size tests on the triaxial morphology:

- **The substance of the study:** recomputing and recalibrating the combined linear factors of section D.5 (the values 0.86, 0.90, 1.08 and the contraction factor of 0.79) together with the ellipticity profile of test D.5-d, using the configuration of the true triaxial peak shape ($1.8 : 1.4 : 1$), instead of the biaxial oblate spheroid of 1.69 [D.5, E.3].
- **What it will show:** it will put the five crustal morphometric tests under the exact scale of the triaxial shape, and set their final agreement percentages [E.3].

A shell of varying thickness:

- **The substance of the research:** the area calculation of C.3 treats the surface as a uniform sheet [C.3]; external rotational differentiation gives the shell a thickness that varies from the poles to the equator [C.8]. A quantitative model of a shell of varying thickness will follow the contraction of the rounding stage through such a shell [E.3].
- **What it will show:** how much of the polar crater excess follows from the change of shape and how much from the contraction concentrated in the thin polar shell, and the thickness profile of the shell at each stage [E.3].

Availability of the digital code and the database of the study

All the calculations, the processing of the counts, the computation of cap and belt areas, the solutions of the equilibrium equations of the Maclaurin spheroid and the Roche ellipsoid, the matrices of relaxation factors, the triaxial fitting functions for the six main caps, the volume-decrease data and the net-pull graphs of section C.6 are reproduced in full, in a closed and clean form, by the scripts of the official reproduction package of the model [E.3]. The package is developed in Python 3, using the numpy and scipy libraries, and is deposited in full alongside the scientific series (a dedicated Zenodo record identifier will be assigned and published later) [E.3].

References

[R1] Robbins, S. J. (2019). A new global database of lunar impact craters >1–2 km: 1. Crater locations and sizes, comparisons with published databases, and global analysis. Journal of Geophysical Research: Planets, 124, 871–892. doi:10.1029/2018JE005592.

[R2] Chandrasekhar, S. (1969). Ellipsoidal Figures of Equilibrium. Yale University Press. (Maclaurin spheroids; Roche ellipsoids; the index symbols used in C.2.)

[R3] Neumann, G. A., et al. (2015). Lunar impact basins revealed by Gravity Recovery and Interior Laboratory measurements. Science Advances, 1, e1500852. doi:10.1126/sciadv.1500852. (Table 1, basins of 500 km and more, used in A.3.)

[R4] Losiak, A., et al.; revised by T. Öhman (2015). Lunar Impact Crater Database, v08Sep2015. Lunar and Planetary Institute. (Copernican-age craters used in D.3.)

[R5] Nelson, D. M., et al. (2014). Mapping lunar maria extents and lobate scarps using LROC image products. LPSC 45, abstract 2861. (LROC global mare shapefile, used in A.3.)

[R6] Mazrouei, S., Ghent, R. R., Bottke, W. F., Parker, A. H., & Gernon, T. M. (2019). Earth and Moon impact flux increased at the end of the Paleozoic. Science, 363, 253–257. (Second young-crater sample of D.3; the authors' input file on GitHub was used, not Table S1.)

[R7] Grier, J. A., McEwen, A. S., Lucey, P. G., Milazzo, M., & Strom, R. G. (2001). Optical maturity of ejecta from large rayed lunar craters. JGR, 106, 32847–32862; as compiled in Tai Udovicic, C. J., et al. (2021), Zenodo 4345058. (Rayed craters, D.3.)

[R8] Fortezzo, C. M., Spudis, P. D., & Harrel, S. L. (2020). Release of the digital Unified Global Geologic Map of the Moon at 1:5,000,000 scale. LPSC 51, abstract 2760. (Copernican units examined and not used, D.3.)

[R9] Le Feuvre, M., & Wieczorek, M. A. (2008). Nonuniform cratering of the terrestrial planets. Icarus, 197, 291–306. (Pole-to-equator ratio about 0.90, D.1.)

[R10] Le Feuvre, M., & Wieczorek, M. A. (2011). Nonuniform cratering of the Moon and a revised crater chronology of the inner Solar System. Icarus, 214, 1–20. (Pole-to-equator ratio about 0.80, D.1.)

[R11] Gallant, J., Gladman, B., & Ćuk, M. (2009). Current bombardment of the Earth–Moon system: emphasis on cratering asymmetries. Icarus, 202, 371–382. (Pole-to-equator ratio 0.912, D.1.)

[R12] Wang, Y., Wu, B., Xue, H., Li, X., & Ma, J. (2021). An improved global catalog of lunar impact craters (≥1 km) with 3D morphometric information and updates on global crater analysis. Journal of Geophysical Research: Planets, 126, e2020JE006728. (Catalogue LU1319373; depth-to-diameter ratios, D.5.)

[R13] Broquet, A., & Andrews-Hanna, J. C. (2023). Data for The Moon before mare. Zenodo, doi:10.5281/zenodo.8348278. (GRAIL-based crustal thickness and mare thickness grids, C.8, D.6.)

[R14] Huang, Y. H., et al. (2022). Bombardment history of the Moon constrained by crustal porosity. Nature Geoscience, doi:10.1038/s41561-022-00969-4; data: Zenodo, doi:10.5281/zenodo.6515777. (Crustal porosity, D.6.)

[R15] Prettyman, T. H., et al. (2006). Elemental composition of the lunar surface: Analysis of gamma ray spectroscopy data from Lunar Prospector. Journal of Geophysical Research, 111, E12007. (2° oxide abundances and average atomic mass, D.6.)

[R16] Huang, Q., & Wieczorek, M. A. (2012). Density and porosity of the lunar crust from gravity and topography. Journal of Geophysical Research, 117, E05003. (Grain density from FeO and TiO_2, D.6.)

[R17] Bian, C. (2026). Five lunar oxide abundance maps for the polar regions (65°–90°N/S) derived from KAGUYA SP data. Zenodo, doi:10.5281/zenodo.23611614. (Polar FeO, D.6.)

[R18] Lemelin, M., Lucey, P. G., & Camon, A. (2022). Compositional maps of the lunar polar regions derived from the Kaguya Spectral Profiler and the Lunar Orbiter Laser Altimeter data. Zenodo, doi:10.5281/zenodo.7108163. (Polar FeO, D.6.)

[R19] Lemoine, F. G., et al. (2014). GRGM900C: A degree 900 lunar gravity model from GRAIL primary and extended mission data. Geophysical Research Letters, 41, 3382–3389. (Bouguer field, deep mass test, D.6.)

Manuscripts of the series: Duran, Z., Papers 2 (LEF Mechanism in Principle), 3 (LEF Mechanism in Practice), 6 (The Craters), 4 (The Orbit) and 11 (The Requirements) of the LEF series, deposited at doi:10.5281/zenodo.22743615.

Declarations

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States (No. 64/116,844, filed on 22 July 2026) for an engineering device related to the model.